

PSMtags improve peptide sequencing and throughput in sensitive proteomics

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Mass spectrometry-based proteomics enables comprehensive characterization of protein abundance, function, and interactions. Label-free approaches are simple to implement but challenging to scale to thousands of samples per day. Multiplexed techniques, such as plexDIA, can address these limitations but remain restricted by the lack of mass tags optimized for data-independent acquisition (DIA) workflows. Here, we present a systematic approach screening a library of 576 compounds that identifies several small molecules that, when conjugated to peptides, improve their detection and sequence identification by mass spectrometry. The lead molecule, PSMtag, substantially increases the detection of fragment b-ions, which increases the confidence of sequence identification and enhances *de novo* sequencing. PSMtags allow 9-plexDIA, using only stable isotopes of carbon, oxygen and nitrogen. As a result, it allows simultaneously increasing proteome coverage and sample throughput for plexDIA workflows without compromising quantitative accuracy. We demonstrate 240 samples-per-day with 9-plexDIA, while acquiring 28,359 protein data points in the same time label-free methods acquire 4,340. Our approach constitutes an expandable framework for designing mass tags to overcome existing limitations in multiplexed proteomics and provides plexDIA reagents capable of analyzing over 1,000 samples per day when using 10 minute runs. By facilitating higher throughput and improved identification, this innovation holds significant potential for accelerating proteomic studies across diverse biological and clinical applications.

Introduction

Mass spectrometry-based proteomics has a long history of using isotopes for increasing sample throughput via isotopic labeling. Isotopes can be introduced via metabolic labeling^{1,2} or as mass tags that covalently modify proteins or peptides, as reviewed by Pappireddi, Martin, and Wuhr³. Mass tags can be used with many types of samples and have been adopted for clinical studies, such as the National Cancer Institute's Clinical Proteomic Tumor Analysis Consortium (CPTAC).

Early examples of mass tags include isotope-coded affinity tags⁴ and tandem mass tags⁵. The advantages of mass tags motivated successful commercialization, including isobaric and non-isobaric tags for relative and absolute quantitation (iTRAQ and mTRAQ, respectively)^{6,7} that allowed for combining up to four different proteome samples for simultaneous analysis. More recently, tandem mass tags (TMT, TMTpro)^{5,8–10} multiplex up to 35 samples. Some labeling strategies have been achieved with readily available reagents to achieve dimethylation^{11,12} or diethylation^{13,14} of tryptic or LysC-derived peptides. Other mass tags and amino acid reagents for multiplexing have highlighted do-it-yourself synthesis (DiLeu)^{15,16}, low energy fragmentation of isobaric tags (EASI-tag)¹⁷ or increased isotopic multiplexing by using mass defects^{18,19}. These chemical labeling strategies have allowed researchers to compare multiple biological conditions simultaneously, offering deeper insights into cellular processes, disease mechanisms, and biomarker discovery.

Despite these benefits, labeling with popular tags, such as mTRAQ and TMT/TMTpro, reduces peptide identification^{20–22}. In some cases, labeling favors the detection of b-ions²⁰, but it generally reduces the number of confidently identified peptides^{20,21}. Since previous reports support that at least some types of labeling alter peptide fragmentation, we reasoned other chemical structures might hold the potential to further enhance peptide sequence identification through improved production of annotatable fragment ions. Towards this goal, we started exploring the large chemical space that has not yet been evaluated for mass tags.

Advances in algorithms interpreting data from peptides collected in parallel by LC-MS/MS (data-independent acquisition, or DIA) now allow the broadest proteome coverage per unit of LC-MS/MS time, but mass tags compatible with this powerful approach remain limited^{21,23}. Simultaneously increasing the number of protein data points measured per sample per unit time by LC-MS/MS requires engineering new mass tags that incorporate more isotopes and preserve or enhance the quality of peptide sequencing. Using combinatorial chemistry and high-throughput screening, we developed a mass tag, called PSMtag, with the potential to allow the simultaneous increase of proteome coverage and sample throughput for DIA workflows without compromising quantitative accuracy. For instance, the 5-plex version of PSMtag allows 19,665 protein data points to be acquired in the same time a label-free workflow acquires 4,826 protein data points from single-cell level proteomes.

Results

Combinatorial synthesis

We performed a systematic evaluation of a range of chemical structures of mass tags for their effect on peptide sequencing quality when conjugated to peptides. Using readily-available building blocks, we employed on-peptide split-pool compound synthesis²⁴ to create hundreds of candidates. We synthesized these candidates directly on two model peptides (AVQVAHWASNEK, TLESS-GENLRLQK) immobilized on resin to limit the impact of synthetic yield and leftover reagents on analysis. The split-pool approach used sixteen trifunctional cores, five boronic acids, and five carboxylic acids as shown in Fig. 1a and Supplementary Fig. S1. This design allowed a combinatorial synthesis of 576 compounds (including expected products of partial synthesis) directly on the two model peptides.

Compound screen for beneficial properties

Of the 576 compounds, 116 were identified on both model peptides and evaluated for their effect on the confidence of matching mass spectra to peptide sequences (Fig. 1b). This confidence was measured by the Andromeda score, a value summarizing the similarity between empirical mass spectra and predicted mass spectra for a given peptide sequence²⁹. A lead candidate tag was identified (Fig. 1b, red) to maximize the Andromeda scores for both model peptides. We named it Parallel Squared Multiplexing tag (PSMtag) and synthesized multiple isotopologues by incorporating heavy atoms in the positions shown in Fig. 1c. These isotopologues allow for 9 tags spaced by 2 Da, and 5 tags spaced by 4 Da.

Validation on the human proteome

To further evaluate the properties of PSMtag beyond the model peptides, we characterized its effect on a broad range of endogenous peptides from the human proteome. To facilitate further characterization, the enantiomers of the racemic synthesis of PSMtags were separated by chiral chromatography. Only enantiopure tags were used for all following analysis in this study. To reliably label many peptides, PSMtag was functionalized with NHS-ester chemistry to label peptides derived from human proteome samples. NHS-functionalized PSMtags allowed for efficient and specific labeling of N-terminal and lysine amine groups, Supplementary Fig. S2.

Byproducts of NHS-ester labeling can be separated from tagged peptides in ion mobility and m/z space as shown in Fig. 1g. All data in this study were acquired with solutions by MS manufacturers originally developed to remove similar, unrelated byproducts from proteomic analysis by ion mobility and m/z . These solutions include high field asymmetric waveform ion mobility spectrometry (FAIMS) and trapped ion mobility spectrometry (TIMS). TIMS allows selective acquisition of peptide ions from tag byproducts acquiring only the signal for fragmentation and sequencing shown in the black boxes ("diaPASEF windows"). FAIMS effectively reduces the signal from tag byproducts several orders of magnitude as shown in Supplementary Fig. S3. Peptide ion transmission can also be impacted, but signal-to-noise improvements with FAIMS have been thoroughly discussed^{30,31}.

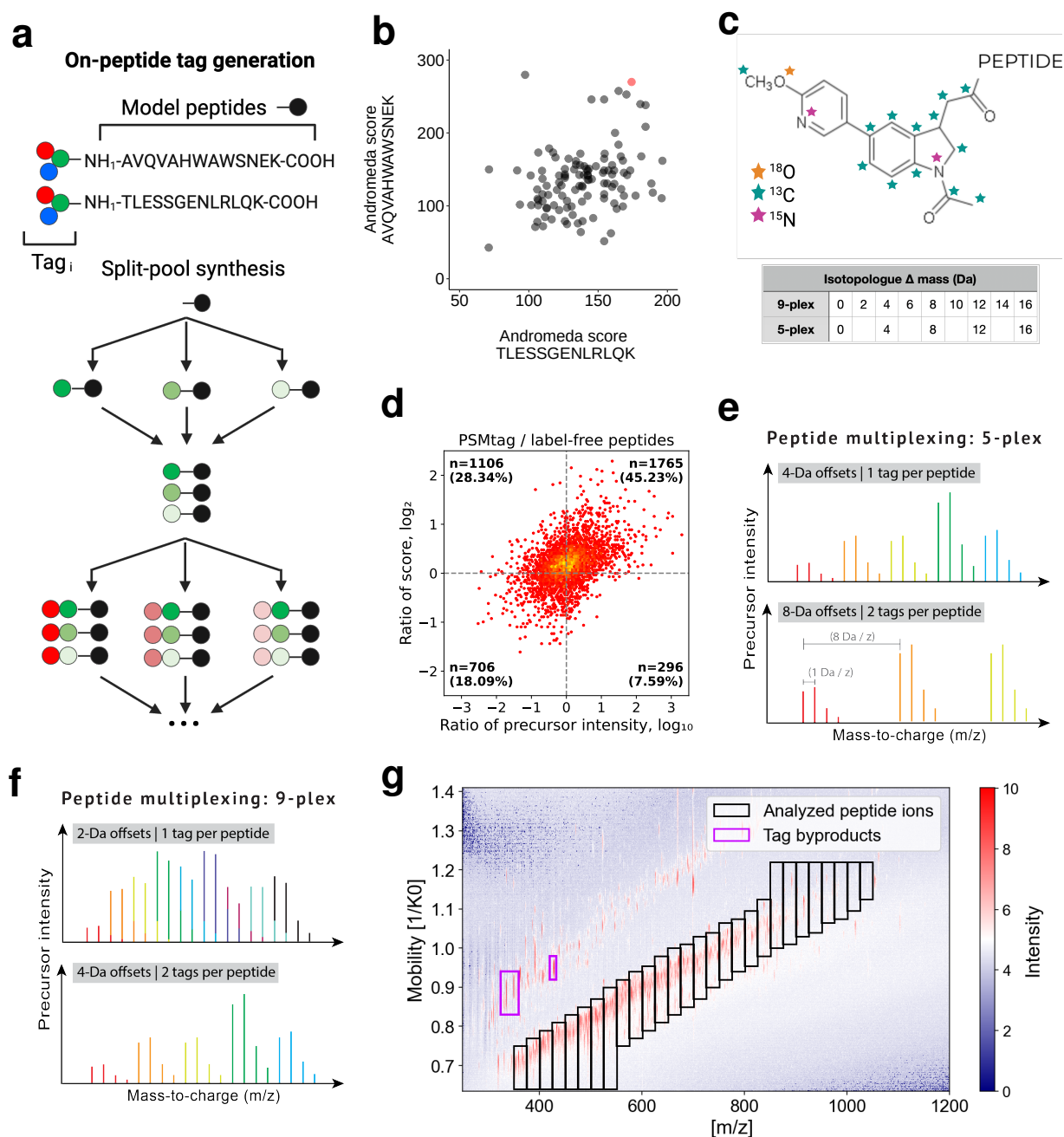


Figure 1 | A systematic approach to find tags that improve peptide sequencing and throughput in proteomics workflows. **a**, Two model peptides were synthesized on resin and tag candidates were combinatorially synthesized directly on the model peptides using a split-pool approach. **b**, Pools of tagged peptides were analyzed by LC-MS/MS in data-dependent acquisition mode and searched with search engine MaxQuant^{25,26}. A lead candidate was identified for further evaluation (data point colored red). **c**, The structure of the lead candidate, with sites of isotopic doping of carbon, oxygen, and nitrogen indicated. A 9-plex set of reagents separated by 2 Da and a 5-plex subset of reagents separated by 4 Da were synthesized. **d**, The lead candidate was synthesized as an NHS-ester and applied to a human proteome (K-562 cancer cell line, tryptic digest) for comparison of peptide sequence identification, as measured by ratio of score (Sage²⁷ hyperscore), conditioned on the ratio of precursor intensity. **e, f**, Illustration of isotopic envelopes for peptides from different samples multiplexed by PSMtag. The naturally-occurring isotopic envelopes for a given peptide are shown using the same color and the same peptide in different samples are shown with different colors. Overlapping signal may be deconvolved *in silico* by search algorithms that model the linear superposition, such as JMod²⁸. **g**, The excess tag remaining after conjugation to peptides (as a quenched or hydrolyzed product, shown in pink box) is separable from peptides by ion mobility as implemented by trapped ion mobility (TIMS) (shown) or field asymmetric ion mobility spectrometry (Supplementary Fig. S3).

To facilitate reliable comparisons of spectral quality, we examined the same peptide ions with and without tag while controlling for changes in their abundance (as estimated by precursor intensity). This systematic view is presented in Fig. 1d. The majority of peptides scored more favorably with the tag attached, with 73% of all peptides having higher scores when tagged across all peptide abundances. Even when the tagged peptides were apparently less abundant than their label-free counterparts, still more peptides scored higher when tagged (28% vs. 18%).

PSMtags improve the identification of peptide spectra

Database-matching sequence identification

We sought to identify the mechanism increasing scores for tagged peptides compared to label-free peptides. Given that peptide sequencing relies on detecting peptide fragments, we had the expectation that the improved peptide sequencing could be driven by increased observation of peptide fragments for tagged peptides. Tagged and label-free human proteome standards (K-562 cancer cell line, tryptic digest) were diluted to 2 ng input levels and analyzed by data-dependent acquisition on an Orbitrap Astral mass spectrometer. Tagged and label-free spectra for peptide ion FEELNADLFR (2+) exemplified an increase in score (Sage hyperscore²⁷) for the tagged peptide ion, primarily through the stabilization of the b-ion series (Fig. 2a). Scores for both the label-free and labeled peptide allowed confident assignment of the peptide sequence to the spectra.

To systematically characterize the effect of the tag on peptide spectra, we examined the percent of theoretical fragments expected per peptide that were observed empirically in the tagged and label-free spectra for the same peptides in Fig. 2b. The tagged peptides exhibited greater representation of expected fragments. Indeed, this allowed for greater overall fragmentation site coverage across all peptide lengths, as shown in Fig. 2c. We conditioned on peptide length for this examination because longer peptides require more unique fragments to be produced for confident sequencing than short peptides. For the set of peptides confidently assigned from both tagged and label-free samples, an average of 2.77 more observed fragments per spectra were annotated to an expected fragment when tagged (Fig. 2d). A primary difference in spectra between tagged and label-free peptides was the greater rate of observation of b-ions for tagged peptides, as shown in Fig. 2e.

De novo sequence identification

We also sought to determine if the mechanism increasing scores for database-matching approaches to peptide spectra identification would extend to modern *de novo* sequencing approaches³². We evaluated this possibility with peptides produced by GluC and ArgC as they better model immunopeptides that demand *de novo* sequencing for scientific and practical reasons. These peptides are more challenging to *de novo* sequence than peptides produced by trypsin because they do not have a positively charged basis amino acid at the C-terminus. Furthermore, on average, they are longer (ArgC and GluC) and/or contain more negatively charged residues (GluC)³³. For this reason, we chose proteomes digested by these alternative proteases to compare the efficacy of *de novo* algorithms to identify label-free and PSMtagged spectra.

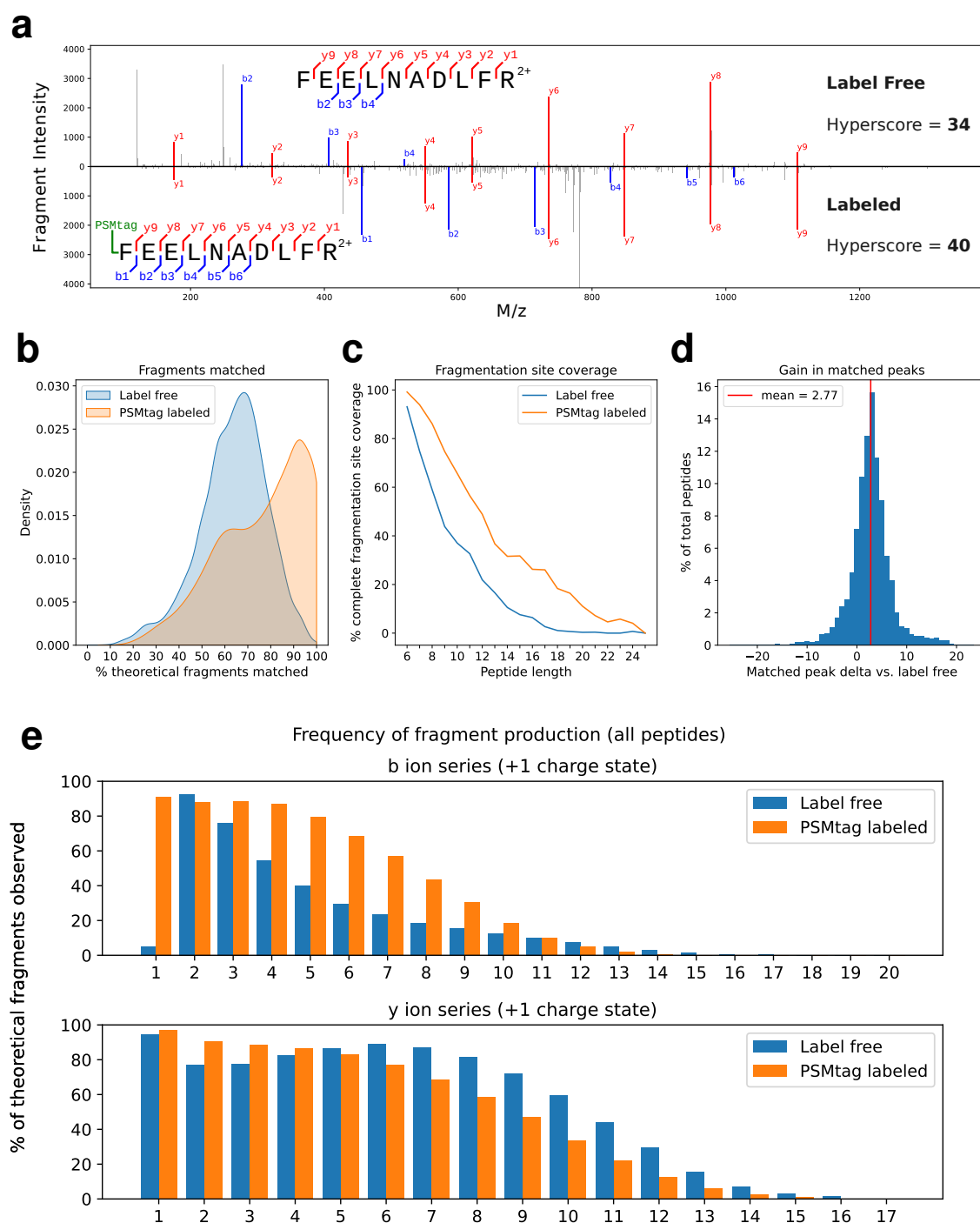


Figure 2 | Exploring improvements to peptide sequencing. **a**, Tagged and label-free spectra for peptide ion FEELNADLFR (2+) exemplifying an increase in score (Sage hyperscore) for the tagged peptide ion, primarily through the stabilization of the b-ion series. **b**, Distribution of the percent of theoretically-expected fragments from peptide sequences observed in their empirical spectra for both tagged and label-free samples. **c**, Percent of predicted fragmentation sites covered for tag and label-free spectra conditioned on peptide length. **d**, Distribution of the relative number of peaks in individual peptide spectra that can be assigned to theoretically-expected fragments. **e**, For all peptides confidently assigned in tagged and label-free populations, the percent of theoretical fragments observed according to their ion type: b-ions (containing N-terminus of peptide) and y-ions (containing C-terminus of peptide). The number series indicates the number of amino acids in that ion. 6

A set of high-confidence peptide sequences in the samples was determined by taking the peptide sequences identified by a database matching approach for each protease-digested proteome, both labeled and label-free (Sage search engine, q -value < 0.01). This set of high-confidence peptide sequences for each sample type and their intersect is depicted in Fig. 3a,b. We used the *de novo* search engine PointNovo³⁴ trained alternately on label-free and PSMtag-labeled tryptic peptide spectra to identify sequences from the same samples that were processed by database matching. The numbers of peptide spectra that were assigned identical sequences by both *de novo* sequencing and database search for each sample type are shown in Fig. 3a,b in dashed lines for different peptide length bins. To control for labeling-specific sampling effects on the peptide spectra acquired, peptide-spectral matches were additionally subsetted to include only peptides identified by database search in both labeled and label-free samples. Numbers of matching *de novo* and database-search sequence assignments for this intersected peptide subset are shown in solid lines. When not restricting sequences to those identified both labeled and label-free, more short and long peptides were identified *de novo* by application of PSMtag for both protease digests. For the GluC digest, more medium-length peptides were observed to correspond to database results. The subset of sequences from the intersect of both labeled and label-free database results showed more similar rates of correspondence, but the quantity and quality of *de novo* sequence assignments were improved for both intersected and non-intersected comparisons, as indicated in Fig. 3c,d. Similar to the mechanism improving database-matching results, the improvements are likely driven by increased b-ion production, as visualized in Fig. 3e,f. The relative gain in accuracy for individual amino acid assignments increased with proximity to the N-terminus, where a PSMtag is conjugated. Results were qualitatively similar using a different *de novo* search engine, Novor³⁵ via [novor.cloud](#), shown in Supplementary Fig. S4.

Descriptive statistics of DDA spectra

Data-dependent workflows for bottom-up proteomics aim to acquire spectra of peptides individually. This allowed for the comparison of spectra for the same peptides tagged and label-free in Fig. 2. Overall, these low-input samples produced comparable numbers of peptide-spectra-matches (PSMs), unique sequence assignments, and proteins supported by the unique sequence assignments as shown in Supplementary Fig. S5a, whether tagged or label-free. The counts are broken down by sequences containing at least one arginine or sequences containing at least one lysine. This breakdown was chosen to support the investigation of the different effects on spectral quality induced by the conjugation of one tag (most arginine-terminating peptides) versus two tags (most lysine-terminating peptides). Small differences between the count statistics for arginine- vs. lysine-terminating sequences were observed for this data acquired by a DDA workflow. The distribution of scores (Supplementary Fig. S5b) was likewise comparable with some benefit afforded by doubly-tagged sequences. The effect on peptide chromatographic retention time of applying one versus two tags was significant and depicted in Supplementary Fig. S5c. DDA workflows, because they do not rely on predicting peptide chromatographic retention time, are well-suited to account for changes in peptide retention time without modification. Along with differences in retention time, the specific peptides yielded from the same sample by tagged and label-free preparations were different. The consistency of peptides and proteins observed in replicate label-free analyses is shown in Supplementary Fig. S5d,e. To account for the different fragmentation energy param-

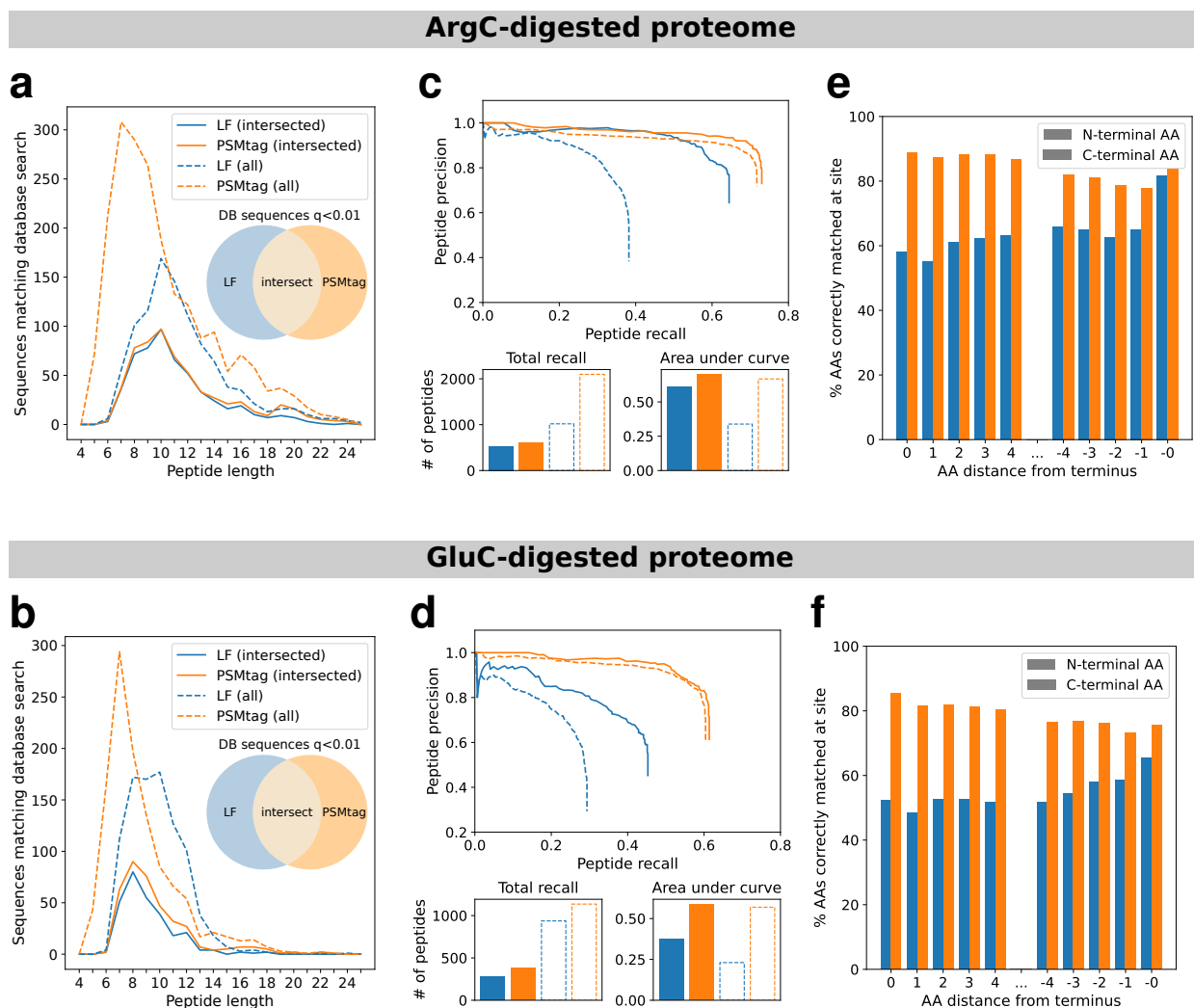


Figure 3 | Improvements to PointNovo *de novo* sequencing with ArgC- and GluC-derived peptides. **a, b.** Total numbers of correct *de novo* peptide identifications conditioned on peptide length. Peptide spectra assigned the same sequence by both PointNovo and Sage (at Sage spectrum q -value < 0.01) are considered correct *de novo* matches. **c, d.** Peptide-level precision-recall curves for tagged and label-free samples with measurements of total numbers of peptides recalled and area under each precision-recall curve. **e, f.** Overall accuracy of *de novo* amino acid assignments for residues with a distance of up to 4 AA from the peptide N- or C-terminus. Amino acids in the *de novo* sequence are considered to be correctly assigned if the database sequence contains a counterpart amino acid within 0.02 Da in mass with a prefix mass (sum of all preceding amino acids) within 0.1 Da.

ter used for the tagged peptides, the label-free replicates were acquired with different normalized collision energies (NCE = 24, NCE = 28), one determined to maximize label-free PSMs (NCE = 28) and the other determined to maximize tagged PSMs (NCE = 24). Even accounting for the different collision energies, the populations of tagged and label-free peptides and proteins were less similar, as shown in [Supplementary Fig. S5f,g](#). Little off-target and other labeling products of PSMtag were found, as shown in [Supplementary Fig. S2](#) and [Supplementary Fig. S6](#), respectively. The relative amino acid frequencies of confidently identified peptides in the label-free vs. PSM-tag data was also examined in [Supplementary Fig. S7](#) and found to be similar. The combination of tagged and label-free replicate preparations may allow greater protein sequence coverage and proteome coverage than label-free replicate preparations of the same input.

Proteome coverage and plexDIA multiplexing with PSMtags

Isotopic doping for increasing throughput

We sought to maximize the number of samples that could be isotopically encoded by PSMtag without constraining the cycle of data acquisition or undermining protein quantitation. Based on previous observations, we had a strong expectation that encoding separated by 4 Da or more would allow this³⁶. Furthermore, 2-Da multiplexing has been demonstrated for data-dependent workflows¹⁴, so we aimed to enable that encoding for plexDIA²⁸ and take advantage of its greater multiplexing. The structure of PSMtag with the implemented sites of isotopic doping is shown in [Fig. 1c](#). The sites of isotopic doping allowed the creation of a set of nine reagents separated by 2 Da. Using a subset of those reagents separated by 4 Da allowed multiplexing five samples, an increase over the state of the art (three) for plexDIA of proteomes digested by trypsin. The masses of the reagents are precisely described in [Supplementary Fig. S8](#). The retention time consistency of the reagents is depicted in [Supplementary Fig. S9](#), where the +16 Da reagent, the only reagent doping the pyridine nitrogen, is observed to elute earlier than other reagents.

Tags are conjugated to peptides in a sample-specific but not peptide-specific manner. Stated another way, all the peptides in a given sample are labeled by the same tag. The naturally-occurring isotopic envelope (attributable largely to natural ¹³C abundance in bio-molecules) for a given peptide is shown in [Fig. 1e,f](#) with offsets induced in the m/z domain by sets of tags, allowing the encoding of 5 or 9 samples, respectively. When the mass offset contributed by the tags is smaller than the width of the isotopic envelope of a peptide, the isotopic peaks of neighboring channels partially overlap. This overlap can be modeled as a linear superposition and deconvolved *in silico*.

More specifically, the 9-plex described in [Fig. 1f](#) induced spacing of 2 Da / z for peptides conjugated to one tag, spacing of 4 Da / z for peptides conjugated to two tags, and 6 Da / z for three, *etc.* Enzymes commonly used in proteomics allow attaching one or two tags per peptide (trypsin) or at least two tags per peptide (LysC). Peptides resulting from incomplete cleavage by either enzyme may allow three or more tags to be conjugated.

Evaluating identification rates and quantitation quality

Next we evaluated the amino acid sequence identification of PSMtagged peptides analyzed by DIA and throughput gains realized by PSMtags used for plexDIA. Sequence identification by DIA is

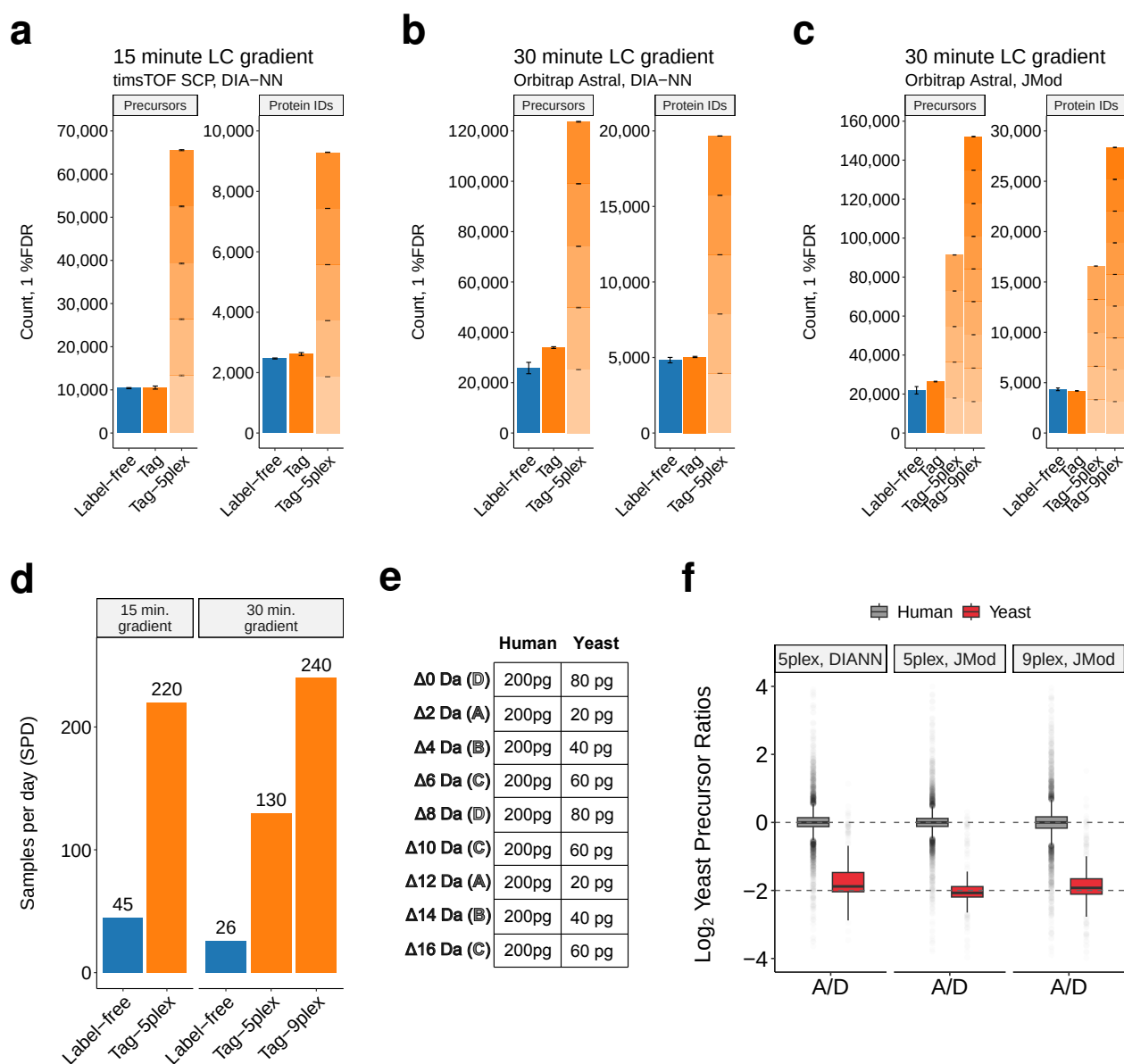


Figure 4 | Identification rates for tagged and label-free single-cell-level (200 pg) proteomes by DIA. **a**, Numbers of human precursors and proteins identified in replicate injections of samples containing 200 pg human proteome (K562 cancer cell line) and yeast proteomes spiked as described in **e**. Samples were unlabeled, labeled with a single PSMtag isotopologue, or with the 5-plex PSMtag set and multiplexed. Label-free and single-plex samples used sample D diluted to 200 pg of human material. Plexed samples were diluted to 200 pg of human material per channel. Data acquired using the timsTOF SCP and analyzed with DIA-NN 2.1. **b**, The same visualization for the same samples acquired with Orbitrap Astral and search with DIA-NN 2.1 (**b**) or JMod²⁸ (**c**), including a PSMtag 9-plex set. **d**, The number of samples per day (SPD) that can be analyzed using each approach. Full run-to-run time, including sample loading, active gradient, wash, and equilibration steps, is considered in the calculation. **e**, The experimental design for samples in this figure, prepared in bulk quantities and diluted to 200 pg of human input and 20-80 pg of yeast input. **f**, The measured versus expected ratio for yeast spike-ins in samples comprising 200 pg of human proteome and 20-80 pg of yeast proteome, summarizing the results of McDonnell *et al.*²⁸. More ratios also are examined in [Supplementary Fig. S10](#). Precursors intersected across all three conditions visualized.

highly dependent on having spectral libraries that faithfully represent peptide retention times and fragmentation patterns. Such spectral libraries are available for label-free analysis but not for PSM-tags. Thus, we used DIA data of PSMtagged peptides to train spectra and retention time prediction models for PSMtags with AlphaPeptDeep³⁷. Deeper coverage of the tagged proteome will allow training more accurate models, creating more accurate spectral libraries and better interpreting DIA spectra from peptides labeled with PSMtags.

To rigorously compare sequence coverage for single-cell level samples, we analyzed 200-pg samples from K-562 cells either as label-free samples or labeled with PSMtags using data-independent acquisition on the Bruker timsTOF SCP and Thermo Fisher Orbitrap Astral mass spectrometers. Yeast proteome digests were also spiked in at amounts described in Fig. 4e to allow for simultaneous evaluation of quantitation quality. Only human peptides and proteins were considered for evaluation of identification rates in Fig. 4a,b,c. Sample D was used for label-free and single-channel evaluation. Protein FDR was computed separately within each sample type (label-free, single-plex, 5-plex and 9-plex).

When searched with DIA-NN³⁸ or JMod²⁸, singly-tagged, arginine-terminating peptides had similar or improved rates of identification, while the identification rate for doubly-tagged, lysine-terminating peptides was preserved or diminished, Supplementary Fig. S11a,b,c. The number of confident peptide and protein identifications was maintained or increased relative to label-free counterparts when considering a single tag applied to the proteome, and coverage by sample decreased slightly in the plexed samples. Using the 5-plex, PSMtags increased the number of quantified human protein data points from 4,826 (label-free) to 19,665 (PSMtag-5plex) using 30-minute LC gradients on the Orbitrap Astral (Fig. 4b) or from 2,468 to 9,292 using 15-minute LC gradients on timsTOF SCP (Fig. 4a). Using the 9-plex, PSMtags increased the number of human protein data points from 4,340 (label-free) to 28,359 (PSMtag-9plex), as shown in Fig. 4c. The applied gradients demonstrated a number of different throughputs. The 5-plex tags demonstrated a 220 samples-per-day (SPD) rate on 15 minute gradients and the 9-plex tags demonstrated a 240 SPD on 30 minute gradients that allow greater depth of proteome coverage, as shown in Fig. 4d. The 9-plex PSMtags and 10 minute run-to-run times allowed a theoretical, but achievable, 1,296 SPD.

The quantitation accuracy achieved by plexDIA using PSMtags was benchmarked using the proteome mixing ratios described in Fig. 4e and found comparable to that achieved by matched label free analysis of the same samples, as shown in Fig. 4f and Supplementary Fig. S10. The results indicate that PSMtags can achieve high accuracy and precision across a range of mixing ratios, Supplementary Fig. S10. The quantification accuracy and its dependence on the mass offset between the tags is thoroughly examined by McDonnell *et al.*²⁸. These results indicate that PSM-tags increase throughput without significant adverse impact on depth of coverage or quantitative accuracy.

Discussion

We present what may be the first systematic approach to discovering and validating new mass tags for bottom-up proteomics by LC-MS/MS. From an initial screen of hundreds of compounds (Fig. 1a,b), we arrive at and demonstrate a new mass tag (Fig. 1c that improves peptide sequencing (Fig. 1f) and throughput (Fig. 4b) for sensitive proteomics applications. Specifically, the 5-plex

and 9-plex sets of reagents allow, with suboptimal spectral libraries, analyzing 16,580 and 28,359 protein data points in the same time a label-free experiment analyzes 4,340 proteins (Fig. 4c).

Peptide sequencing is improved by tag conjugation, as demonstrated in Fig. 2b,c, controlling for both protein input amount and peptide ionization (Fig. 1f). Controlling for protein input amount reveals that tagged peptides are delivered and/or converted into ions more efficiently, which itself may improve sequencing by elevating low-abundance peptides and fragments above the signal-to-noise threshold. Controlling additionally for the abundance of ions inside the mass spectrometer, the peptide sequencing quality is still improved. Improvement in peptide sequencing for this particular mass tag, as quantified in Fig. 2b,c, is driven by stabilization of b-ions (Fig. 2e for both lysine- and arginine-terminating peptides (Supplementary Fig. S12) and higher charge state fragments (Supplementary Fig. S13)).

Mechanisms for greater b-ion production

Although increased intensity of a precursor can lead to improved scoring by virtue of better MS/MS fragment sensitivity, PSMtag boosts peptide scores even in the case of decreased precursor intensity as seen in Fig. 1f. We suggested enhanced b-ion generation accounts for this increase, as shown in Fig. 2e. This observation has been noted for mTRAQ-labeled peptides and attributed to stabilization of b-ions via increased gas-phase basicity of mTRAQ relative to the unlabeled N-terminus²⁰. This justification is supported by the mobile proton model (MPM) that describes the relationship of a proton's mobility with the proton affinity of peptide functional groups^{39,40}. Interestingly, any one site of protonation on PSMtag should not substantially increase the proton affinity of the N-terminus, yet we observe a similar stabilization. It is possible however, that the increased number of similarly basic sites on the N-terminus (from one label-free to three with PSMtag) provide a greater likelihood of detecting a b-ion due to the increased probability of retaining a charge. When C-terminal lysine residues are labeled, the gas-phase basicity of the C-terminus is lowered, allowing for greater proton mobility in accordance with the MPM. This model also offers an explanation for why the optimized collision energy for labeled peptides is lower than label-free overall.

Further potential for multiplexing with isotopic doping

Not all possible sites for stable isotopic incorporation for this molecule were used due to budget and time constraints. Additionally, this study restricted itself to multiplexing using peptides derived from biological samples using the enzyme trypsin to digest proteins due to its near universal adoption by the bottom-up proteomics field. However, less-frequently used enzymes, such as LysC and LysN⁴¹, create multiple sites per peptide for tag conjugation. Guaranteeing multiple tags per peptide allows tags with smaller mass offsets to achieve larger mass offset once conjugated. For example, PSMtags can be synthesized with 1-Da spacing between individual tags and achieve an 18-plex with 2-Da spacing if conjugated at least twice to every peptide. Such increase can be realized for LysC digests and does not use any additional sites of isotopic doping beyond those already described. Deuterium incorporation was not considered to achieve multiplexing in this study, but recent advances start to explore its promise and limitations¹⁰ for use in the context of mass tags. Approaches like JMod that can accurately deconvolve overlapping peptide signal (like label-free

fragment ions, such as y-ions from arginine-terminating peptides) despite variable retention time shifts. Modeling these shifts may allow deuterium incorporation in future versions of PSMtags and thus expansion of its plex without significant detriment for quantification.

Limitations and opportunities in discovering new mass tags for proteomics

Predicting spectral libraries for new mass tags

While spectral libraries can be adequately predicted for label-free, tryptic peptides^{42,43}, tagging peptides alters both spectra and retention times as shown in Fig. S5c. Current *in silico* approaches to library prediction – modeled on label-free data – are insufficient to account for these changes. However, this restriction can be overcome with libraries generated from empirical data acquired by data-dependent acquisition (DDA). Because DDA search algorithms do not rely on peptide fragmentation predictions, the enhanced spectral qualities imparted by tag moieties can be evaluated (at high throughput) without bias from models that have been trained on millions of label-free spectra. The improved sequencing of tagged peptides in Fig. 1f is one such demonstration. Spectra derived from DDA of tagged peptides can be directly used as empirical libraries to search DIA experiments or used to train models of spectra and retention time prediction, such as AlphaPeptDeep used herein.

Compound library represented a limited range of properties

The 576 compounds studied represent a limited range of chemical properties, for example: logP (Supplementary Fig. S14). While the screen resulted in at least one compound to improve peptide sequencing, the quenched and hydrolysis products of the compound (logP approx. 1.5) coelute with peptides on standard C18 reverse-phase chromatography (Supplementary Fig. S3a). While the two compounds are chromatographically well-resolved and omitted from accumulation in the mass spectrometer by nature of being lower *m/z* than most tagged peptides, the peptides with which they coelute may experience ionization suppression and thus decreased quality of sequencing and quantitation. The degree to which this affects peptides is mostly dependent on their relative abundance: highly abundant ions may tolerate ionization suppression without loss of sequencing or quantitative accuracy, while less abundant ions will experience a more deleterious effect.

A larger screen profoundly changes what analysis is possible. Decreasing the compound logP value, which correlates positively with elution on the typical reverse-phase chromatography used for proteomics studies, will allow complete chromatographic separation of a compound's quenched and hydrolyzed products from tagged peptides. The low logP values of mTRAQ and TMTpro, -0.73 and -0.51 of the respective hydrolyzed products (ChemDraw, v22.2.0), and their demonstrated weak binding to C18 material support building screens that include compounds with lower logP values to mitigate the possibility of peptide ionization suppression.

Impact of heavy isotope doping on retention time

Peptide analytes incorporating ¹⁵N can elute earlier than the same peptide analytes incorporating ¹⁴N^{44,45}. The broad adoption and utility of multiplexing reagents which incorporate ¹⁵N motivated

approaches to accurately quantify peptide analytes with such isotopic enrichment^{46–48}. Similarly, isotopologues of PSMtag with ¹⁵N incorporated into the indoline-like portion of the structure show some elution time differences in [Supplementary Fig. S9a](#), while the isotopologue ($\Delta 16$) with ¹⁵N incorporated in the pyridine ring structure shows a larger effect. The effect size is also larger with two tags per peptide than with one, as shown in [Supplementary Fig. S9b,c](#). Future tag designs can consider the incorporation of ¹⁵N at moieties that induce minimal elution time differences, in turn, minimizing the number of coeluting peptide peaks.

Impact of increased hydrophobicity on experimental procedures

Due to its increased hydrophobicity, the optimal recovery of tagged peptides in this study is achieved using a nonionic detergent, n-dodecyl maltoside (DDM), previously used by others for solubilizing membrane proteins and increasing proteome coverage for low input samples [49, 50]. Initial rounds of experimentation yielded an optimal concentration of 0.18% n-dodecyl maltoside (DDM) in 30% acetonitrile in the tagged sample injection buffer ([Supplementary Fig. S15](#)), significantly higher than the optimal concentration found for label-free peptides [49, 50]. The impact of this amount of DDM on LC performance is untested in long-term studies (scale of 1000s of samples).

Minimization or separation of tag byproduct is desirable. We took the approach of minimizing PSMtag byproduct by using anhydrous peptide labeling conditions. Anhydrous conditions minimize tag hydrolysis during peptide labeling, allowing for a lower molar ratio of tag-to-peptide to be used, and thus, decreasing the abundance of coeluting tag byproducts. Additionally, ion mobility separation of the byproduct from the peptides has been used to achieve the results in this study, but not all mass spectrometers are equipped with such ion mobility separation devices (FAIMS or TIMS).

Alternatively, the relatively small size and different charge density of tag byproducts in comparison to tagged peptides may be exploited by different offline clean-up strategies. For instance, strong cation exchange (SCX)⁵¹ may be able to separate tag byproducts and tagged peptides on the basis of their charge densities at low pH. Alternatively, size exclusion chromatography (SEC) may be able to separate tag byproducts from tagged peptides, taking advantage of their size difference.

Conclusion

With the first screen of 576 compounds, we uncovered insights into characteristics that may lend to improved LC-MS sequencing. Such insights provide guidance for designing new compounds and interrogating new chemical spaces, enabling a cycle of discovery that will allow strides to proteomics depth and throughput in parallel.

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Data availability: Data, raw and processed, for reproducing figures is available through data repositories MassIVE [MSV000097968](#) and ProteomeXchange [PXD064191](#)

Code availability: The code used for data analysis and figures is at: github.com/ParallelSquared/tag

Competing Interests: H.S., M.Y., S.S., D.B.S., M.A., W.V., A.Y., M.A.L., B.P. and N.S. are listed as inventors on a patent application for the tags described in this paper.

Author contributions

Experimental design: H.S., M.Y., S.S., B.P. and N.S.

Data analysis: H.S., M.A., M.Y., S.S. and K.M.D.

De novo data analysis: M.A.

LC-MS/MS: S.S., M.Y. and C.F.

Isotopic incorporation route design: D.B.S.

Compound synthesis: D.B.S., W.K.P., Y.H., K.S.R., W.V., A.Y. and M.A.L.

Proteomic sample preparation: S.S., M.Y. and M.P.A.

Raising funding: N.S. and H.S.

Writing: All authors approved the final manuscript.

Methods

Model peptide synthesis

Model peptides were synthesized via manual flow solid phase peptide synthesis (SPPS) as previously described⁵². For each peptide, this involved the use of 500 mg of NH₂-Tentagel resin (500 mg, 0.23 mmol/g loading, 300 μ m monosized). The sequences synthesized using this resin consisted of either:

H-Ala-Val-Gln-Val-Ala-His-Trp-Ala-Trp-Ser-Asn-Glu-Lys-OH

H-Thr-Leu-Glu-Ser-Ser-Gly-Glu-Asn-Leu-Arg-Leu-Gln-Lys-OH

The resin was first prepared and functionalized for use via attachment of a 4-(4-hydroxymethyl-3-methoxyphenoxy)-butyric acid (HMPB) linker and Fmoc-Lys(Boc)-OH (fluorenylmethoxycarbonyl and tert-butoxycarbonyl protecting groups, respectively) as follows: NH₂-Tentagel resin (500 mg, 0.23 mmol/g loading, 300 μ m monosized) was initially swelled using N,N-dimethylformamide (DMF) in a fritted Torviq syringe. Next, a solution consisting of HMPB (276.28 mg, 0.575 mmol, 5.0 equiv) dissolved in a hexafluorophosphate azabenzotriazole tetramethyl uronium (HATU) solution (0.38 M in DMF, 2.878 mL, 4.75 equiv) and activated with diisopropylethylamine (DIPEA) (0.601 mL, 15.0 equiv) was added. The reaction mixture was allowed to react for 1 hour prior to draining, and the resin was washed with DMF (3 x 10 mL). Next, a solution of Fmoc-Lys(Boc)-OH (538.82 mg, 1.15 mmol, 10 equiv) and DMAP (1.405 mg, 0.0115 mmol, 0.1 equiv) in DMF (2.875 mL) was added to the resin, followed by addition of N,N'-diisopropylcarbodiimide (DIC) (90.03 μ L, 0.575 mol, 5 equiv). The syringe was then capped and allowed to mix overnight on a nutating mixer. The solution was subsequently drained, and the resin washed with DMF (3 x 10 mL).

After resin preparation, peptide synthesis was conducted in the following manner: The remaining residues (as appropriate based on sequence) were sequentially coupled in a C-to-N direction via flow SPPS, which included an initial deprotection step to remove the Fmoc group from Lys(Alloc) (Allyloxycarbonyl protecting group)⁵².

On-peptide compound synthesis

Small molecule probes were attached to the free N-terminus of the peptide via procedures based on previous peptide encoded library efforts²⁴. The resin batch was evenly split into 16 different fritted Torviq syringes. To each portion of resin was coupled a different trifunctional building block (10 equiv each) as detailed in [Supplementary Fig. S1](#).

Each trifunctional building block was dissolved in a solution of HATU in DMF (0.38 M, 239.58 μ L, 9.58 μ mol, 9.5 equiv) and activated with DIPEA (50.08 μ L, 0.2875 mmol, 30 equiv) prior to addition to resin. The reaction was then allowed to proceed for 1 h. Afterwards, specific tubes were combined separately due to the isobaric nature of some of the building blocks. Recombination was performed based on the color coding shown in [Supplementary Fig. S16](#), referencing the compound IDs in [Supplementary Fig. S1](#), generating four separate mini-libraries which were then each washed with DMF (3 x 3 mL). Trifunctional building blocks A and C were omitted. Fmoc deprotection was then performed using a solution of 20% piperidine in DMF (3 x 3 mL, 5 min

each) before being washed again with DMF (3 x 3 mL). Each mini-library was then individually subjected to the following procedures: Each batch of resin was equally split into five portions and transferred to smaller fritted Torviq syringes. To each portion of resin was coupled a different carboxylic acid (10 equiv each, twice) as detailed in [Supplementary Fig. S1](#). Each carboxylic acid was dissolved in a solution of HATU in DMF (0.38 M, 143.75 μ L, 54.625 μ mol, 9.5 equiv) and activated with DIPEA (30.05 μ L, 0.1725 mmol, 30 equiv) prior to addition to resin. The reaction was then allowed to proceed for 1 h before being drained, and the coupling step repeated once more. The resin portions were then pooled and washed with DMF (3 x 3 mL), then THF (3 x 3 mL). For Pd-mediated cross-coupling, microcentrifuge tubes (5 x 1.5 mL) were each charged with a different boronic acid (5 equiv each) as detailed in [Supplementary Fig. S1](#).

To each tube was added Pd G4 XPhos (5.937 mg, 6.9 μ mol, 1.2 equiv) and XPhos (3.29 mg, 6.9 μ mol, 1.2 equiv). Next, resin suspended in THF (2.59 mL) was then thoroughly mixed and evenly dispensed into each microcentrifuge tube in 12 equal portions, each containing an identical final volume of THF (517.5 μ L). Finally, a solution of K₃PO₄ (0.5 M, 57.5 μ L, 28.8 μ mol, 5 equiv) was added to each microcentrifuge tube and thoroughly mixed via pipetting. The headspace of each microcentrifuge tube was then purged with a stream of N₂ and left to react on a nutating mixer for 24 h at 37 °C. Finally, each portion of resin was combined, washed with THF (3 x 3 mL), DCM (3 x 3 mL), DMF (3 x 3 mL), H₂O (3 x 3 mL), DMF (3 x 3 mL), diethyldithiocarbamate (DTC) solution until original resin color restored (0.022 M DTC in DMF, prepared via 100 mg DTC per 20 mL DMF, 3 x 3 mL), and DMF (3 x 3 mL). The resin was then washed with DCM (3 x 3 mL) and dried under vacuum prior to cleavage from the resin and global deprotection, furnishing the desired peptide-bound small-molecule library.

Each batch of resin was transferred to a 50 mL conical vial and cleaved and deprotected by the addition of 5 mL of cleavage cocktail (95/2.5/2.5 mixture of TFA/H₂O/TIPS), followed by agitation using a nutating mixer for 3 h at room temperature. The desired material was subsequently precipitated by the addition of cold Et₂O (-78 °C, to a total volume of 50 mL) then vortexed to ensure thorough mixing. This suspension was then centrifuged at 20,800 x g for 10 minutes to yield the desired material as a pellet. The supernatant was decanted and the pellet resuspended with an additional portion of cold Et₂O (-78 °C, 25 mL) prior to being centrifuged once more at 20,800 x g for 10 minutes. The supernatant was discarded, and the pellet containing both resin and peptide was dissolved in a solution of 50/50 H₂O/ACN + 0.1% TFA (15 mL) prior to filtration through a fritted syringe to remove the resin. The filtrate was then frozen and lyophilized to furnish the desired material as a crude white powder.

The crude material was weighed to determine the crude mass prior to loading onto Discovery® DSC-18 SPE Tubes (aiming for sample loading of max 10% of SPE bed wt.). The peptide sample was first dissolved in a minimal volume of DMF (approx. 200 μ L), prior to diluting with 5/95 H₂O/ACN + 0.1% TFA (with a final DMF content of no more than 2% v/v). SPE was then performed using the following procedure as outlined:

Step	Action	Solvent	Column volumes
1	Condition	ACN	2
2	Equilibrate	5/95 ACN/H ₂ O	3
3	Load	Peptide Solution (see above)	As required
4	Buffer Wash	5/95 ACN/H ₂ O	6
5	Elute	50/50 H ₂ O/ACN	4

Eluted volumes were then transferred into pre-weighed 15 mL conical vials before being flash frozen and lyophilized to furnish the desired mini-library.

Compound-peptide library data acquisition with Thermo Scientific Orbitrap Exploris 240

A pool of 576 initial compounds synthesized by split-pool chemistry were conjugated to the model peptides and separated via online nanoflow liquid chromatography (nLC) with a Thermo Vanquish Neo set in trap-and-elute mode and analyzed by a Thermo Scientific Orbitrap Exploris 240 mass spectrometer. Approximately 10 ng of the sample was loaded onto a 0.5 x 0.3 cm trapping column (Thermo Scientific, 174500) and separated with a 25 cm x 75 μ m IonOpticks Aurora Series UHPLC analytical column (IonOpticks, AUR2-25075C18A). Mobile phase A (MPA) was 0.1% formic acid (FA) in water and mobile phase B (MPB) was 80% acetonitrile (ACN) with 0.1% FA. A constant flow rate of 300 nL/min was used throughout sample loading and separation with a column temperature of 55°C. Samples were loaded onto the column at 6% MPB. The active gradient was ramped to 29% MPB over 23 min then increased to 44% B over 7 min. MPB increased to 99% over 0.2 min and was maintained for 4 min before returning to 6% MPB in 0.1 min for column reequilibration. Using the Exploris 240, electrospray voltage was set to 1900 V. The temperature of the ion transfer tube was 275°C. All MS1 scans were collected at a resolution of 60,000 over m/z 380-900 with a maximum fill time of 100 ms, and an AGC of 1e6. The RF lens was set to 80% for MS1 and MS2. Following the MS1 scan, the top 20 ions above the intensity threshold of 5e3 with charge states 2-4 were selected for MS2 fragmentation and measurement. Dynamic exclusion was used with a count of 2 and duration of 60 s with a mass tolerance of 10 ppm. For MS2 scans, ions were isolated with a 1 m/z window, 0.7 Da offset, and fragmented with 33% normalized collision energy (NCE) based on “Standard” AGC and “Auto” max injection time mode settings. The fragments were analyzed at 30,000 resolving power with the first mass fixed at m/z 120.

Compound-peptide identification

Mass tag candidates synthesized directly onto model peptides were identified and scored by the dependent-peptide approach described by Mordret *et al.*⁵³ using 10 ppm tolerance to match the expected mass of the synthesized candidates.

Cell culture

Human K562, myelogenous leukemia cells (ATCC, CCL-243) were grown as suspension cultures in RPMI high glucose media (Sigma-Aldrich D5796) with the addition of 100 U/mL Penicillin-Streptomycin (Sigma-Aldrich P0781) and supplemented with 10% fetal bovine serum (FBS, Millipore Sigma F4135). Cells were passaged when a density of 106 cells/mL was reached, approximately every 2 days.

Cell harvest

Cultured cells were collected in 15 mL conical tubes by centrifugation at 500 g for 10 minutes. An aliquot of 20 μ L was taken before centrifugation and used for counting. The cells were washed twice with PBS in the 15 mL tube then transferred in PBS to a 1.5 mL Eppendorf tube, spun in the microcentrifuge at 500 g for 5 min then quickly washed one time with LC-MS water and spun for 2 minutes. Supernatant was removed by vacuum suction, and the cell pellet disrupted and resuspended in LC-MS water at 2000 cells/ μ L then stored at -80°C until ready for use.

Cell lysis and digestion with trypsin

Intact human and yeast proteins (Promega V6941 and V7341, respectively) were digested via manufacturer's suggestions. Briefly, proteins that are presolubilized by the manufacturer (in 6.5 M urea in 50 mM Tris-HCl, pH 8 at a protein concentration of 10 mg/mL), were reduced with dithiothreitol (DTT, Pierce A39271) to a final concentration of 5 mM for 30 min at 37°C, and then alkylated using iodoacetamide (IAA, Pierce A39270) to a final concentration of 15 mM for 30 min in the dark at room temperature. The urea solution was diluted with 100 mM triethylammonium bicarbonate (TEAB, Thermo Scientific P90114) and trypsin was added at a mass ratio of 1:40 enzyme:protein and allowed to digest overnight at 37°C. The digestion was quenched to 1% FA, and desalted using C18 solid phase extraction with Oasis HLB columns (Waters WAT094225) using manufacturer's protocol.

Frozen, harvested K562 cells were lysed and digested using the Minimal ProteOmic sample Preparation (mPOP) method. Briefly, frozen cells taken directly from -80°C were heated to 90°C in a thermal cycler for 10 minutes to lyse. TEAB was added to a final concentration of 100 mM (pH 8.5) for buffering and Trypsin Gold (Promega V5280) was added to a final trypsin concentration of 10 ng/ μ L, followed by digestion at 37°C for 4 h in a thermal cycler. Digested proteins were then quenched to 1% FA, dried *in vacuo* and stored until ready for future use.

Anhydrous labeling

Labeling of human and yeast peptides was carried out anhydrously to minimize competition of hydrolysis with aminolysis at the primary amines of peptide N-termini and lysine residues. Dried samples were resuspended in a mixture of DIPEA and PSMTAG in DMSO for a final ratio of 1.5:1 tag to peptide mass in 4% v/v DIPEA. The solution was mixed via pipetting to ensure sample is completely dissolved, spun down, and incubated at room temperature (RT) for 2 hours. Finally, the mixture was quenched by the addition of hydroxylamine (Sigma-Aldrich 467804) to a final

volume of 0.25% and incubated at RT for 2 h. Samples were either stored at -20 C or further diluted in an aqueous solution of 30% acetonitrile, 0.18% DDM, 0.1% formic acid and analyzed using LC-MS/MS.

Single-cell level standard data acquisition with Bruker timsTOF SCP

Samples were analyzed on a Bruker timsTOF SCP mass spectrometer coupled to a Vanquish Neo (Thermo Scientific) nLC system. Labeled peptides were diluted with 30% ACN, 0.18% dodecyl beta-D-maltoside (DDM), 2% FA and 200 pg were injected to mimic a single-cell level input. The samples were separated on a 25 cm x 75 μ m IonOpticks Aurora Ultimate CSI C18 column over a 19 min gradient. MPA remained at 0.1%FA (in water) and MPB at 80% ACN, 0.1% FA. The flow rate was kept constant at 250 nL/min with a column temperature of 55°C. The active gradient started at 30% MPB and 300 nL/min flow. %B was increased to 38% over 0.1 min and 62% over 12.9 min. At this point, the flow rate was reduced to 250 nL/min. MPB was then increased to 75% over 2 min and maintained for 3.1 min while the flow rate was increased to 300 nL/min again. Finally, MPB was ramped up to 99% over 0.1 min and maintained for 0.8 min.

Unlabeled peptides were reconstituted in 0.015%DDM, 2% FA. The label-free samples were separated on a 25 cm x 75 μ m IonOpticks Aurora Ultimate CSI C18 column over a 19 min gradient (15 min active gradient). MPA and MPB are identical to description above. The active gradient started at 6% MPB and 300 nL/min flow. %B was increased to 10% over 0.1 min and 35% over 12.9 min. At this point, the flow rate was reduced to 250 nL/min. MPB was then increased to 42% over 2 min and to 75% over 0.1 min. Here it maintained for 3.1 min while the flow rate was increased to 300 nL/min again. Finally, MPB was ramped up to 99% over 0.1 min and maintained for 0.8 min.

The timsTOF SCP was operated in dia-PASEF mode with the following settings: MS1 m/z range of 100 - 1700, mobility scan range of 0.64 - 1.42 V.s/cm², ramp and accumulation times of 100 ms. Capillary voltage was 1500 V, dry gas flow 3 L/min, and dry temp was 200°C. The minimum m/z that passes the quadrupole (low mass) was set to 380 and high sensitivity mode was enabled. The dia-PASEF windows covered an m/z range from 350 - 1050 and mobility range from 0.64 - 1.22 V.s/cm². The collision energy was set to 24 - 64 eV as a function of the inverse mobility of the precursor. Each cycle consisted of 1 MS1 scan, 4 dia-PASEF cycles, 1 MS1 scan, 4 more dia-PASEF cycles for a total cycle time of 1.07 s. The dia-PASEF windows were 25 Th wide.

Data acquisition with Thermo Scientific Orbitrap Astral

Samples were separated via online nLC with a Thermo Vanquish Neo configured for trap-and-elute. Eluted peptides were introduced to an Orbitrap Astral mass spectrometer (Thermo Scientific) fitted with a FAIMS Duo Pro interface and an Easy Spray electrospray ionization (ESI) source. For all experiments, FAIMS was operated at -50 CV with a carrier gas flow set to 3.6 L/min, a spray voltage of 1900 V, and an ion transfer tube temperature 275°C. Samples (prepared identically as described above) were loaded onto the 5 mm x 0.3 mm trapping column and analyzed with a 25 cm x 75 IonOpticks Aurora Series UHPLC column (IonOpticks AUR3-25075C18-XT). MPA and MPB are identical to description above.

For all label-free samples, both DDA and DIA, initial conditions were 6% MPB at 300 nL/min. Flow rate was reduced to 100 nL/min, then ramped to 10% B over 0.1 min. The active gradient began at 10% MPB and increased to 35% over 24.5 min, followed by ramping to 42%B over 5.5 min. The column was then washed with 99% MPB for 4 min, whereupon the flow rate was returned to 300 nL/min prior to column re-equilibration. For the PSMtag samples, the same LC method structure was used as for the label free samples with the following exceptions. For PSMtag samples, initial conditions were 30% MPB at 300 nL/min and the active gradient occurred from 38% to 62% MPB with a 100 nL/min flow rate over 24.5 min, followed by an increase to 75%B over 5.5 min.

Data-dependent acquisition

For DDA experiments, 2 ng of human proteome and 0.8 ng yeast proteome were injected for analysis. MS1 scans were collected at a resolution of 240,000 over an m/z range of 380-900 with a maximum fill time of 100 ms, and an AGC of $3e6$. The RF lens was set to 50% throughout. Following the MS1 scan, ions with charge states 2-5 and m/z 400-900 for label-free or m/z 500-1000 for labeled peptides were selected based on a cycle time of 1 second for MS2 fragmentation and measurement in the Astral detector. Dynamic exclusion was implemented, set to the “auto” setting and the MS2 fragment scan range was set to 120-1800 m/z . Ions were isolated with a 1 m/z window, 0.2 m/z offset. AGC was set to $2e4$ and maximum injection time of 50 ms. Peptides were fragmented with 24, 26, 28, or 30 NCE for determining optimal MS2 parameters.

Data-independent acquisition

For all DIA injections, 200 pg of human proteome and 80 pg yeast proteome were injected for analysis. MS1 scans were collected at a resolution of 240,000 over an m/z range of either 380-900 (label-free) or 480-1000 (labeled) with a maximum fill time of 150 ms and an AGC of 300%. The RF lens was set to 50%. DIA spectra were collected with the Astral detector using 20 windows of variable size. Window sizes for label-free and labeled peptides were identical, but shifted by 100 m/z . The fragment scan range was set to 150-2000 with an acquisition loop set to 0.6 sec. Collision energy was set to 28% (label-free) or 24% NCE (labeled) with an AGC set to 500% and maximum injection time of 60 ms. LC parameters were identical to DDA injections.

Data analysis

Data analysis was performed using the programming languages R and Python. Code for reproducing figures available at: github.com/ParallelSquared/tag

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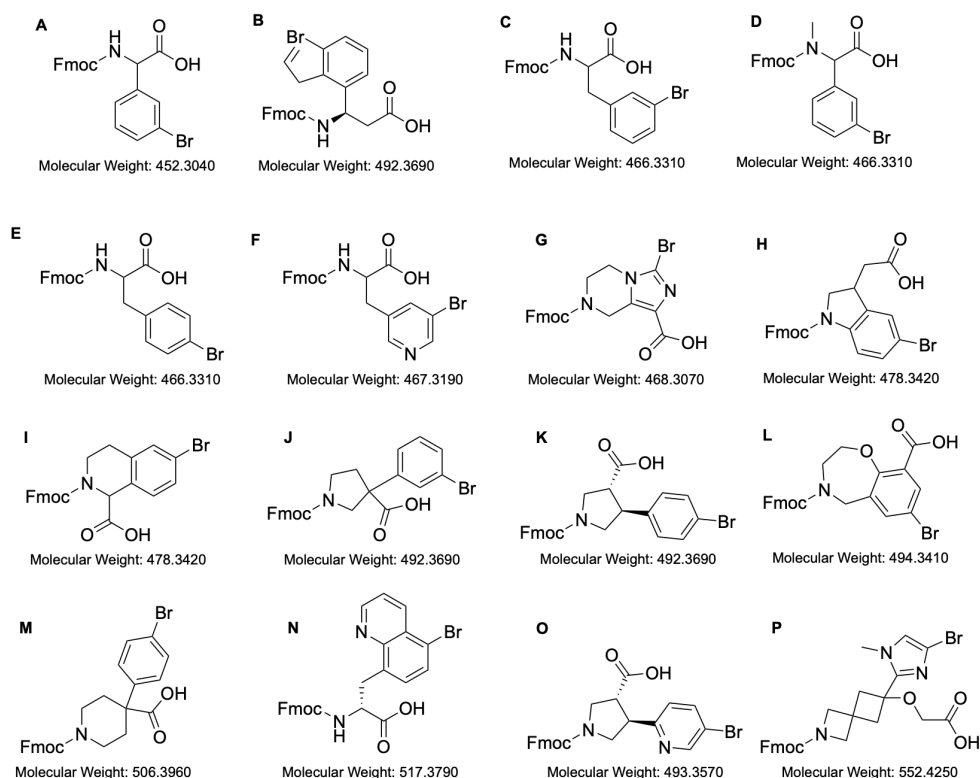
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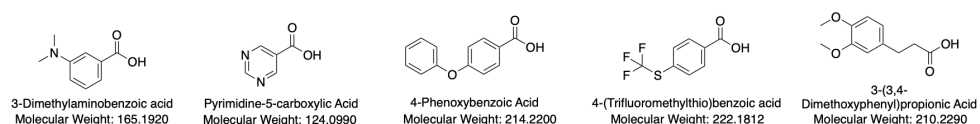
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Supplemental



Carboxylic Acid Building Blocks



Boronic Acid Building Blocks

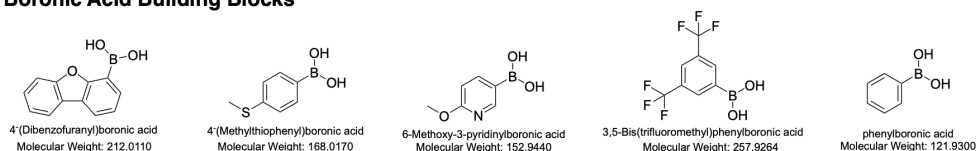


Figure S1 | Building plots for for split-pool synthesis of mass tag candidates The boronic acids, carboxylic acids and trifunctional cores used.

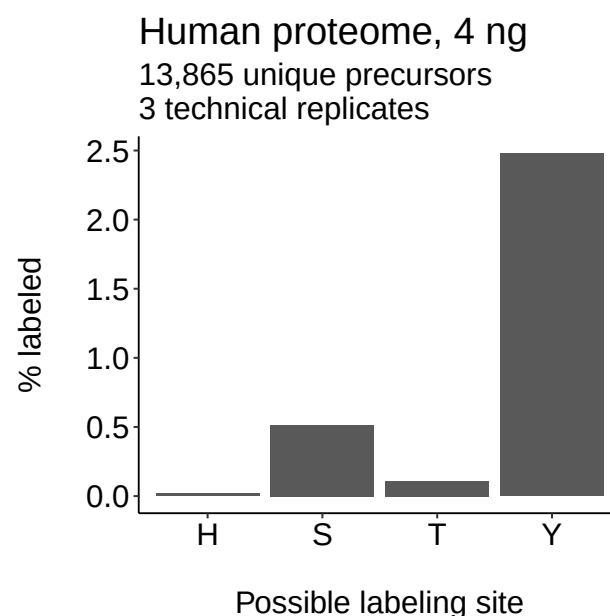


Figure S2 | Off-target labeling by PSMtag Three replicate injections of PSMtag labeled human proteomes (K-562 cell lines) of 4 ng were acquired by the Orbitrap Astral and analyzed by MaxQuant using fixed modification of PSMtag at lysine and N-terminus and variable modifications of PSMtag at histidine (H), serine (S), threonine (T), tyrosine (Y).

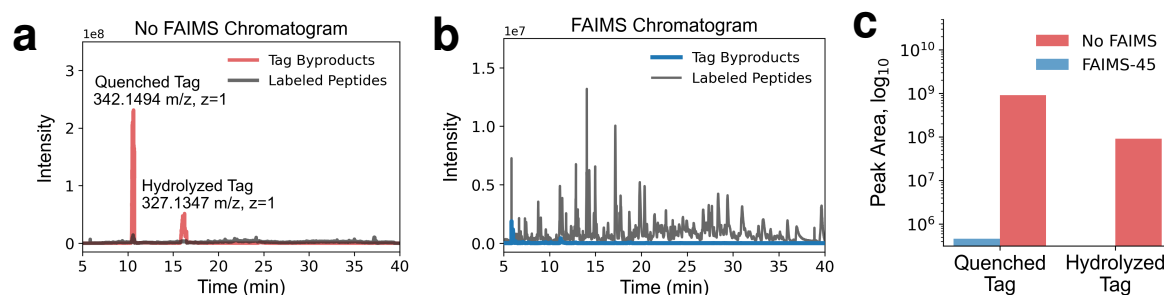


Figure S3 | Ion mobility properties allow separation of co-eluting tag byproducts and tagged peptides. **a**, The relative extracted total ion currents for quenched (m/z 342.15) and hydrolyzed (m/z 327.13) tag byproducts (red) and tagged peptides (grey) without using FAIMS Pro. **b**, The relative extracted total ion currents for tag byproducts (red) and tagged peptides (grey) using FAIMS Pro at -45 V. **c**, The total extracted total ion currents for tag byproducts without using FAIMS (red) and using FAIMS (blue).

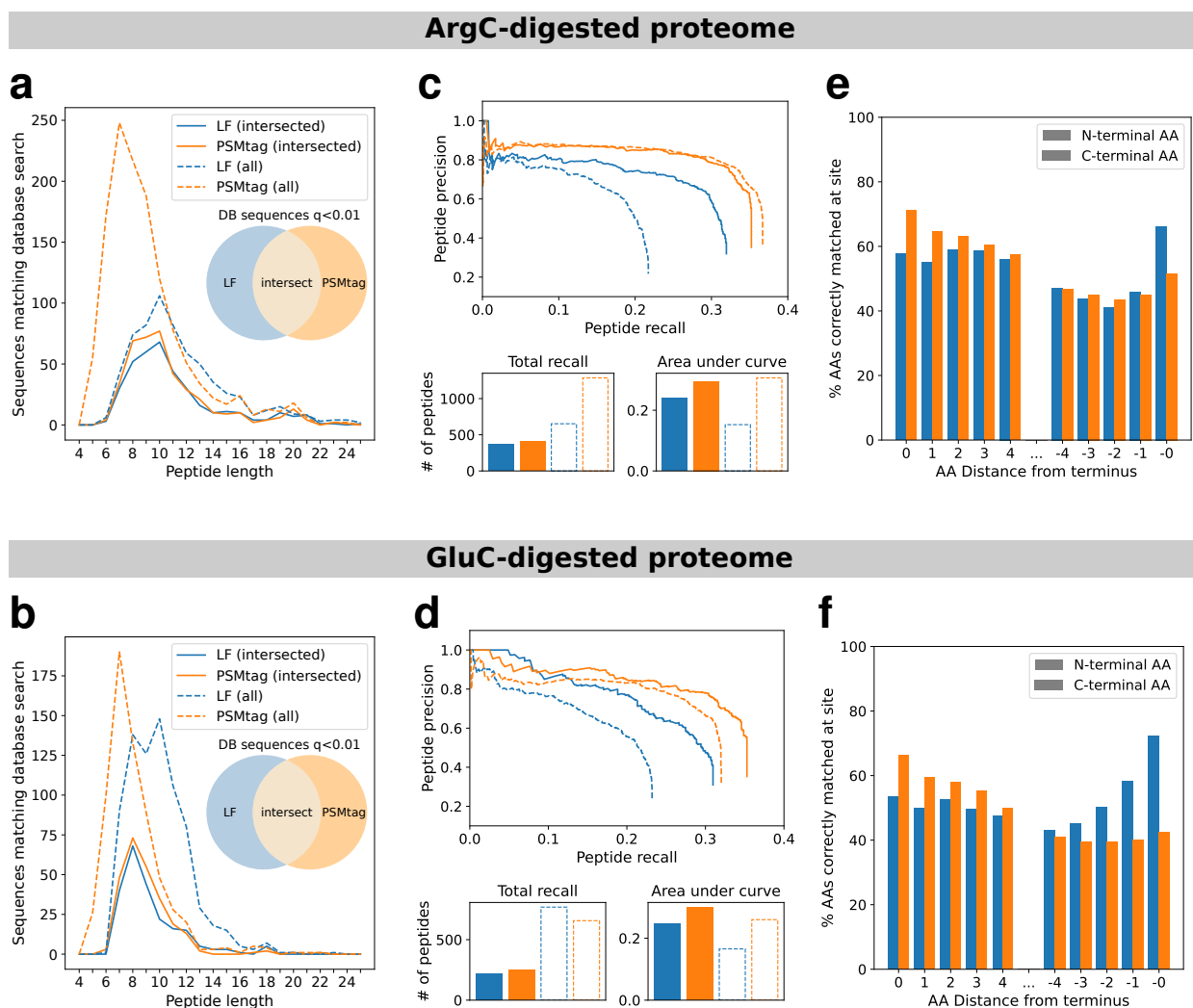


Figure S4 | Improvements to Novor *de novo* sequencing with ArgC- and GluC-derived peptides. **a, b.** Total numbers of correct *de novo* peptide identifications, conditioned on peptide length. Peptide spectra assigned the same sequence by both Novor and Sage (at Sage spectrum q-value <0.01) are considered correct *de novo* matches. **c, d.** Peptide-level precision-recall curves for tagged and label-free samples, with measurements of total numbers of peptides recalled and area under each precision-recall curve. **e, f.** Overall accuracy of *de novo* amino acid assignments for residues with a distance of up to 4 AA from the peptide N- or C-terminus. Amino acids in the *de novo* sequence are considered to be correctly assigned if the database sequence contains a counterpart amino acid within 0.02Da in mass, with a prefix mass (sum of all preceding amino acids) within 0.1Da.

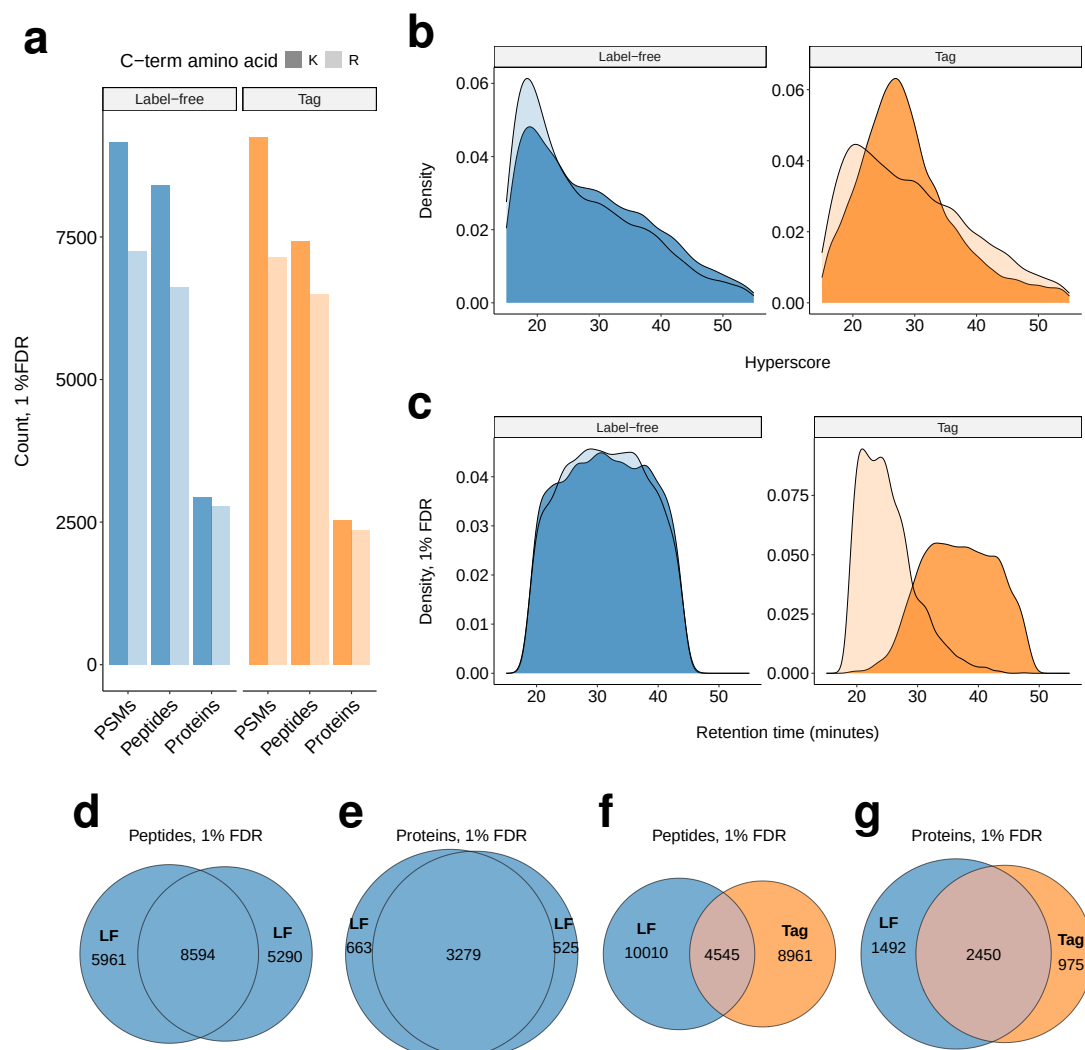


Figure S5 | Descriptive statistics on peptides and proteins identified from 2ng tagged and label-free samples by DDA. **a**, The counts at 1% FDR (spectrum Q-value calculated by Sage): peptide-spectra-matches (PSMs), peptides identified with or without quantitative information extracted by Sage, peptides identified with quantitative information extracted by Sage, and likewise for proteins. Counts supported by a sequence with at least one arginine are lighter shades. Counts supported by a sequence with at least one lysine are darker shades. **b**, The density distributions of peptides' scores. Peptides with arginine on the C-terminus are the lighter shades. Peptides with lysine on the C-terminus are darker shades. **c**, The density distributions of peptide retention times. While most arginine and lysine terminating peptides elute together when label-free, the lysine peptide elute significantly later due to carrying an extra tag. **d**, Peptides observed in common or uniquely for two label-free replicates, one replicate at normalized collision energy (NCE) 28, the other other at NCE = 24. **e**, Proteins observed in common or uniquely for two label-free replicates, one replicate at normalized collision energy (NCE) 28, the other other at NCE = 24. **f**, Peptides observed in common or uniquely for tagged (NCE=24) and label-free samples (NCE=28). **g**, Proteins observed in common or uniquely for tagged (NCE=24) and label-free samples (NCE=28).

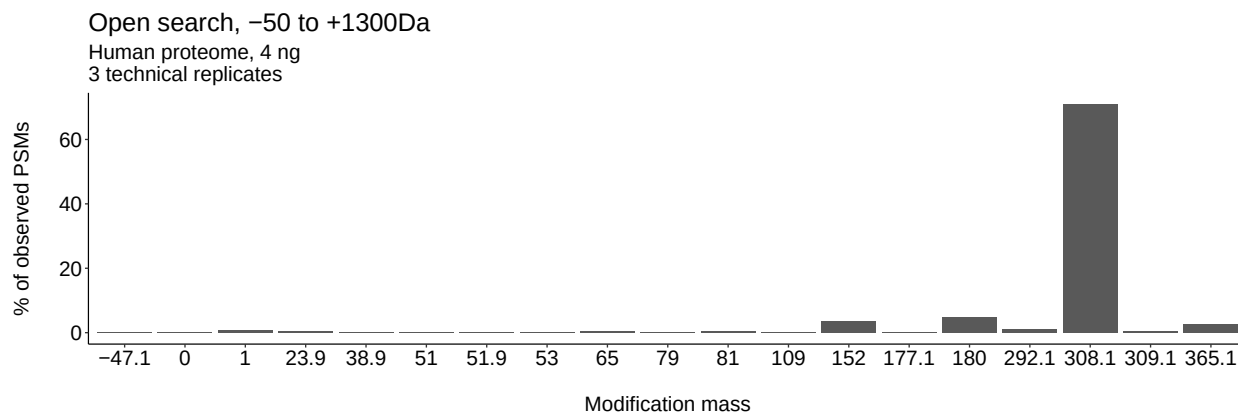


Figure S6 | Other labeling products of PSMtag Three replicate injections of PSMtag labeled human proteomes (K-562 cell lines) of 4 ng were acquired by the Orbitrap Astral and analyzed by MSFragger using the open search workflow over a range of -50 Da to 1300 Da.

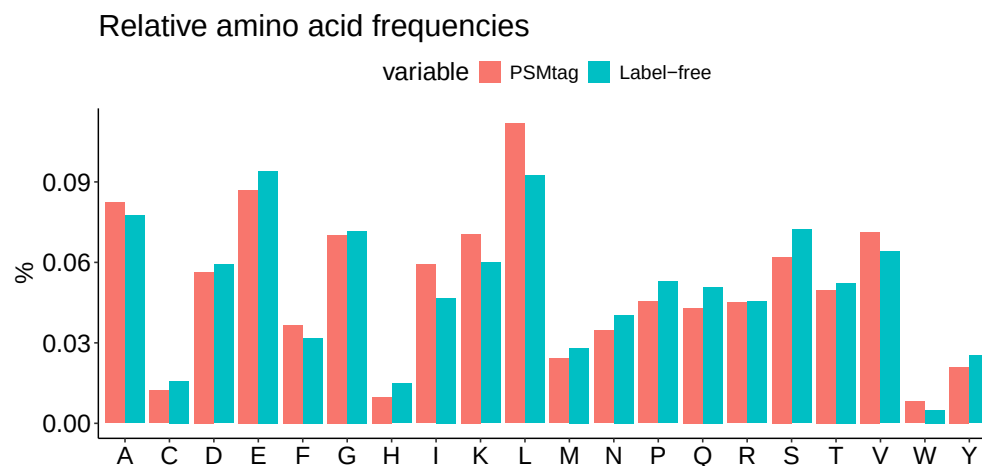


Figure S7 | The relative observation of amino acids in PSMtag vs. label-free proteomes The DDA data used in Fig. S5 was examined for the relative observation frequency of amino acids and labeled and label-free peptides to understand the source of different peptides observations Fig. S5f.

Formula (NHS-ester)	Mass	Mass delta	Adduct formula	Adduct mass
C ₂₂ H ₂₁ N ₃ O ₆	423.1430	0.0000	C ₁₈ H ₁₆ N ₂ O ₃	308.11609
C ₂₀ ¹³ C ₂ H ₂₁ N ₃ O ₆	425.1497	2.0067	C ₁₆ ¹³ C ₂ H ₁₆ N ₂ O ₃	310.12280
C ₂₀ ¹³ C ₂ H ₂₁ N ₃ O ₅ ¹⁸ O	427.1540	4.0110	C ₁₆ ¹³ C ₂ H ₁₆ N ₂ O ₂ ¹⁸ O	312.12704
C ₁₆ ¹³ C ₆ H ₂₁ N ₃ O ₆	429.1632	6.0202	C ₁₂ ¹³ C ₆ H ₁₆ N ₂ O ₃	314.13622
C ₁₄ ¹³ C ₈ H ₂₁ N ₃ O ₆	431.1699	8.0269	C ₁₀ ¹³ C ₈ H ₁₆ N ₂ O ₃	316.14293
C ₁₄ ¹³ C ₈ H ₂₁ N ₃ O ₅ ¹⁸ O	433.1741	10.0311	C ₁₀ ¹³ C ₈ H ₁₆ N ₂ O ₂ ¹⁸ O	318.14717
C ₁₁ ¹³ C ₁₁ H ₂₁ N ₂ ¹⁵ NO ₆	435.1770	12.0340	C ₇ ¹³ C ₁₁ H ₁₆ N ¹⁵ NO ₃	320.15003
C ₉ ¹³ C ₁₃ H ₂₁ N ₂ ¹⁵ NO ₆	437.1837	14.0407	C ₅ ¹³ C ₁₃ H ₁₆ N ¹⁵ NO ₃	322.15674
C ₁₀ ¹³ C ₁₂ H ₂₁ N ¹⁵ N ₂ O ₅ ¹⁸ O	439.1879	16.0449	C ₆ ¹³ C ₁₂ H ₁₆ ¹⁵ N ₂ O ₂ ¹⁸ O	324.15466

Figure S8 | Empirical formulas and masses for tag reagents The empirical formulas for the isotopically-doped reagents functionalized with NHS-ester allowing 5-plex and 9-plex with their corresponding monoisotopic masses (Mass) and differences in mass from the lowest mass isotopomer (Mass delta).

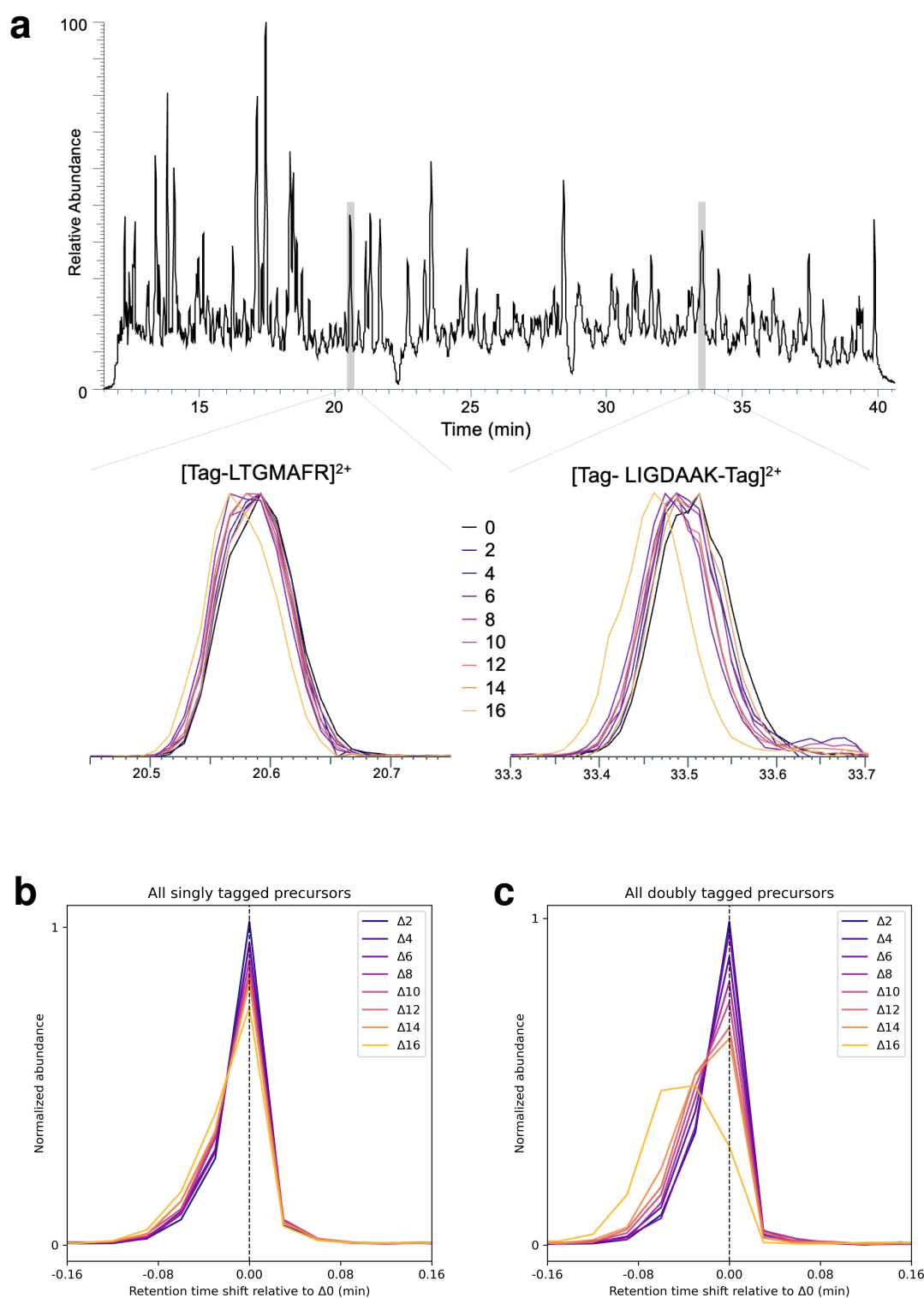


Figure S9 | The effect of isotopic doping on tagged peptide retention time **a**, The extracted signal for select high-abundance tagged peptides and their isotopologues. **b**, The differences in retention time for all confidently-identified singly tagged peptide isotopologues from their respective $\Delta 0$ channel. **c**, The differences in retention time for all confidently-identified doubly tagged peptide isotopologues from their respective $\Delta 0$ channel.

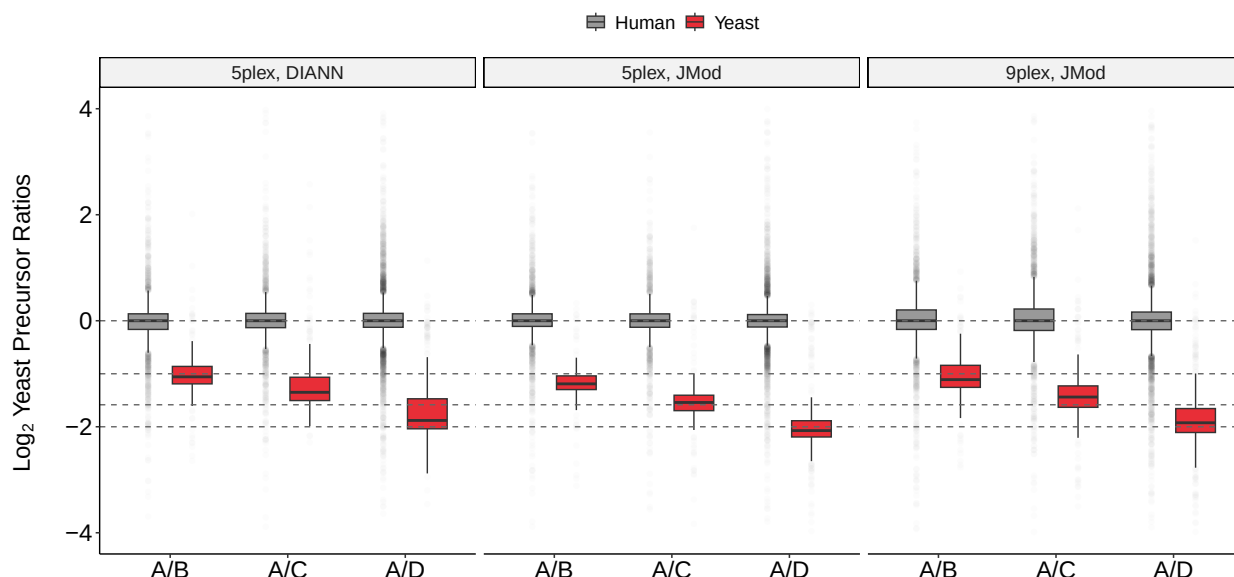


Figure S10 | Extended examination of yeast spike in ratios Extension of Fig. 4f, examining more mixing ratios as described in Fig. 4e.

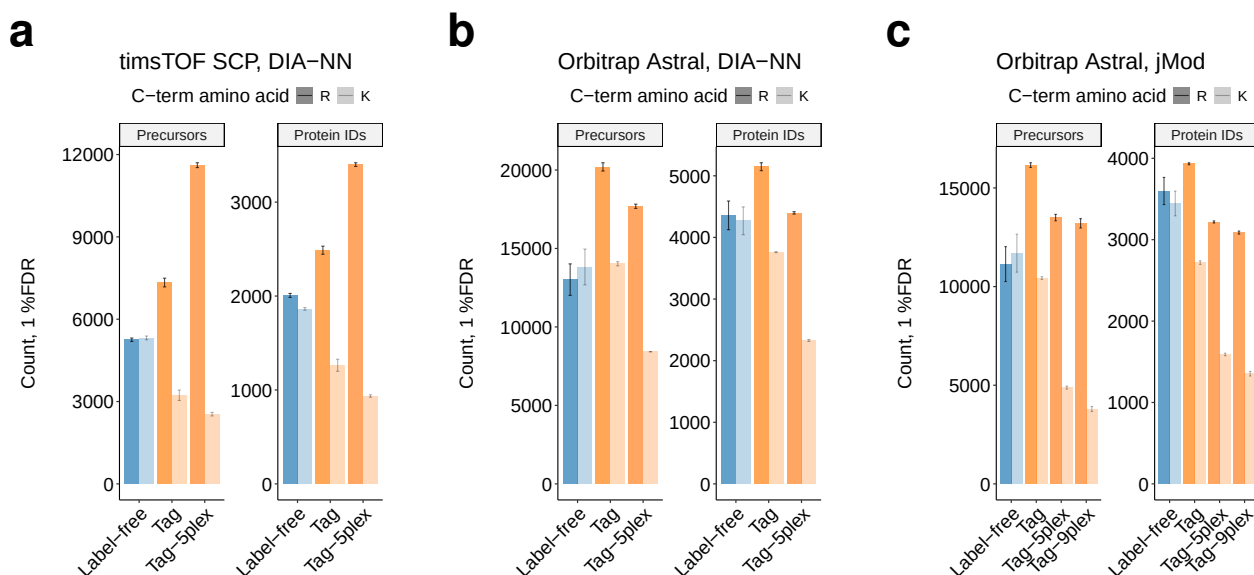


Figure S11 | Identification rates for tagged and label-free single-cell-level (200pg) proteomes by DIA. **a**, The number precursors containing at least one lysine or arginine and proteins supported by these precursors identified in replicate injections of a 200pg human proteome labeled, unlabeled, and labeled with 5-plex reagents. Data acquired using the Orbitrap Astral mass spectrometer and analyzed with search engine DIA-NN 2.1. **b**, The same, but data acquired with timsTOF SCP mass spectrometer. **c**, The same, but data (Astral) analyzed with JMod.

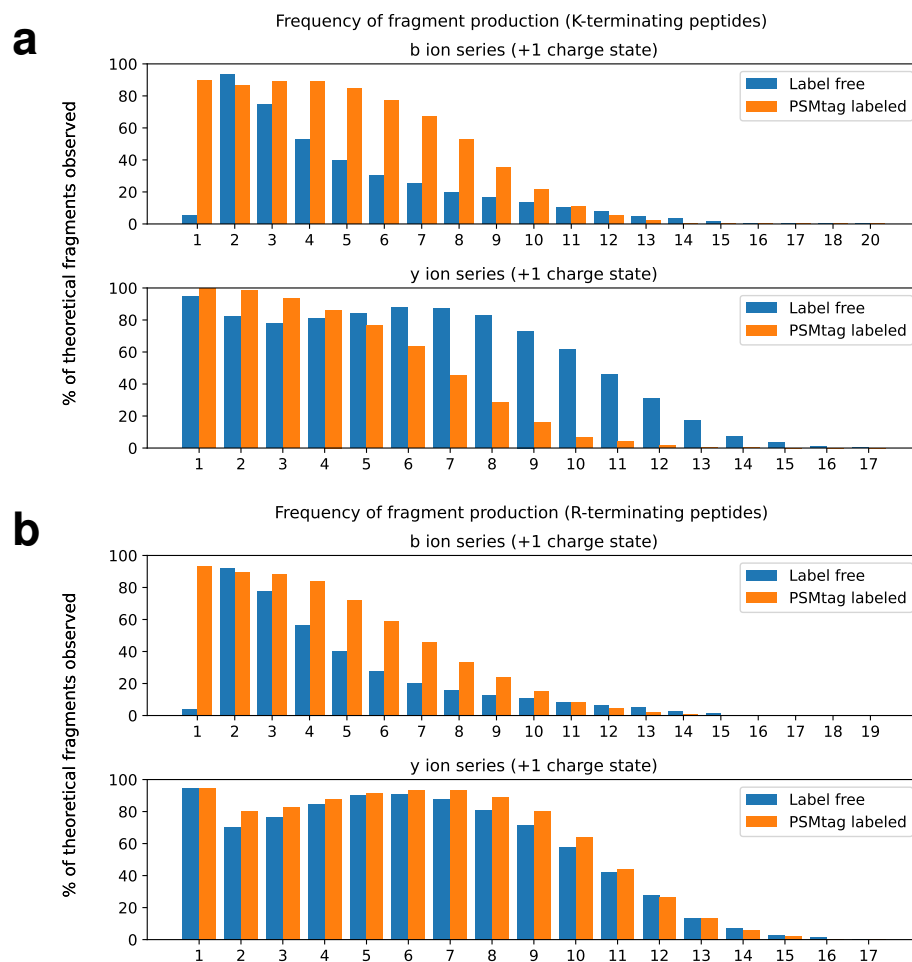


Figure S12 | Fragment ion series with one positive charge for tagged and label-free peptides **a**, The frequency of observing b-ions and y-ions with two positive charges for all peptides confidently assigned to terminate in lysine. **b**, The frequency of observing b-ions and y-ions with two positive charges for all peptides confidently assigned to terminate in arginine.

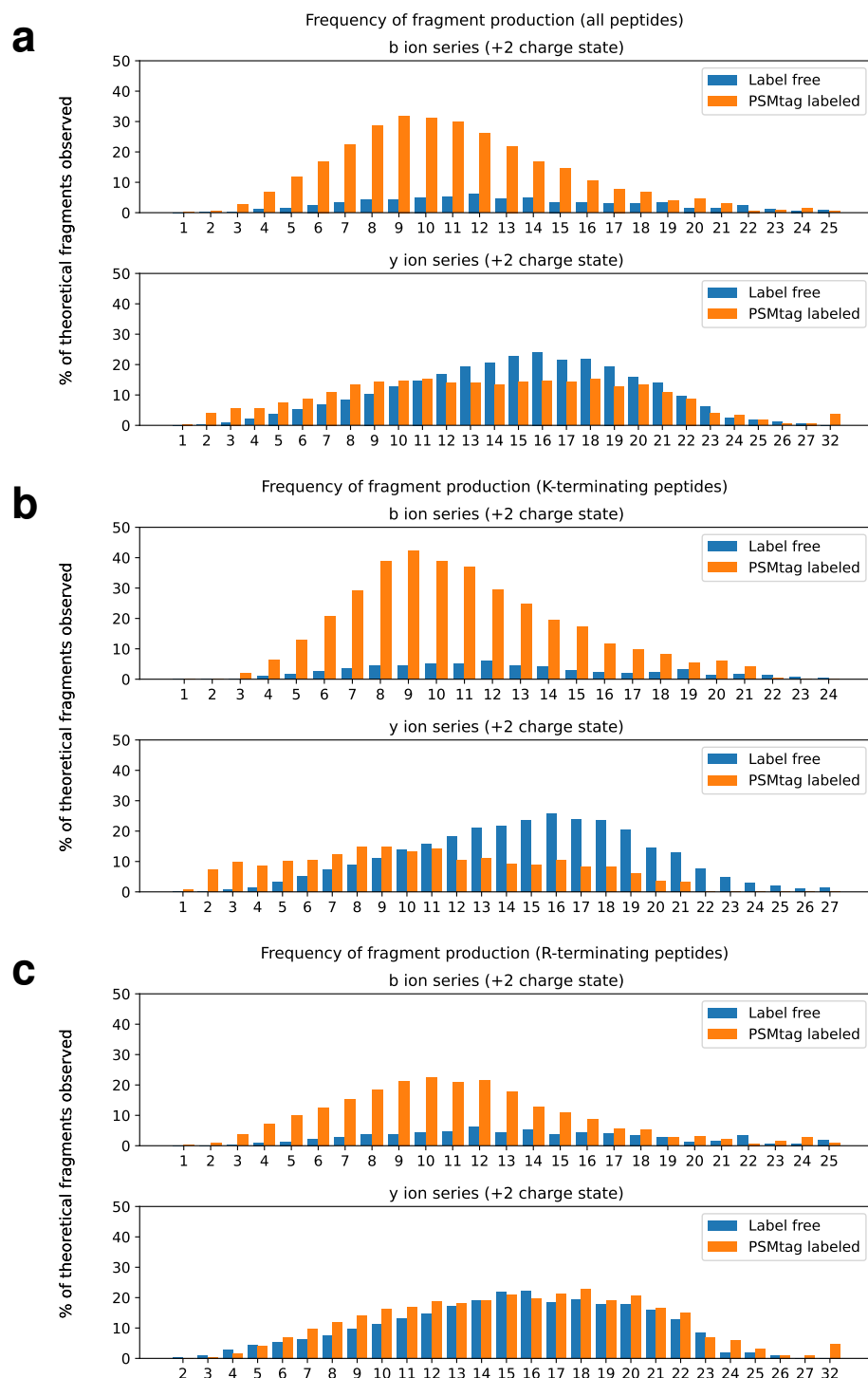


Figure S13 | Fragment ion series with two positive charges for tagged and label-free peptides **a**, The frequency of observing b-ions and y-ions with two positive charges for all peptides confidently assigned. **b**, The subset terminating in lysine. **c**, The subset terminating in arginine.

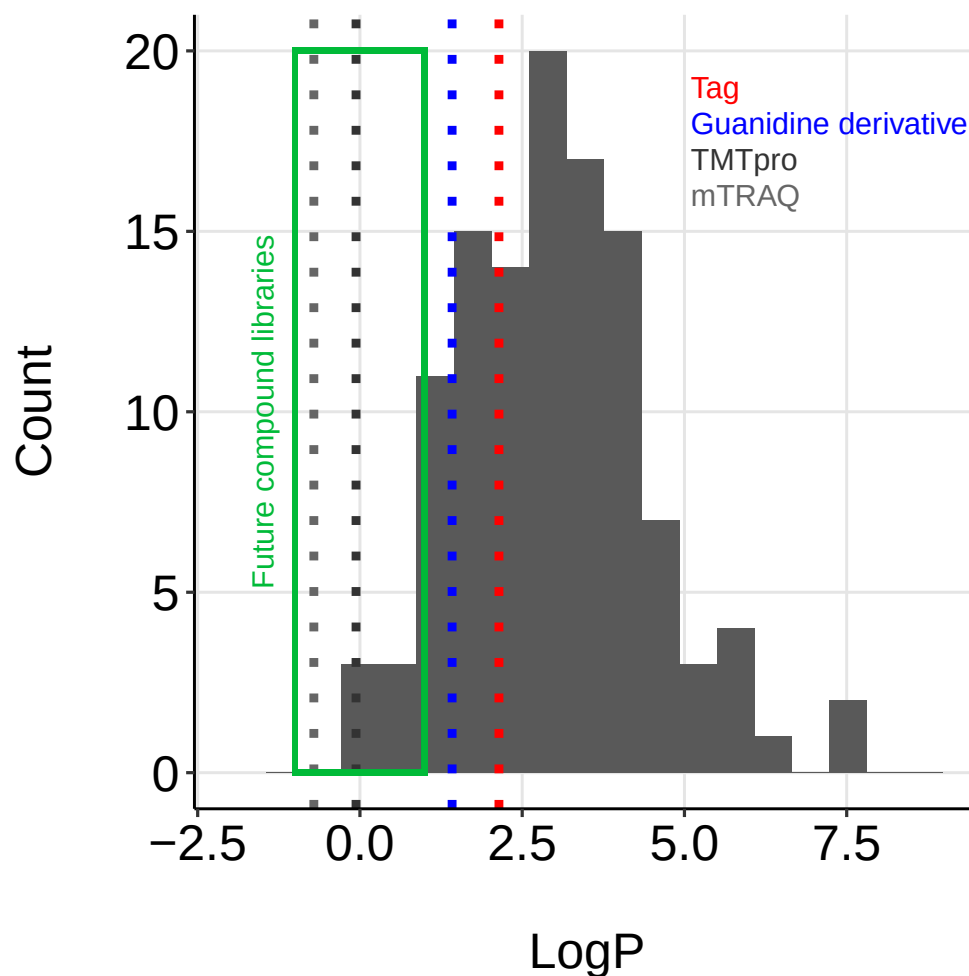


Figure S14 | The distribution of logP values for mass tag candidates and commercial mass tags The logP values for compounds were estimated from SMILES strings using the python package *rdkit*.

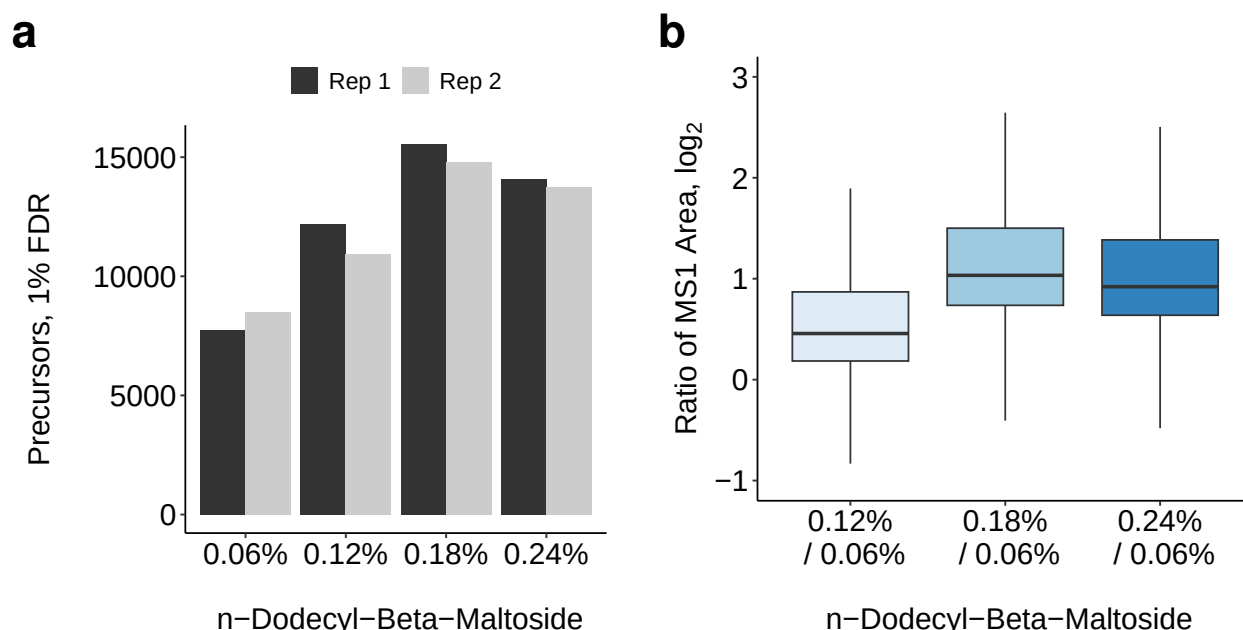


Figure S15 | Determining optimal reconstitution conditions for tagged peptides **a**, The number of precursors identified by the search engine DIA-NN for 200pg inputs of tagged peptides from a human proteome from reconstitution in varying level of n-dodecyl-beta-maltoside (DDM). **b**, For precursors intersected across conditions, the distribution of relative intensities, relative to the 0.06% DDM condition. All pairwise combinations of precursor replicates are considered.

ID	MW (mg/mmol)	mass req (mg)
A	452.304	32.51
B	466.331	33.52
C	466.331	33.52
D	466.331	33.52
E	466.331	33.52
F	467.319	33.59
G	468.307	33.66
H	478.342	34.38
I	478.342	34.38
J	492.369	35.39
K	492.369	35.39
L	494.341	35.53
M	506.396	36.40
N	517.379	37.19
O	493.357	35.46
P	552.425	39.71

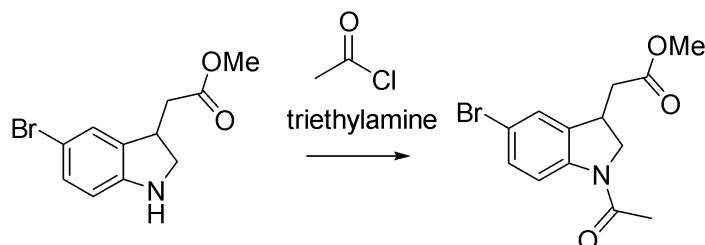
Figure S16 | The pooling-scheme for increasing LC-MS/MS throughput of combinatorially-synthesized mass tag candidates To increase throughput of analyzing mass tag-model peptide conjugates by LC-MS/MS those candidates with different masses were combined.

Supplemental Note 1:

The tag described to this study is referred to as "Tag 6" in the descriptions on the following pages.

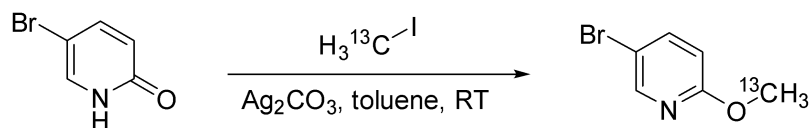
LCMS method: column: Agilent Zorbax SB-C18, 3.5 μ m, 4.6x50 mm, flow rate 1 mL/min, 0.1% Formic acid ACN/H₂O. Gradient: 0.5 min at 2% ACN, linear ramp to 100% ACN over the next 7 min, hold at 100% ACN for 2.5 min. Return to 2% ACN over the next 0.1 min.

Synthesis schemes for isotopologs and solution phase synthesis of analogs of Tag 6



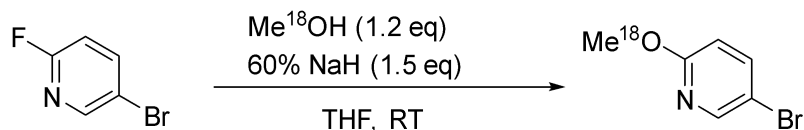
methyl 2-(1-acetyl-5-bromoindolin-3-yl)acetate

To a stirred mixture of methyl 2-(5-bromo-2,3-dihydro-1H-indol-3-yl)acetate hydrochloride (0.5 g, 1.63 mmol), triethylamine (455 μ L, 3.26 mmol) in CH₂Cl₂ (8 mL) at 0 °C acetyl chloride (139 μ L, 1.96 mmol) in CH₂Cl₂ (2 mL) was added dropwise over 10 min. The reaction mixture was stirred for 2 hours under a nitrogen atmosphere. After completion of the reaction as indicated by TLC, the reaction was quenched with cold water, and saturated NaHCO₃. (10 mL) The organic layer was extracted with CH₂Cl₂ (3 x 30 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na₂SO₄. The solvent was removed under reduced pressure to give crude product. The crude residue was purified by column chromatography (Hexane/EtOAc, 5:5) to provide methyl 2-(1-acetyl-5-bromoindolin-3-yl)acetate (0.47 g, 92.3%) as off-white solid. ¹H NMR (300 MHz, CDCl₃) δ 8.10 (d, *J* = 8.4 Hz, 1H), 7.34 (dd, *J* = 8.4 Hz, 2H), 4.32 (t, *J* = 8.7 Hz, 1H), 3.78–3.73 (m, 5H), 2.85–2.78 (m, 1H), 2.61–2.52 (m, 1H), 2.21 (s, 3H); [M+H]⁺: 312.



Preparation of 5-bromo-2-(methoxy-¹³C)pyridine

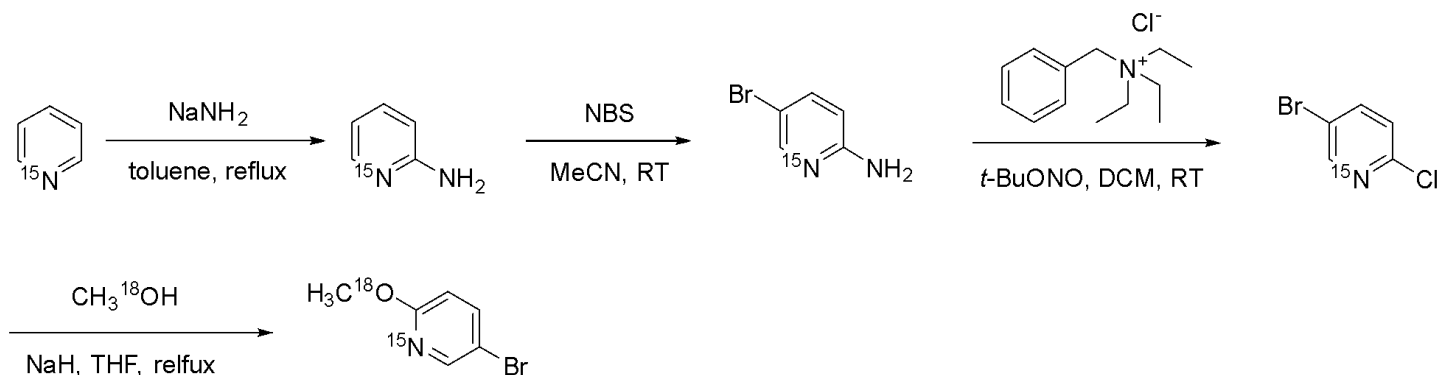
To the suspension of 5-Bromo-1H-Pyridin-2-one (550.0 mg, 1 Eq, 3.161 mmol) and silver carbonate (958.8 mg, 1.1 Eq, 3.477 mmol) in Toluene (10 mL) was added Iodo(¹³C)methane (497.0 mg, 217 μ L, 1.1 Eq, 3.477 mmol), and stirred at RT for 3 days. The reaction mixture was filtered through celite to remove the silver salt, and the filter cake was flushed with DCM. The filtrate was concentrated under reduced pressure (~10 bar) to remove the solvent. The residue was purified by silica gel flash chromatography using 0–3% DCM in hexane as the eluent to obtain 5-bromo-2-(methoxy-¹³C)pyridine (228.8 mg, 38.2% yield) as a colorless oil. ¹H NMR (300 MHz, CDCl₃-d): δ 8.20 (d, *J* = 2.4 Hz, 1H), 7.63 (dd, *J* = 9 Hz, 2.7 Hz, 1H), 6.66 (dd, *J* = 9 Hz, 0.6 Hz, 1H), 3.90 (d, *J* = 146.1 Hz, 3H). ¹³C NMR (75 MHz, CDCl₃-d): δ 163.17, 147.69, 141.28, 112.84, 111.91, 53.97.



Preparation of 5-bromo-2-(methoxy-¹⁸O)pyridine

To a solution of methanol-¹⁸O (578 mg, 690 μ L, 1.20 Eq, 17.0 mmol) in anhydrous THF (20 mL) under nitrogen with vigorous stirring was added the NaH (852.3 mg, 60% Wt, 1.5 Eq, 21.308 mmol) at 0 °C. After stirring at

RT for 15 min, 5-bromo-2-fluoropyridine (2500.0 mg, 1 Eq, 14.205 mmol) was added dropwise. The resulting mixture was stirred at RT for 4h, then poured into saturated aq. NH_4Cl and extracted with EtOAc (5×10 mL). The combined organic layers were dried over Na_2SO_4 , filtered, and concentrated under reduced pressure to obtain a crude product, which was purified by silica gel flash chromatography by using 0-3% DCM in Hexanes as the eluent to afford 5-bromo-2-(methoxy- ^{18}O)pyridine (2405.1 mg, 89.1 % yield) as a colorless oil. ^1H NMR (300 MHz, CDCl_3 - d): δ 8.20 (d, $J = 2.7$ Hz, 1H), 7.63 (dd, $J = 8.7$ Hz, 2.7 Hz, 1.5 Hz, 1H), 6.66 (d, $J = 9.0$ Hz, 1H), 3.90 (s, 3H). ^{13}C NMR (75 MHz, CDCl_3 - d): δ 163.25, 147.76, 141.22, 112.78, 111.87, 53.85.



Preparation of pyridin-1- ^{15}N -2-amine

To the suspension of sodium amide (1.22 g, 99% Wt, 2.5 Eq, 30.9 mmol) in Toluene (4 mL) in a 40 mL reaction vial at 115 °C with vigorous stirring under Argon atmospheric was slowly added Pyridine- ^{15}N (990 mg, 1.01 mL, 1 Eq, 12.4 mmol) over 5 min, kept heating at 115 °C overnight. After cooling to RT, the residue was slowly quenched by 2 mL of water dropwise, then heated to 70 °C for 20 min. Observed a lot of bubbles during the water addition. Cooled to RT again, separated the lower aqueous solution, and extracted with toluene (6×2 mL). Combined the organic layers, dried over Na_2SO_4 , filtered, and removed the solvent under reduced pressure to obtain an orange-brown solid as the crude product. The crude product was sublimated under reduced pressure at 90-110°C to obtain pyridin-1- ^{15}N -2-amine (898.3 mg, 76.4 % yield) as a light yellow solid. ^1H NMR (300 MHz, CDCl_3 - d): δ 8.12 – 7.99 (m, 1H), 7.47 – 7.36 (m, 1H), 6.68 – 6.59 (m, 1H), 6.49 (dd, $J = 8.4$ Hz, 0.6 Hz, 1H), 4.45 (br s, 2H). ^{13}C NMR (75 MHz, CDCl_3 - d): δ 158.65, 148.40, 137.85, 114.23, 108.72. HRMS m/z calc for $\text{C}_5\text{H}_7\text{N}^{15}\text{N}$ ($\text{M}+\text{H}$) $^+$: 96.06; found: 95.89.

Preparation of 5-bromopyridin-1- ^{15}N -2-amine

To the solution of pyridin-1- ^{15}N -2-amine (890.0 mg, 1 Eq, 9.358 mmol) in dry Acetonitrile (20 mL) in an ice water bath was added a solution of NBS (1.699 g, 1.02 Eq, 9.545 mmol) in Acetonitrile (10 mL) dropwise. Slowly warmed up to RT, and stirred the reaction mixture for 2 h at RT. Removed the solvent under reduced pressure, then diluted with sat. NaHCO_3 aq (35 mL), extracted with EtOAc (4×15 mL). Combined the organic layers, dried over Na_2SO_4 , filtered, and removed the solvent under reduced pressure. The residue was purified by silica gel flash chromatography using 0-15% EtOAc in DCM as the eluent to obtain 5-bromopyridin-1- ^{15}N -2-amine (1.7007 g, 104.4 % yield) as a yellow solid. ^1H NMR (300 MHz, CDCl_3 - d): δ 8.09 (dd, $J = 10.8$ Hz, 2.4 Hz, 1H), 7.49 (dd, $J = 8.4$ Hz, 2.4 Hz, 1H), 6.41 (d, $J = 9.0$ Hz, 1H), 4.58 (br s, 2H). ^{13}C NMR (75 MHz, CDCl_3 - d): δ 157.30, 148.83, 140.36, 110.23, 108.50. HRMS m/z calc for $\text{C}_5\text{H}_6\text{BrN}^{15}\text{N}$ ($\text{M}+\text{H}$) $^+$: 173.97; found: 174.03.

Preparation of 5-bromo-2-chloropyridine-1- ^{15}N

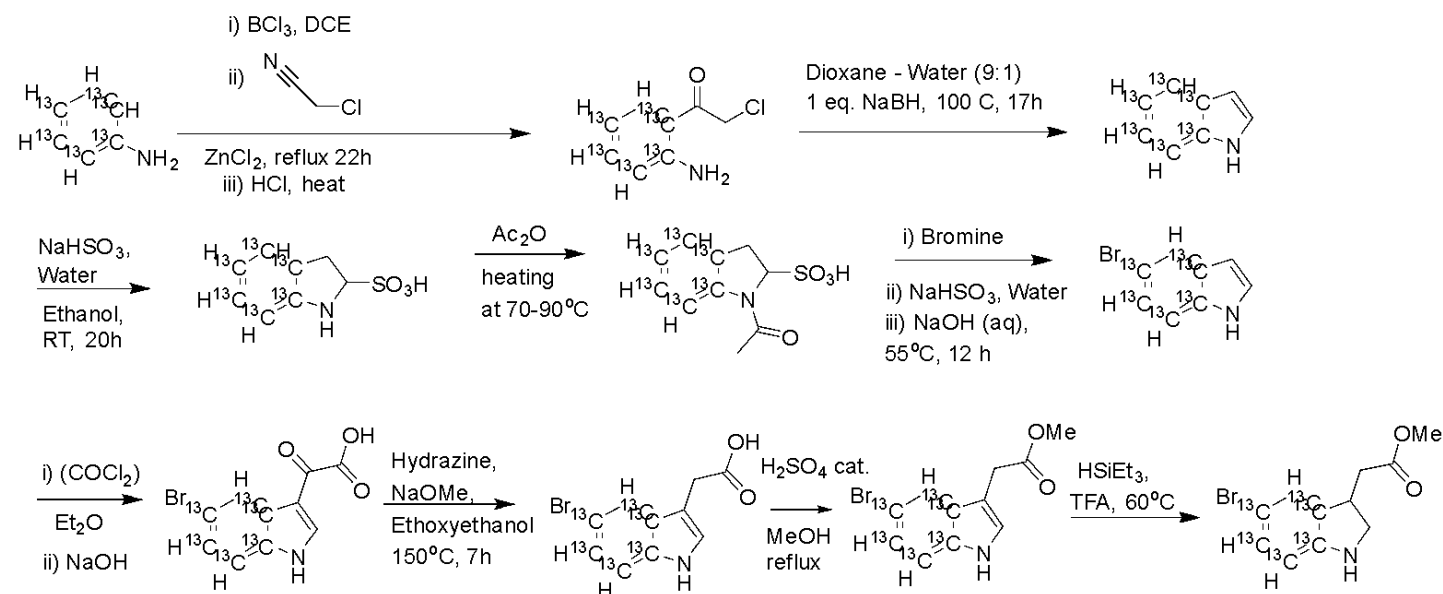
To a solution of 5-bromopyridin-1- ^{15}N -2-amine (1000 mg, 1 Eq, 5.747 mmol) in dry DCM (20 mL) was added benzyl(triethyl)ammonium chloride (5.891 g, 4.5 Eq, 25.86 mmol), followed by the addition of tert-Butyl nitrite (5.926 g, 6.84 mL, 10 Eq, 57.47 mmol) in an ice-water bath under an argon atmosphere. After stirring the solution at RT overnight, the reaction mixture was quenched by sat. aqueous NaHCO_3 solution (35 mL), then extracted with DCM (3×10 mL). Combined the organic layers, dried over Na_2SO_4 , filtered, and removed the

solvent under reduced pressure. The residue was purified by silica gel flash chromatography using 0-5% DCM in hexane as the eluent to obtain 5-bromo-2-chloropyridine-1-¹⁵N (484.3 mg, 43.57 % yield) as a white solid. ¹H NMR (300 MHz, CDCl₃-d): δ 8.46 (dd, *J* = 11.7 Hz, 2.7 Hz, 1H), 7.77 (dd, *J* = 8.4 Hz, 2.7 Hz, 1H), 7.24 (d, *J* = 8.4 Hz, 1H). ¹³C NMR (75 MHz, CDCl₃-d): δ 150.93, 150.39, 141.40, 125.83, 119.32.

Preparation of 5-bromo-2-(methoxy-¹⁸O)pyridine-1-¹⁵N

To a solution of methanol-18O (84.47 mg, 101 μL, 1.2 Eq, 2.482 mmol) in anhydrous THF (6 mL) under nitrogen with vigorous stirring was added NaH (124.1 mg, 60% Wt, 1.5 Eq, 3.102 mmol) at 0 °C. After stirring at 0 °C for 15 min, a solution of 5-bromo-2-chloropyridine-1-¹⁵N (400.0 mg, 1 Eq, 2.068 mmol) in THF (2 mL) was added dropwise. The resulting mixture was stirred at 85 °C for 41h. After cooling to RT, the reaction mixture was poured into saturated aq. NH₄Cl (30 mL) and extracted with Et₂O (2×10 mL). Combined the organic layers, dried over Na₂SO₄, filtered, and removed the solvent under reduced pressure. The residue was purified by silica gel flash chromatography by using 0-5% DCM in hexanes to afford 5-bromo-2-(methoxy-¹⁸O)pyridine-1-¹⁵N (317.6 mg, 80.40 % yield) as a colorless oil. The sample solidified at -20°C during storage. ¹H NMR (300 MHz, CDCl₃-d): δ 8.19 (ddd, *J* = 11.4 Hz, 2.7 Hz, 0.6, 1H), 7.63 (dd, *J* = 8.7 Hz, 2.7 Hz, 1H), 6.66 (dd, *J* = 9 Hz, 0.9 Hz, 1H), 3.90 (s, 3H). ¹³C NMR (75 MHz, CDCl₃-d): δ 163.26, 147.73, 141.22, 112.76, 111.90, 53.85.

Intermediate for T6, T8, and T10:



1-(2-aminophenyl-1,2,3,4,5,6-¹³C₆)-2-chloroethan-1-one was prepared according to the protocol described in Wager, C.A.B; Miller, S.A; J. Label Cmpd Radiopharm 2006; 49; 615-622, LCMS RT 5.23 min, (M+H)⁺: Expected 176.1; observed 176.0

1H-indole-3a,4,5,6,7,7a-¹³C₆ was prepared according to the protocol described in Wager, C.A.B; Miller, S.A; J. Label Cmpd Radiopharm 2006; 49; LCMS RT 5.47 min, (M+H)⁺ Expected MW: 124.1; observed 124.2

5-bromo-1H-indole-3a,4,5,6,7,7a-¹³C₆ was prepared via a sulfonate according to the protocol described in Chinese Patent Suzhou Hande Chuanghong Biochemical Technology Co., Ltd.;Mao Zhongping;Ma Dongxu;Jiang Xiangsheng CN110396059, 2019. LCMS RT 6.08 min, (M-H)⁻: Expected 200.0/202.0; observed 200.2/202.1

2-(5-bromo-1H-indol-3-yl-3a,4,5,6,7,7a-¹³C₆)-2-oxoacetic acid was prepared according to the protocol described in the Japanese Patent (2022) 7125719 B2, WO2018164214 A1 2018-09-13, LCMS RT 4.75 min, (M-H)⁻: Expected 272.0, 274.0, observed 272.0, 274.0

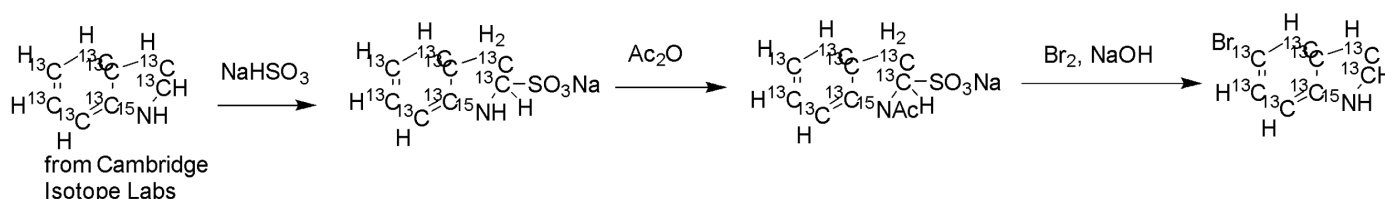
2-(5-bromo-1H-indol-3-yl-3a,4,5,6,7,7a-¹³C₆)acetic acid was prepared according to the protocol described in the Japanese Patent (2022) 7125719 B2, WO2018164214 A1 2018-09-13, LCMS RT 5.08 min, (M-H)⁻: Expected 258.0, 260.0; observed 258.1, 260.1

Methyl 2-(5-bromo-1H-indol-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate methyl
2-(5-bromo-1H-indol-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate

The indole carboxylic acid was dissolved in methanol and catalytic amount of sulfuric acid, and the resultant solution was heated at reflux overnight and then evaporated to dryness and used directly in the next step. LCMS RT 6.00 min, (M-H)⁻: expected 272.0, 274.0; observed 272.1, 274.3

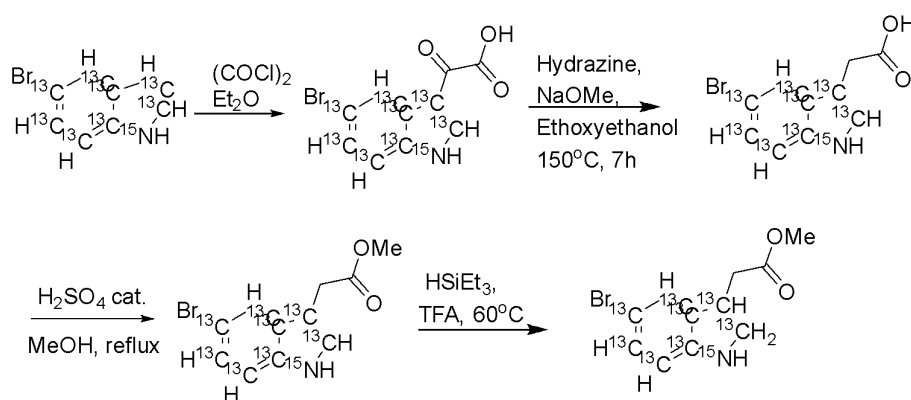
Methyl 2-(5-bromoindolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate was prepared according to the protocol described in Phytochemistry 68 (2007) 2512–2522. LCMS RT 5.37 min, (M+H)⁺: expected 276.0, 278.0; observed 276.0, 278.0

Intermediate for T12, T14, and T16



5-bromo-1H-indole-¹³C₈-¹⁵N was prepared via a sulfonate according to the protocol described in Chinese Patent Suzhou Hande Chuanghong Biochemical Technology Co., Ltd.;Mao Zhongping;Ma Dongxu;Jiang Xiangsheng CN110396059, 2019.

Intermediate for T12



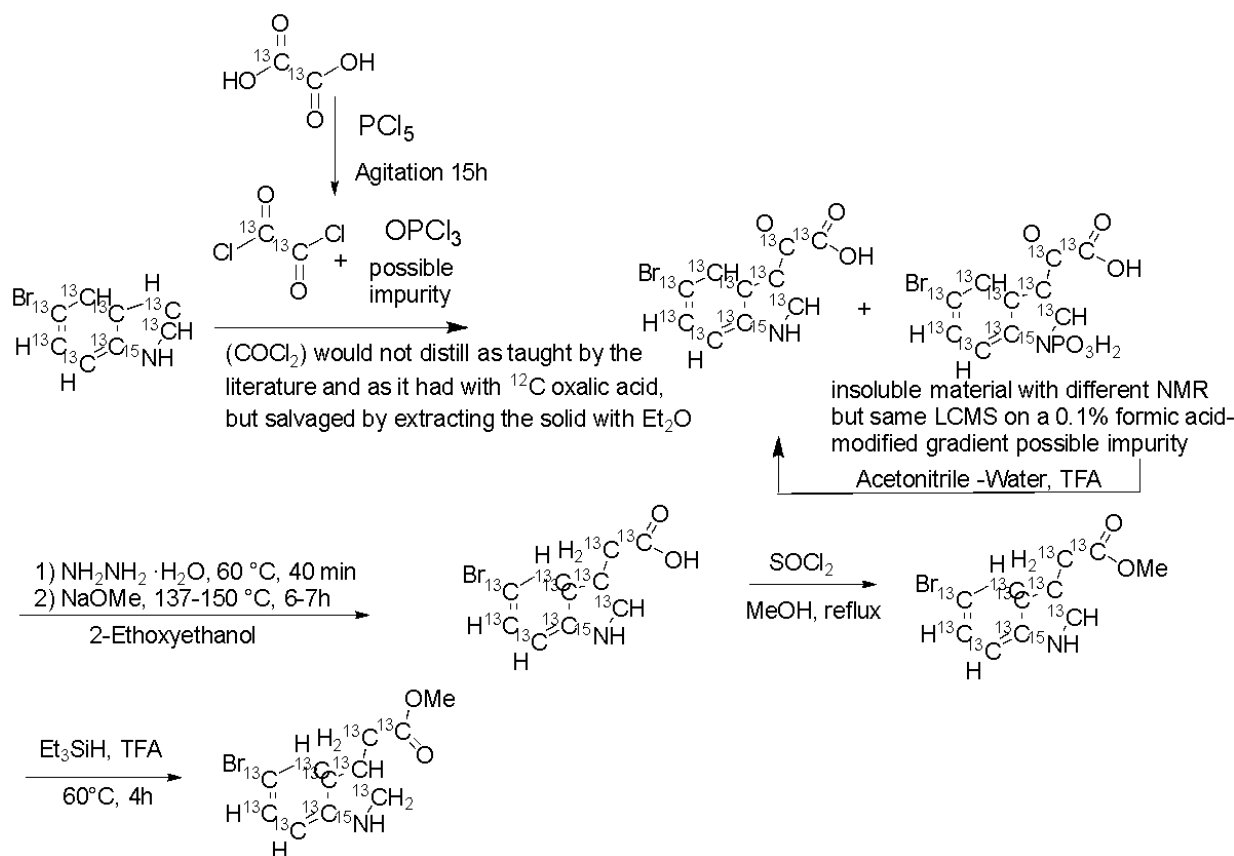
2-(5-Bromo-1H-indol-3-yl-¹³C₈-¹⁵N)-2-oxoacetic acid was prepared according to the protocol described in the Japanese Patent (2022) 7125719 B2, WO2018164214 A1 2018-09-13. LCMS RT 4.78 min, (M-H)⁻: 275.0, 277.0; observed 275.2, 277.1

2-(5-Bromo-1H-indol-3-yl-¹³C₈-¹⁵N)acetic acid was prepared according to the protocol described in the Japanese Patent (2022) 7125719 B2, WO2018164214 A1 2018-09-13. LCMS RT 5.15 min, (M-H)⁻: Expected 261.0, 263.0, observed 261.1, 263.1

Methyl 2-(5-bromo-1H-indol-3-yl-¹³C₈-¹⁵N)acetate. The indole carboxylic acid was dissolved in methanol and catalytic amount of sulfuric acid, and the resultant solution was heated at reflux overnight. The mixture was evaporated to dryness and used directly in the next reaction. LCMS RT 5.90 min, (M-H)⁻: Expected 275.0, 277.0; observed 275.1, 277.0

methyl 2-(5-bromoindolin-3-yl- $^{13}\text{C}_8$ - ^{15}N)acetate was prepared according to the protocol described in *Phytochemistry* 68 (2007) 2512–2522. LCMS RT 5.30 min, $(\text{M}+\text{H})^+$: Expected 279.0, 281.0, observed 279.0, 281.0

Intermediate for T14 and T16



2-(5-bromo-1H-indol-3-yl- $^{13}\text{C}_8$ - ^{15}N)-2-oxoacetic- $^{13}\text{C}_2$ acid

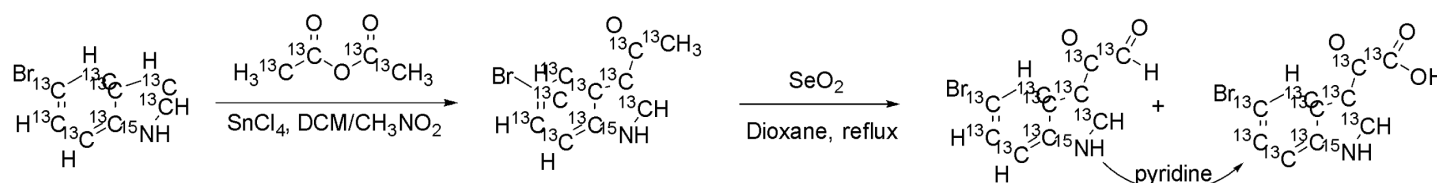
^{13}C -Oxalic acid (purchased from Cambridge isotopes lab) and Phosphorus pentachloride (5.4 g) was mixed in a 5 ml sealed vial and gently agitated with a vortex at RT for 15 hours. Hickman head distillation set up for distillation of labelled Oxalyl chloride failed, even though the reaction mixture liquified and this procedure worked multiple times reliably using ^{12}C -oxalic acid. Oxalyl chloride is volatile but nothing was collected in the Hickman head receiver. Ether 5 ml was added into the reaction mixture to extract the ^{13}C oxalyl chloride, and this was cooled to 0 C and added to 350 mg of 5-bromo-1H-indole- $^{13}\text{C}_8$ - ^{15}N , in 10 mL diethyl ether. The resulting reaction mixture was allowed to warm to room temperature and stirred overnight. A precipitate formed and was collected by filtration. Both the precipitate and the filtrate showed an LCMS profile consistent with the desired product. The precipitate was hypothesized to contain an N-P bond resulting from residual POCl_3 from the oxalyl chloride formation step, and the precipitate was treated with 10% water, 70% acetonitrile, 20% TFA, resulting in a solution. The mixture was stirred 4h and evaporated to dryness. 110 mg (23%) LCMS RT 4.88 min, $(\text{M}-\text{H})^-$: Expected 277.0, 279.0; observed 277.2, 279.2

2-(5-bromo-1H-indol-3-yl- $^{13}\text{C}_8$ - ^{15}N)acetic- $^{13}\text{C}_2$ acid was prepared according to the protocol described in Japanese Patent (2022) 7125719 B2 WO2018164214 A1 2018-09-13 LCMS RT 5.09 min, $(\text{M}-\text{H})^-$: expected 263.0, 265.0; observed 263.1, 265.1

Methyl 2-(5-bromo-1H-indol-3-yl- $^{13}\text{C}_8$ - ^{15}N)acetate- $^{13}\text{C}_2$. The indole carboxylic acid was dissolved in methanol and catalytic amount of sulfuric acid, and the resultant solution was heated at reflux overnight and then

evaporated to dryness and used directly in the next step. LCMS RT 5.90 min, (M-H)⁻: expected MW: 277.0, 279.0; observed 277.1, 279.1

Methyl 2-(5-bromoindolin-3-yl-¹³C8-¹⁵N)acetate-¹³C2 was prepared according to the protocol described in Phytochemistry 68 (2007) 2512–2522. LCMS RT 5.51 min, (M+H)⁺: Expected MW: 281.0, 283.0, observed (ES⁺) 283.1



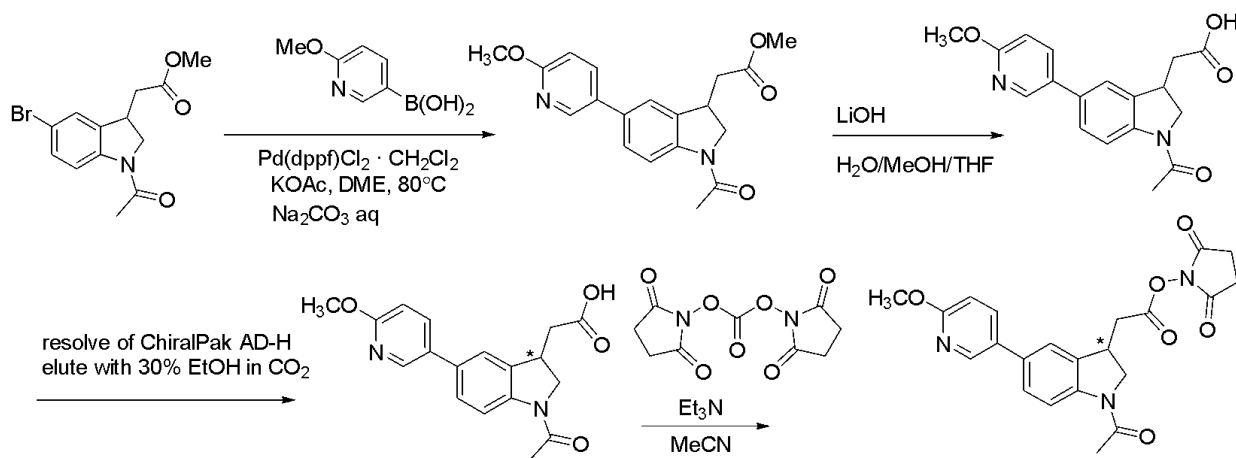
1-(5-bromo-1H-indol-3-yl-¹³C8-¹⁵N)ethan-1-one-1,2-¹³C2

To a stirring solution of 5-bromo-1H-indole-¹³C8-¹⁵N (400 mg, 1 Eq, 1.95 mmol) in DCM (4 mL) under argon at 0 °C was added tin(IV) chloride (610 mg, 275 μ L, 1.2 Eq, 2.34 mmol) in a single portion via syringe, and the mixture was stirred at 0 °C for 5 min. A light yellow precipitate formed. After the ice bath was removed, the mixture was stirred at room temperature for 30 min, at which point the mixture became a green solution. A solution of acetic anhydride-1,1,2,2-¹³C4 (207 mg, 185 μ L, 1 Eq, 1.95 mmol) in nitromethane (3 mL) was added dropwise. The mixture was stirred for 2 h at room temperature. After being quenched with ice (10 g), the mixture was extracted with ethyl acetate (5x6 mL). The organic phase was dried over Na₂SO₄ and concentrated at reduced pressure to give the crude product. The crude product was slurried in Et₂O and filtered to collect the off-white solid. The solid and filtrate were purified separately by on C18 silica using 3-100% MeCN in water with 0.1% FA as the eluent to afford 1-(5-bromo-1H-indol-3-yl-¹³C8-¹⁵N)ethan-1-one-1,2-¹³C2 (331.1 mg, 1.330 mmol, 68.1 %). LCMS rt 5.09 min, [M+H]⁺ expected 249.0/251.0, observed 249.2/251.2

2-(5-bromo-1H-indol-3-yl-¹³C8-¹⁵N)-2-oxoacetic-¹³C2 acid

To a suspension of 1-(5-bromo-1H-indol-3-yl-¹³C8-¹⁵N)ethan-1-one-1,2-¹³C2 (330 mg, 1 Eq, 1.33 mmol) in 1,4-Dioxane (30 mL) was added selenium dioxide (221 mg, 1.5 Eq, 1.99 mmol), and the mixture was heated at reflux overnight. LC-MS indicated predominant formation of 2-(5-bromo-1H-indol-3-yl-¹³C8-¹⁵N)-2-oxoacetaldehyde-¹³C2 to the reaction mix was added 1 mL of pyridine, and the mixture was heated at reflux overnight. After cooling to room temperature, the mixture was filtered to remove the SeO₂, and the filtrate was concentrated under reduced pressure to obtain the crude product as a pink solid. The crude product was dissolved in 100 mL of 1N NaOH aq, extracted with ethyl acetate (3 $\times$ 15 mL). The aqueous layer was acidified with 6 N HCl solution to pH=3 and extracted with ethyl acetate (5 $\times$ 20 mL). The aqueous layer was then acidified with 12 N HCl solution to pH=1 and extracted with ethyl acetate (2 $\times$ 20 mL). All the organic layers were combined, dried over anhydrous Na₂SO₄ and concentrated under reduced pressure to obtain 2-(5-bromo-1H-indol-3-yl-¹³C8-¹⁵N)-2-oxoacetic-¹³C2 acid (320 mg, 1.15 mmol, 86.5 %) as a yellow solid. LCMS rt 4.63 min, ES⁻ expected 277.0/279.0, observed 277.0/279.0

T0:



Methyl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate

In a 50 mL glass vial containing a magnetic stir bar, methyl 2-(1-acetyl-5-bromoindolin-3-yl)acetate **2** (0.25 g, 0.801 mmol) and 2-methoxy-5-pyridinylboronic acid **3** (0.147g, 0.961 mmol) were dissolved in 4 mL of DME (dimethoxyethane) and 1.2 mL of water. The vial was purged with nitrogen for 10 minutes.

[1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium(II)ComplexWithDichloromethane (0.0327 g, 0.0401 mmol), and sodium carbonate (0.170 g, 1.60 mmol) were then added, and the vial was purged with nitrogen for an additional 10 minutes. The resultant solution was stirred at room temperature for 5 minutes, then the vial was capped, heated to 80°C, and stirred for 60 minutes. LCMS analysis of the reaction mixture showed a major peak corresponding to the desired product, indicating that the starting material was completely consumed. The reaction mixture was filtered through a celite pad, and the solvent was concentrated under reduced pressure using a rotary evaporator. The crude residue was purified by column chromatography (Hexane/EtOAc, 2:8) to provide Methyl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate (0.240 g, 88%) as off- white solid. ¹H NMR (300 MHz, CDCl₃) δ 8.33 (s, 1H) 8.26 (d, *J* = 8.1 Hz, 1H), 7.74 (d, *J* = 8.4 Hz, 1H), 7.39 (d, *J* = 8.7 Hz, 1H), 7.29 (s, 1H), 6.80 (d, *J* = 9.0 Hz, 1H), 4.37 (t, *J* = 9.3 Hz, 1H), 3.97 (s, 3H) 3.90-3.74 (m, 5H), 2.92 – 2.87 (m, 1H), 2.66 – 2.57 (m, 1H), 2.25 (s, 3H)

2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid

Methyl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate (0.24 g, 0.705 mmol) was dissolved in methanol (6 mL) and water (2 mL), and LiOH (0.57g, 2.12 mmol) was added. The reaction mixture was stirred at room temperature for 1 hour. LCMS analysis indicated that the starting material was consumed, and the desired acid peak was observed. The methanol was removed under reduced pressure using a rotary evaporator, and 3 mL of water was added. The solution was neutralized with 4 N HCl until the pH reached 4. A solid formed, which was filtered and dried under vacuum to give 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid (0.206 g, 90%) as off-white solid. ¹H NMR (300 MHz, DMSO-D₆) δ 8.24 (s, 1H) 8.06 (d, *J* = 8.1 Hz, 1H), 7.95 (d, *J* = 8.7 Hz, 1H), 7.59 (s, 1H), 7.46 (d, *J* = 8.1 Hz, 1H), 6.87 (d, *J* = 8.7 Hz, 1H), 4.34 (t, *J* = 9.6 Hz, 1H), 3.86-3.70 (m, 5H), 2.97 – 2.91 (m, 1H), 2.61 – 2.55 (m, 1H), 2.15 (s, 3H); LCMS rt 4.87 min, [M+H]⁺: expected 327.1, observed 327.1

2,5-Dioxopyrrolidin-1-yl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate (racemic)

2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid **5** (10 mg, 0.031 mmol), N,N'-Disuccinimidyl Carbonate (8.7 mg, 0.034 mmol), and TEA (13 µL, 0.096 mmol) were dissolved in 0.5 mL of CH₃CN. The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of DCM and 10 mL of 0.1 M KH₂PO₄ solution, followed by extraction with DCM (3 × 20 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na₂SO₄. The solvent was removed under reduced pressure to give crude product. The crude

residue was purified by column chromatography (DCM/EtOAc, 5:5) to provide **2**. The product was dissolved in 5 mL of acetonitrile, lyophilized, and give 2,5-dioxopyrrolidin-1-yl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate (0.0105 g, 80%) as off -white solid. ¹H NMR (300 MHz, CDCl₃) δ 8.35 (s, 1H) 8.28 (d, *J* = 8.4 Hz, 1H), 7.76 (dd, *J* = 8.4 Hz, 1H), 7.40 (d, *J* = 8.1 Hz, 1H), 7.37 (s, 1H), 6.81 (d, *J* = 8.7 Hz, 1H), 4.37 (t, *J* = 9.6 Hz, 1H), 3.98 (s, 3H) 3.97-3.88 (m, 2H), 3.2 – 3.12 (m, 1H), 2.94 – 2.91 (m, 1H), 2.90 – 2.88 (m, 4H), 2.26 (s, 3H); LCMS rt 5.25 min, [M+H]⁺: expected 424.1, observed 424.0

Resolution: ChiralPak AD-H 21x250 mm, with 30% ethanol in CO₂ and a flow rate of 70 mL/min. Sample dissolved at 2 mg/mL in 50:50 dichloromethane:methanol and loaded at 2 mL per injection to afford 22.4 mg of peak 1, >99.5% purity and ee, 23.2 mg of peak 2 with >99.5% purity and 99% ee.

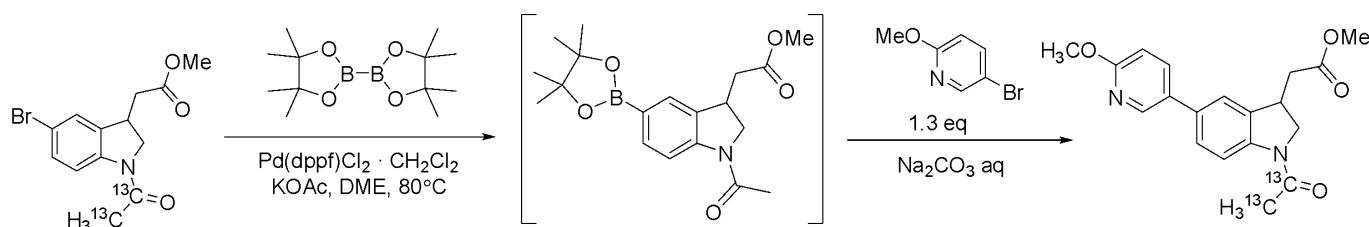
Peak 1: 2,5-Dioxopyrrolidin-1-yl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate

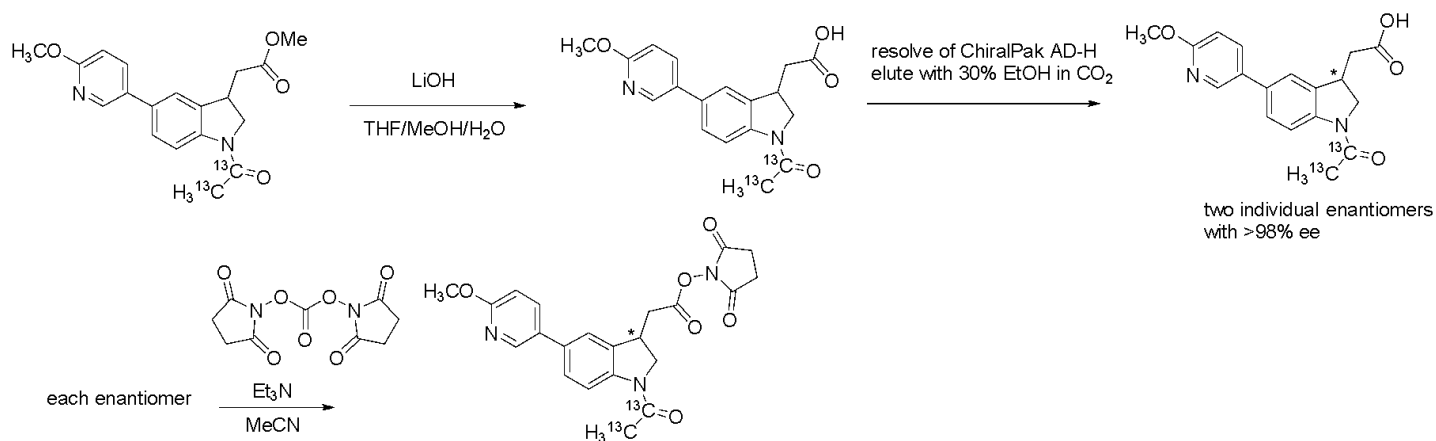
To a suspension of 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid (22.4 mg, 1 Eq, 68.6 μmol) in MeCN (0.5 mL) in a 40 mL vial was added N,N'-Disuccinimidyl Carbonate (21.1 mg, 0.02 mL, 1.2 Eq, 82.4 μmol) followed by N,N'-Disuccinimidyl Carbonate (21.1 mg, 0.02 mL, 1.2 Eq, 82.4 μmol). The reaction mixture became a colorless clear solution after adding the E3N. The mixture was stirred at room temperature for 1h, and LC-MS showed ~13% of acid remaining. 3.6 mg of N,N'-Disuccinimidyl Carbonate was added, stirred at RT for 30min. LC-MS showed ~40% of SM remained. 15 mg of N,N'-Disuccinimidyl Carbonate was added, stirred at RT for 1h. The mixture was diluted with 1 mL of brine and 1 mL of NaHCO₃ aq in the vial, extracted with DCM (4*1 mL, centrifuged to get a better separation in the vial). A pipet was used to remove the DCM out of the vial and all the organic layers were combined in a new vial, dried over Na₂SO₄, rotavap to dry. The residue was purified on silica gel (dry loading) eluting with 0-100% DCM in hexane then 0-35% EtOAc in DCM to get the pure product 10.7 mg as a white solid after freeze-drying. LCMS rt 5.53 min, [M+H]⁺: expected 424.1, observed 424.3

Peak 2: 2,5-Dioxopyrrolidin-1-yl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate

2-(1-Acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid (23 mg, 0.070 mmol), N,N'-Disuccinimidyl Carbonate (20 mg, 0.077 mmol), and TEA (31 μL, 0.22 mmol) were dissolved in 0.5 mL of CH₃CN. The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of water 1 mL of saturated NaHCO₃ solution, followed by extraction with DCM (3 × 20 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na₂SO₄. The solvent was removed under reduced pressure to give crude product. The crude residue was purified by column chromatography (DCM/EtOAc, 5:5) to provide **T0-peak2**. The product was dissolved in 5 mL of acetonitrile, lyophilized, and afforded **T0-peak2** (0.015 g, 51%) as off -white solid. ¹H NMR (300 MHz, CDCl₃) δ 8.35 (s, 1H) 8.28 (d, *J* = 8.1 Hz, 1H), 7.77 (dd, *J* = 8.4 Hz, 1H), 7.42 (d, *J* = 8.1 Hz, 1H), 7.37 (s, 1H), 6.81 (d, *J* = 8.7 Hz, 1H), 4.37 (t, *J* = 8.7 Hz, 1H), 3.98 (s, 3H) 3.97-3.88 (m, 2H), 3.2 – 3.12 (m, 1H), 2.94 – 2.91 (m, 1H), 2.90 – 2.88 (m, 4H), 2.27 (s, 3H); LCMS rt 5.38 min, [M+H]⁺: expected 424.1, observed 424.3

T2:





Preparation of Methyl 2-(1-(acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate

To the solution of methyl 2-(1-(acetyl-¹³C₂)-5-bromoindolin-3-yl)acetate (150.0 mg, 1 Eq, 477.5 μmol) in DME (3.6 mL) was added KOAc (93.72 mg, 2 Eq, 955.0 μmol) and Bis(pinacolato)diborane (181.9 mg, 1.5 Eq, 716.2 μmol), degassed and recharged with Argon for 3 times, then added Pd(dppf)Cl₂·CH₂Cl₂ (38.99 mg, 0.1 Eq, 47.75 μmol). The mixture was heated at 80 °C under Argon atmosphere for 70min. After cooling to RT, a solution of Na₂CO₃ (101.2 mg, 2 Eq, 955.1 μmol) in Water (1.8 mL) was added to the reaction mixture, followed by the solution of 5-bromo-2-methoxypyridine (114.5 mg, 1.3 Eq, 609.0 μmol) in DME (1.8 mL). After heating at 80 °C under Argon atmosphere for 2.5 hours, the solvent was removed under reduced pressure. The residue was purified by silica gel flash chromatography by using 0-40% EtOAc in hexane as the eluent to obtain the desired product (128.9 mg, 78.8% yield) as a yellow foamy solid. LCMS rt 5.67 min, [M+H]⁺ expected 343.1, observed 343.1.

Preparation of 2-(1-(Acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid

To the solution of Methyl 2-(1-(acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate (124.6 mg, 1 Eq, 363.9 μmol) in Methanol (0.5 mL), THF (1 mL), and Water (0.5 mL) was added LiOH (34.87 mg, 4 Eq, 1.456 mmol). The reaction mixture was stirred at RT for 3h, then acidified by 3M HCl aq to adjust the pH=2~3 to precipitate the product. The suspension was diluted with 2 mL of water, sonicated for 1min, centrifuged, and removed the clear liquid on the upper layer by pipet. Repeated the water wash twice (2×2mL), then used MeCN (3×2mL) instead of water, repeated 3 times. The solid was suspended in 1:1 MeCN/water, freeze-dried to obtain 2-(1-(Acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid (98.1 mg, 82.1% yield) as an off-white solid. LCMS rt 4.80 min, [M+H]⁺ expected 329.1, observed 329.1.

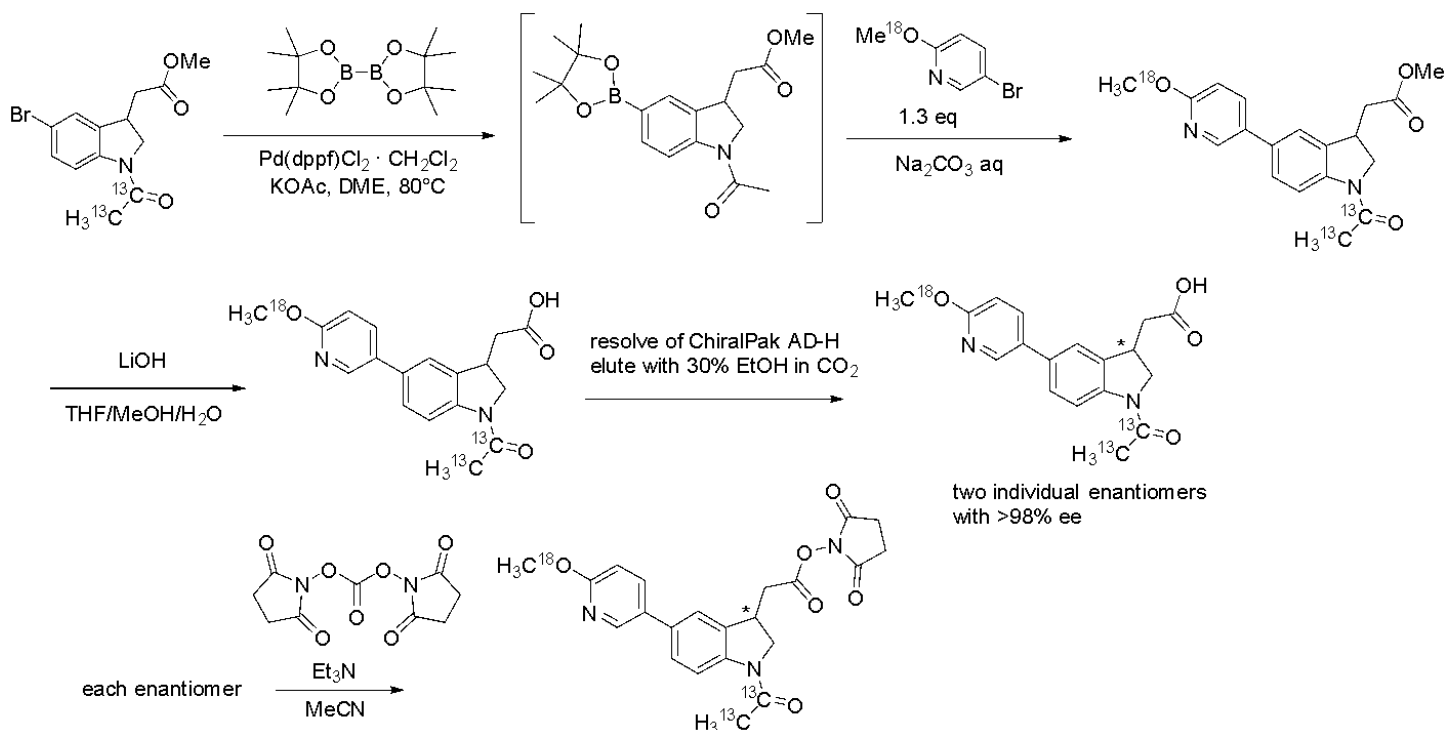
Resolution: ChiralPak AD-H 21x250 mm, with 30% ethanol in CO₂ and a flow rate of 70 mL/min. Sample dissolved at 2 mg/mL in 50:50 dichloromethane:methanol and loaded at 2 mL per injection to afford 21.8 mg of peak 1, >99.5% purity and ee, 26.1 mg of peak 2 with >99.5% purity and 98.98% ee.

Preparation of 2,5-Dioxopyrrolidin-1-yl 2-(1-(acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate

Peak 1: To the suspension of 2-(1-(Acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid (21.8 mg, 1 Eq, 66.4 μmol, Peak-1) in MeCN (0.5 mL) in a 40mL vial was added Et₃N (20.8 mg, 28.7 μL, 3.1 Eq, 206 μmol) followed by *N,N'*-Disuccinimidyl Carbonate (29.9 mg, 1.8 Eq, 117 μmol). The mixture was stirred at room temperature for 3h. The reaction mixture was quenched with 0.1M KH₂PO₄ aq (10mL), extracted with DCM (4×2 mL). The combined organic layers were dried over Na₂SO₄, filtered and removed the solvent under reduced pressure. The residue was purified by silica gel flash chromatography using 0-100% DCM in hexane then 0-40% EtOAc in DCM to obtain 2,5-Dioxopyrrolidin-1-yl 2-(1-(acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetate (18.0 mg, 63.7% yield) as a white solid after freeze-drying. LCMS rt 5.38 min, [M+H]⁺ 426.1 expected, 426.2 observed.

Peak 2: 2-(1-(acetyl- ^{13}C 2)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid (26.1 mg, 0.0795 mmol), N,N'-Disuccinimidyl Carbonate (22.4 mg, 0.0875 mmol), and TEA (34.3 μL , 0.246 mmol) were dissolved in 0.5 mL of CH_3CN . The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of water and 1 mL of saturated NH_4Cl solution, followed by extraction with DCM (3×20 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na_2SO_4 . The solvent was removed under reduced pressure to give crude product. The crude residue was purified by column chromatography (DCM/EtOAc, 5:5) to provide **T2-peak2**. The product was dissolved in 5 mL of acetonitrile, lyophilized to give **T2-peak2** (0.09 g, 27%) as off-white solid. ^1H NMR (300 MHz, CDCl_3) δ 8.35 (s, 1H) 8.28 (d, $J = 8.4$ Hz, 1H), 7.77 (dd, $J = 8.4$ Hz, 1H), 7.41 (d, $J = 8.7$ Hz, 1H), 7.37 (s, 1H), 6.81 (d, $J = 8.7$ Hz, 1H), 4.37 (t, $J = 9.3$ Hz, 1H), 3.98 (s, 3H) 3.97-3.88 (m, 2H), 3.2 – 3.13 (m, 1H), 2.94 – 2.8 (m, 5H), 2.25 (bd, $J = 128.7$ Hz, 3 H); LCMS rt 5.29 min, $[\text{M}+\text{H}]^+$: expected 426.1, observed 426.2

T4:



Preparation of 5-bromo-2-(methoxy- ^{18}O)pyridine

To a solution of methanol- ^{18}O (578 mg, 690 μL , 1.20 Eq, 17.0 mmol) in anhydrous THF (20 mL) under nitrogen with vigorous stirring was added the NaH (852.3 mg, 60% Wt, 1.5 Eq, 21.308 mmol) at 0 °C. After stirring at RT for 15 min, 5-bromo-2-fluoropyridine (2500.0 mg, 1 Eq, 14.205 mmol) was added dropwise. The resulting mixture was stirred at RT for 4h, then poured into saturated aq. NH_4Cl and extracted with EtOAc (5×10 mL). The combined organic layers were dried over Na_2SO_4 , filtered, and concentrated under reduced pressure to obtain a crude product, which was purified by silica gel flash chromatography by using 0-3% DCM in Hexanes as the eluent to afford 5-bromo-2-(methoxy- ^{18}O)pyridine (2405.1 mg, 89.1 % yield) as a colorless oil. ^1H NMR (300 MHz, CDCl_3 - d): δ 8.20 (d, $J = 2.7$ Hz, 1H), 7.63 (dd, $J = 8.7$ Hz, 2.7 Hz, 1.5 Hz, 1H), 6.66 (d, $J = 9.0$ Hz, 1H), 3.90 (s, 3H). ^{13}C NMR (75 MHz, CDCl_3 - d): δ 163.25, 147.76, 141.22, 112.78, 111.87, 53.85.

Preparation of Methyl 2-(1-(acetyl- ^{13}C 2)-5-(6-(methoxy- ^{18}O)pyridin-3-yl)indolin-3-yl)acetate

To the solution of methyl 2-(1-(acetyl- ^{13}C 2)-5-bromoindolin-3-yl)acetate (150.0 mg, 1 Eq, 477.5 μmol) in DME (3.6 mL) was added KOAc (93.72 mg, 2 Eq, 955.0 μmol) and Bis(pinacolato)diborane (144.5 mg, 1.2 Eq, 573.0 μmol), degassed and recharged with Argon for 3 times, then added $\text{Pd}(\text{dppf})\text{Cl}_2 \cdot \text{CH}_2\text{Cl}_2$ (38.99 mg, 0.1 Eq,

47.75 μmol). The mixture was heated at 80 °C under Argon atmosphere for 90min. After cooling to RT, a solution of Na_2CO_3 (101.2 mg, 2 Eq, 955.1 μmol) in Water (1.8 mL) was added to the reaction mixture, followed by the solution of 5-bromo-2-methoxypyridine (120.0 mg, 1.3 Eq, 631.5 μmol) in DME (1.8 mL). After heating at 80 °C under Argon atmosphere for 1h, the solvent was removed under reduced pressure. The residue was purified by silica gel flash chromatography by using 0-40% EtOAc in hexane as the eluent to obtain the desired product (139 mg, 84.5% yield) as a yellow foamy solid. LCMS rt 5.41 min, $[\text{M}+\text{H}]^+$ 346.2 expected, 346.2 observed.

Preparation of 2-(1-(Acetyl- $^{13}\text{C}_2$)-5-(6-(methoxy- ^{18}O)pyridin-3-yl)indolin-3-yl)acetic acid

To the solution of Methyl 2-(1-(acetyl- $^{13}\text{C}_2$)-5-(6-(methoxy- ^{18}O)pyridin-3-yl)indolin-3-yl)acetate (139 mg, 1 Eq, 404 μmol) in Methanol (0.5 mL), THF (1 mL), and Water (0.5 mL) was added LiOH (38.7 mg, 4 Eq, 1.61mmol). The reaction mixture was stirred at RT for 5h, then acidified by 3M HCl aq to adjust the pH=4~5 to precipitate the product. The suspension was diluted with 2 mL of water, sonicated for 1min, centrifuged, and removed the clear liquid on the upper layer by pipet. Repeated the water wash twice (2 $\times$ 2mL), then used MeCN (3 $\times$ 2mL) instead of water, repeated 3 times. The solid was suspended in 1:1 MeCN/water, freeze-dried to obtain 2-(1-(Acetyl- $^{13}\text{C}_2$)-5-(6-(methoxy- ^{18}O)pyridin-3-yl)indolin-3-yl)acetic acid (111 mg, 83.5% yield) as an off-white solid. LCMS rt 4.77 min, $[\text{M}+\text{H}]^+$ 331.1 expected, 331.1 observed

Resolution: ChiralPak AD-H 21 $\times$ 250 mm, with 30% ethanol in CO_2 and a flow rate of 70 mL/min. Sample dissolved at 2 mg/mL in 50:50 dichloromethane:methanol and loaded at 2 mL per injection to afford 33.4 mg of peak 1, >99.5% purity and ee, 35.3 mg of peak 2 with >99.5% purity and 98.72% ee.

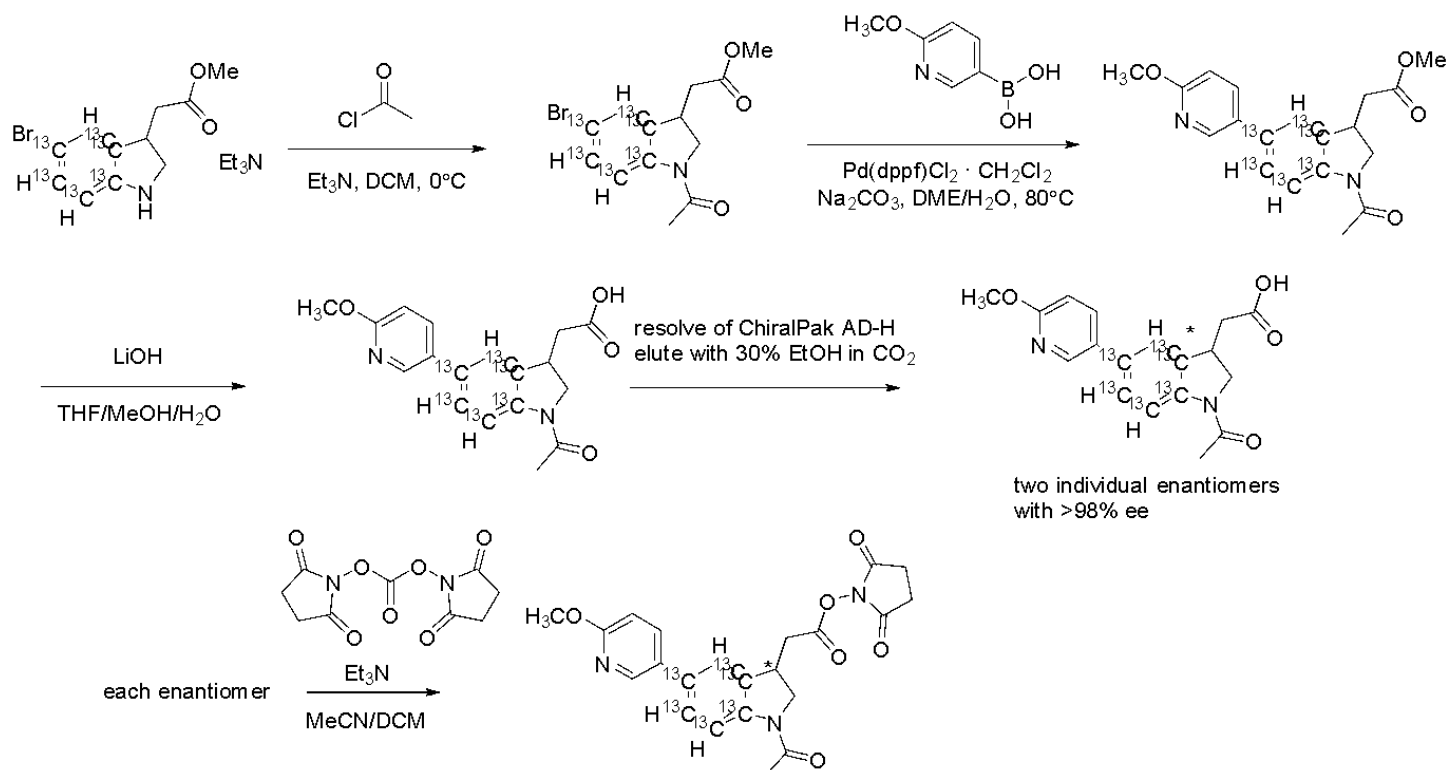
Preparation of 2,5-Dioxopyrrolidin-1-yl 2-(1-(acetyl- $^{13}\text{C}_2$)-5-(6-(methoxy- ^{18}O)pyridin-3-yl) indolin-3-yl)acetate

Peak 1: To the suspension of 2-(1-(acetyl- $^{13}\text{C}_2$)-5-(6-(methoxy- ^{18}O)pyridin-3-yl)indolin-3-yl)acetic acid (26.3 mg, 1 Eq, 79.6 μmol , Peak-1) in MeCN (0.5 mL) in a 40mL vial was added Et_3N (22 mg, 30 μL , 2.7 Eq, 215 μmol). The solvent was removed under reduced pressure and dried in high vacuum overnight. The solid was suspended in MeCN (0.5 mL) in a 20mL vial, and Et_3N (16.9 mg, 23.3 μL , 2.1 Eq, 167 μmol) was added, followed by *N,N'*-Disuccinimidyl Carbonate (29.2 mg, 1.4 Eq, 114 μmol). The mixture was stirred at room temperature for 1h. The reaction mixture was quenched with 0.1M KH_2PO_4 aq (10mL), extracted with DCM (4 $\times$ 2 mL). The combined organic layers were dried over Na_2SO_4 , filtered and removed the solvent under reduced pressure. The residue was purified by silica gel flash chromatography using 0-100% DCM in hexane then 0-40% EtOAc in DCM to obtain 2,5-Dioxopyrrolidin-1-yl 2-(1-(acetyl- $^{13}\text{C}_2$)-5-(6-(methoxy- ^{18}O)pyridin-3-yl) indolin-3-yl)acetate (20.9 mg, 61.4% yield) as a white solid after freeze-drying. LCMS rt 5.29 min, $[\text{M}+\text{H}]^+$ 428.2 expected, 428.1 observed

Peak 2:

2-(1-(acetyl- $^{13}\text{C}_2$)-5-(6-(methoxy- ^{18}O)pyridin-3-yl)indolin-3-yl)acetic acid (34mg, 0.10 mmol), *N,N'*-Disuccinimidyl Carbonate (29 mg, 0.11 mmol), and TEA (44 μL , 0.32 mmol) were dissolved in 0.5 mL of CH_3CN . The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of DCM and 10 mL of 0.1 M KH_2PO_4 solution, followed by extraction with DCM (3 $\times$ 20 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na_2SO_4 . The solvent was removed under reduced pressure to give crude product. The crude residue was purified by column chromatography (DCM/EtOAc, 5:5) to provide **T4-peak2**. The product was dissolved in 5 mL of acetonitrile, lyophilized, and give **T4-peak2** (0.035 g, 80%) as off-white solid. ^1H NMR (300 MHz, CDCl_3) δ 8.35 (s, 1H) 8.28 (d, J = 8.4 Hz, 1H), 7.77 (dd, J = 8.4 Hz, 1H), 7.41 (d, J = 8.7 Hz, 1H), 7.38 (s, 1H), 6.81 (d, J = 8.7 Hz, 1H), 4.37 (t, J = 8.7 Hz, 1H), 4.05-3.88 (m, 5H), 3.2 – 3.13 (m, 1H), 2.94 – 2.8 (m, 5H), 2.25 (bd, J = 128.4 Hz, 3 H); LCMS 5.57 min, $[\text{M}+\text{H}]^+$: 428.2 expected, 428.1 observed

T6:



Preparation of Methyl 2-(1-acetyl-5-bromoindolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate

To a solution of methyl 2-(5-bromoindolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate (200 mg, 1 Eq, 724 μmol) in DCM (5 mL) in an ice bath was added triethylamine (147 mg, 202 μL, 2 Eq, 1.45 mmol) followed by Acetyl chloride (62.6 mg, 56.3 μL, 1.1 Eq, 797 μmol). After stirring for 2h, the reaction mixture was quenched with sat. NaHCO₃ aq (2 mL). The aqueous layer was separated and washed with DCM (3×2 mL). The combined organic layers were dried over Na₂SO₄, filtered and removed the solvent under reduced pressure. The residue was purified by silica gel flash chromatography using 0-40% DCM in hexane then 0-40% EtOAc in Hexane to obtain methyl 2-(1-acetyl-5-bromoindolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate (95.6 mg, 41.5 % yield) as an off-white solid. LCMS rt 5.57 min, 318.0/320.0 expected, 318.2/320.2 observed.

Preparation of Methyl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate

To the suspension of methyl 2-(1-acetyl-5-bromoindolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate (142.6 mg, 1 Eq, 448.3 μmol), 2-Methoxy-5-pyridinylboronic Acid (82.27 mg, 1.2 Eq, 537.9 μmol) and Na₂CO₃ (95.02 mg, 2 Eq, 896.5 μmol) in DME (1.2 mL) and Water (0.4 mL) was added [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium (II) Complex with Dichloromethane (18.30 mg, 0.05 Eq, 22.41 μmol), then degassed and refilled by Argon for 3 times. The mixture was heated at 80 °C for 1h, then removed the solvent under reduced pressure. The residue was purified by silica gel flash chromatography using 0-40%EtOAc in Hexane to obtain Methyl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate (151 mg, 97.27% yield) as a very light yellow foamy semi-solid. LCMS rt 5.57 min, 346.2 expected, 346.1 observed.

Preparation of 2-(1-Acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetic acid

To the solution of Methyl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate (142.0 mg, 1 Eq, 410.0 μmol) in Methanol (0.5 mL), THF (1 mL), and Water (0.5 mL) was added LiOH (42.8 mg, 4.4 Eq, 1.78mmol). The reaction mixture was stirred at RT for 3h, then acidified by 3M HCl aq to adjust the

pH=4~5 to precipitate the product. The suspension was diluted with 4 mL of water, sonicated for 1min, centrifuged, and removed the clear liquid on the upper layer by pipet. Repeated the water wash twice (2×4mL), then used MeCN (3×2mL) instead of water, repeated 3 times. The solid was suspended in 1:1 MeCN/water, freeze-dried to obtain 2-(1-Acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetic acid (116 mg, 85.1% yield) as a white solid. LCMS rt 4.85 min, [M+H]⁺ 333.1 expected, 333.1 observed

Resolution: ChiralPak AD-H 21x250 mm, with 30% ethanol in CO₂ and a flow rate of 70 mL/min. Sample dissolved at 2 mg/mL in 50:50 dichloromethane:methanol and loaded at 2 mL per injection to afford 27 mg of peak 1, >99.5% purity and ee, 28.9 mg of peak 2 with >99.5% purity and ee.

Preparation of 2,5-Dioxopyrrolidin-1-yl

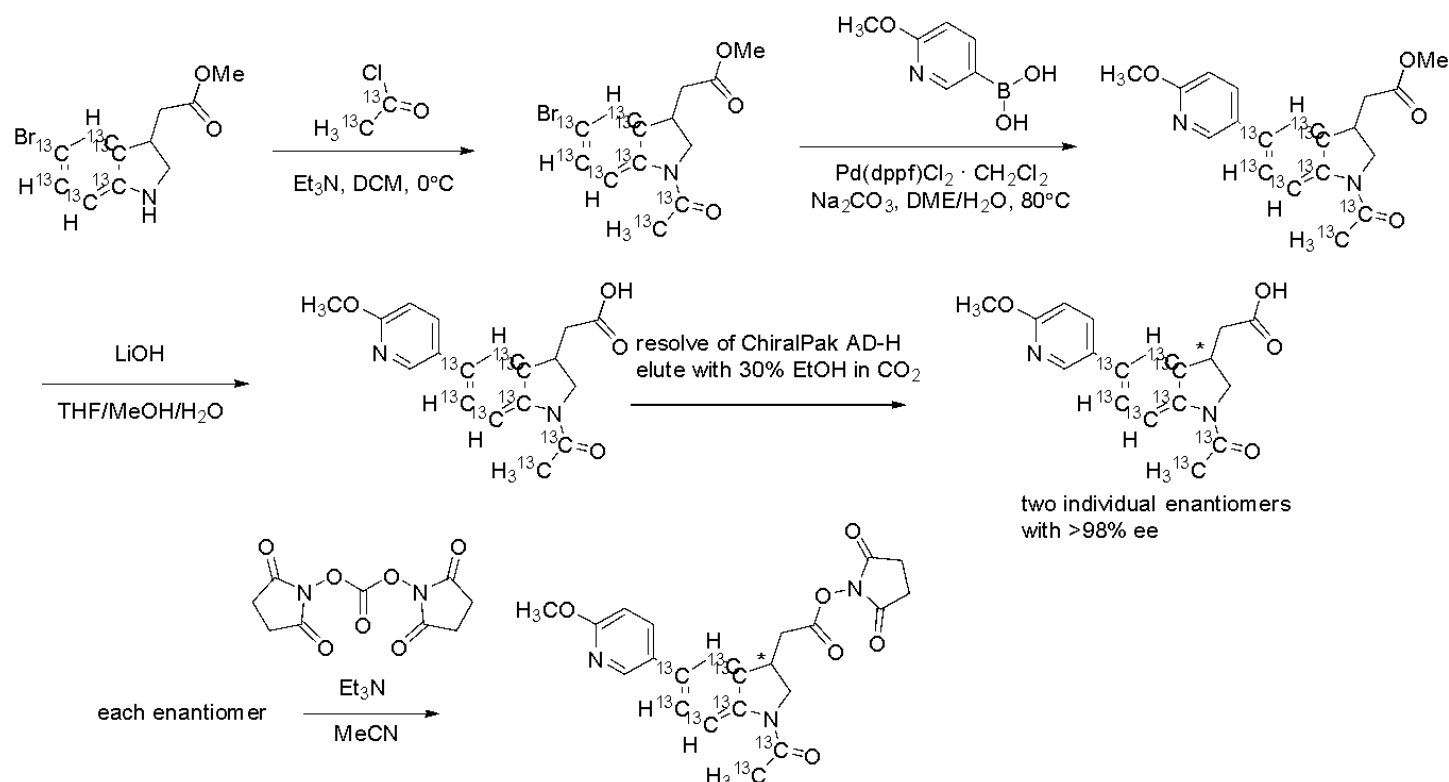
2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate

Peak 1: To the suspension of 2-(1-Acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetic acid (27 mg, 1 Eq, 81 μmol, Peak-1) in MeCN (0.5 mL) in a 20mL vial was added Et₃N (22 mg, 30 μL, 2.6 Eq, 215 μmol). The solvent was removed under reduced pressure and dried in high vacuum for 30 min. The solid was suspended in MeCN (0.5 mL) in a 20mL vial, and Et₃N (22 mg, 30 μL, 2.6 Eq, 215 μmol) was added, followed by *N,N'*-Disuccinimidyl Carbonate (29.7 mg, 1.4 Eq, 116 μmol). The mixture was stirred at room temperature for 1h. The reaction mixture was quenched with 0.1M KH₂PO₄ aq (10mL), extracted with DCM (4×2 mL). The combined organic layers were dried over Na₂SO₄, filtered and removed the solvent under reduced pressure. The residue was purified by silica gel flash chromatography using 0-100% DCM in hexane then 0-40% EtOAc in DCM to obtain 2,5-Dioxopyrrolidin-1-yl

2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetate (19.7 mg, 56.5% yield) as a white solid after freeze-drying. LCMS rt 5.48 min, [M+H]⁺ expected 430.2 observed 430.3

Peak 2: 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C₆)acetic acid (28mg, 0.084 mmol), *N,N'*-Disuccinimidyl Carbonate (24 mg, 0.092 mmol), and TEA (36 μL, 0.26 mmol) were dissolved in 0.5 mL of CH₃CN. The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of DCM and 10 mL of 0.1 M KH₂PO₄ solution, followed by extraction with DCM (3 × 20 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na₂SO₄. The solvent was removed under reduced pressure to give crude product. The crude residue was purified by column chromatography (DCM/EtOAc, 5:5) to provide **T6-peak2**. The product was dissolved in 5 mL of acetonitrile and lyophilized to give **T6-peak2** (0.021 g, 58%) as off-white solid. ¹H NMR (300 MHz, CDCl₃) δ 8.35 (s, 1H), 8.25 (bd, *J* = 167.4 Hz, 1 H), 7.79 – 7.63 (m, 2H), 7.17 – 7.10 (m, 1H), 6.81 (dd, *J* = 8.4 Hz, 1H), 4.37 (t, *J* = 8.4 Hz, 1H), 4.0– 3.88 (m, 5H), 3.2 – 3.13 (m, 1H), 2.94 – 2.8 (m, 5H), 2.28 (s, 3H); LCMS rt 5.55 min [M+H]⁺: expected 430.2 observed 430.3

T8:



Methyl 2-(1-(acetyl-¹³C₂)-5-bromoindolin-3-yl)-3a,4,5,6,7,7a-¹³C₆)acetate was prepared according to the protocol used for methyl 2-(1-acetyl-5-bromoindolin-3-yl)-3a,4,5,6,7,7a-¹³C₆)acetate. 271.5 mg (67%) LCMS rt 5.82 min, [M+H]⁺ expected 320.0/322.0 observed 320.3/322.3

Methyl 2-(1-(acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C₆)acetate was prepared according to the protocol used for methyl 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C₆)acetate. 99.0 mg (76%) LCMS rt 5.71 min, [M+H]⁺ expected 349.2 observed 349.1

2-(1-(Acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C₆)acetic acid was prepared according to the protocol used for 2-(1-Acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C₆)acetic acid 66.0 mg (76%) LCMS rt 4.87 min, [M+H]⁺ expected 335.2 observed 335.0

Resolution: ChiralPak AD-H 21x250 mm, with 30% ethanol in CO₂ and a flow rate of 70 mL/min. Sample dissolved at 2 mg/mL in 50:50 dichloromethane:methanol and loaded at 2 mL per injection to afford 23.6 mg of peak 1, >99.5% purity and ee, 24.7 mg of peak 2 with >99.5% purity and ee.

Peak1: 2,5-Dioxopyrrolidin-1-yl

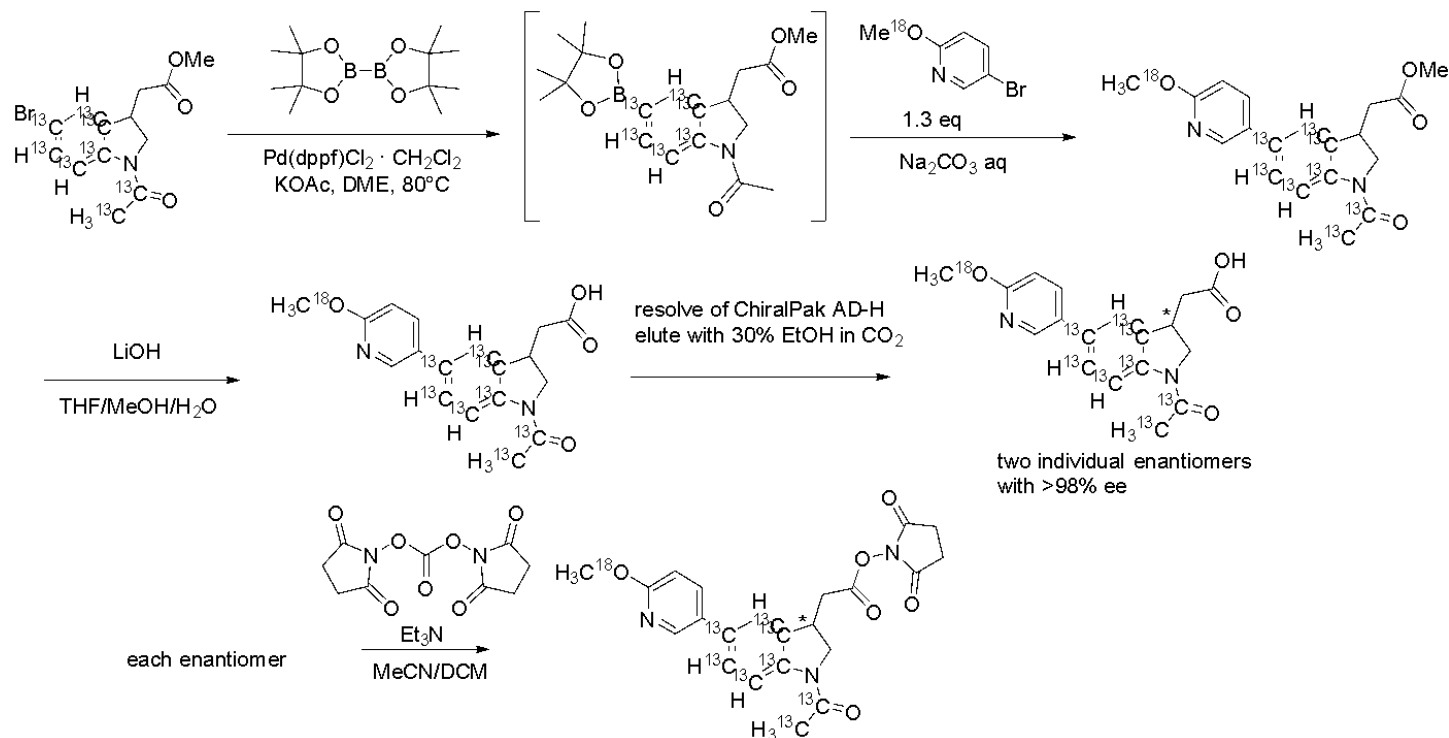
2-(1-(acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C₆)acetate was prepared according to the protocol used for 2,5-Dioxopyrrolidin-1-yl

2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C₆)acetate. 21.2 mg (70%) LCMS rt min, [M+H]⁺ 5.59 expected 432.2 observed 432.3

Peak 2: 2-(1-(acetyl-¹³C₂)-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C₆)acetic acid (24mg, 0.072 mmol), N,N'-Disuccinimidyl Carbonate (20 mg, 0.079 mmol), and TEA (31 μL, 0.22 mmol) were dissolved in 0.5 mL of CH₃CN. The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of DCM and 10 mL of saturated NaHCO₃ solution, followed by extraction with DCM (3 × 20 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na₂SO₄. The solvent was removed under reduced pressure to give crude product. The crude residue was purified by column chromatography (DCM/EtOAc, 5:5) to provide **T8-peak2**.

The product was dissolved in 5 mL of acetonitrile and lyophilized to give **T8-peak2** (0.010 g, 32%) as off-white solid. ¹H NMR (300 MHz, CDCl₃) δ 8.35 (s, 1H), 8.25 (bd, *J* = 168.6 Hz, 1 H), 7.79 – 7.60 (m, 2H), 7.17 – 7.10 (m, 1H), 6.81 (d, *J* = 9.0 Hz, 1H), 4.37 (t, *J* = 9.6 Hz, 1H), 3.98 (s, 3H) 3.97-3.88 (m, 2H), 3.2 – 3.13 (m, 1H), 2.94 – 2.8 (m, 5H), 2.25 (bd, *J* = 128.7 Hz, 3 H); LCMS rt 5.26 min, [M+H]⁺: expected 432.2 observed 432.1

T10:



Methyl 2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C6)acetate was prepared according to the protocol used for methyl 2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl)indolin-3-yl)acetate. 120 mg (75%) LCMS rt 5.38 min, [M+H]⁺ expected 351.1 observed 351.1

2-(1-(Acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C6)acetic acid was prepared according to the protocol used for 2-(1-Acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C6)acetic acid. 76 mg (79%) LCMS rt 4.95 min, [M+H]⁺ expected 337.2 observed 337.1

Resolution: ChiralPak AD-H 21x250 mm, with 30% ethanol in CO₂ and a flow rate of 70 mL/min. Sample dissolved at 2 mg/mL in 50:50 dichloromethane:methanol and loaded at 2 mL per injection to afford 32 mg of peak 1, >99.5% purity and ee, 33.9 mg of peak 2 with >99.5% purity and ee.

Peak 1: 2,5-Dioxopyrrolidin-1-yl

2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C6)acetate was prepared according to the protocol used for 2,5-Dioxopyrrolidin-1-yl

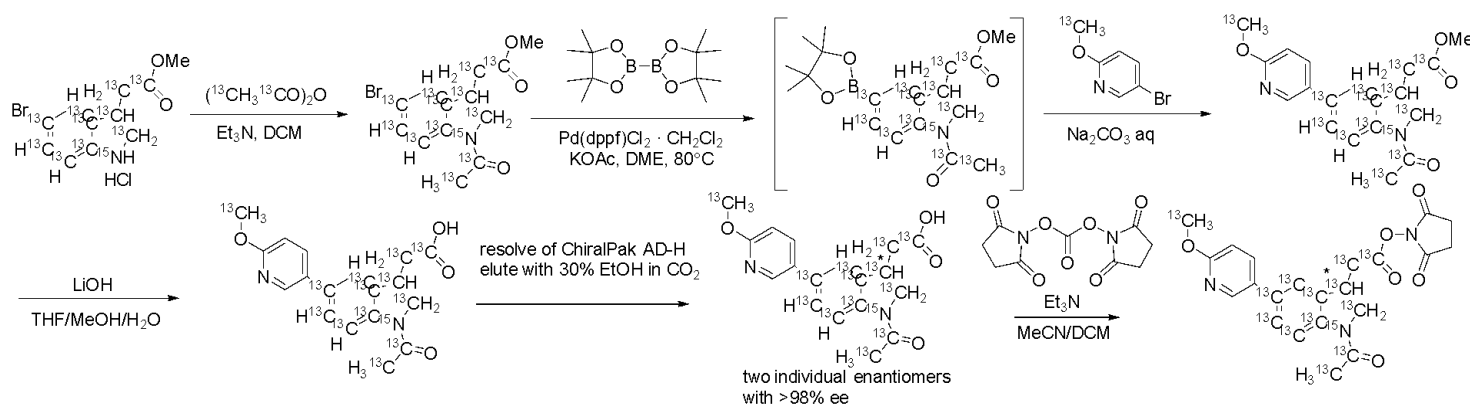
2-(1-(acetyl-¹³C2)-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C6)acetate. 31.8 mg (77%) LCMS rt 5.55 min, [M+H]⁺ expected 434.2 observed 434.2

Peak 2: 2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C6)acetic acid (33 mg, 0.072 mmol), N,N'-Disuccinimidyl Carbonate (28 mg, 0.11 mmol), and TEA (42 μL, 0.30 mmol) were dissolved in 0.5 mL of CH₃CN. The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of DCM and 10 mL of 0.1 M KH₂PO₄ solution, followed by extraction with DCM (3 × 20 mL). The combined organic layers were

2-(1-(acetyl-¹³C2)-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C6)acetate 26.3 mg (81%) LCMS rt 5.51 min, [M+H]⁺ expected 436.2 observed 436.2

Peak 2: 2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹³C)pyridin-3-yl)indolin-3-yl)-¹³C8-¹⁵N)acetic acid (27 mg, 0.080 mmol), N,N'-Disuccinimidyl Carbonate (23 mg, 0.088 mmol), and TEA (35 μ L, 0.25 mmol) were dissolved in 0.5 mL of CH₃CN. The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of DCM and 10 mL of 0.1 M KH₂PO₄ solution, followed by extraction with DCM (3 $\times$ 20 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na₂SO₄. The solvent was removed under reduced pressure to give crude product. The crude residue was purified by column chromatography (DCM/EtOAc, 5:5) to provide **T12-peak2**. The product was dissolved in 5 mL of acetonitrile and lyophilized to give **T12-peak2** (0.0253 g, 73 %) as off-white solid. ¹H NMR (300 MHz, CDCl₃) δ 8.35 (s, 1H), 8.25 (bd, *J* = 169.5 Hz, 1 H), 7.79 – 7.63 (m, 2H), 7.17 – 7.10 (m, 1H), 6.81 (dd, *J* = 8.4 Hz, 1H), 4.37 (t, *J* = 9.3 Hz, 1H), 4.29– 3.90 (m, 5H), 3.2 – 3.13 (m, 1H), 2.94 – 2.8 (m, 5H), 2.25 (bd, *J* = 128.1 Hz, 3 H); LCMS rt 5.44 min, [M+H]⁺: expected 436.2, observed 436.1

T14:



methyl 2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹³C)pyridin-3-yl)indolin-3-yl)-¹³C8-¹⁵N)acetate-¹³C2 was prepared according to the protocol used for methyl

2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl)indolin-3-yl)acetate 97.1 mg (89%) LCMS RT: 5.38 min, [M+H]⁺ 355.2 expected, observed 355.1

2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹³C)pyridin-3-yl)indolin-3-yl)-¹³C8-¹⁵N)acetic-¹³C2 acid was prepared according to the protocol used for 2-(1-(acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C6)acetic acid 79.1 mg LCMS rt 4.84 min, [M+H]⁺: expected 341.2, observed 341.0

Resolution: ChiralPak AD-H 21x250 mm, with 30% ethanol in CO₂ and a flow rate of 70 mL/min. Sample dissolved at 2 mg/mL in 50:50 dichloromethane:methanol and loaded at 2 mL per injection to afford 36.5 mg of peak 1, >99.5% purity and ee, 33.3 mg of peak 2 with >99.5% purity and ee.

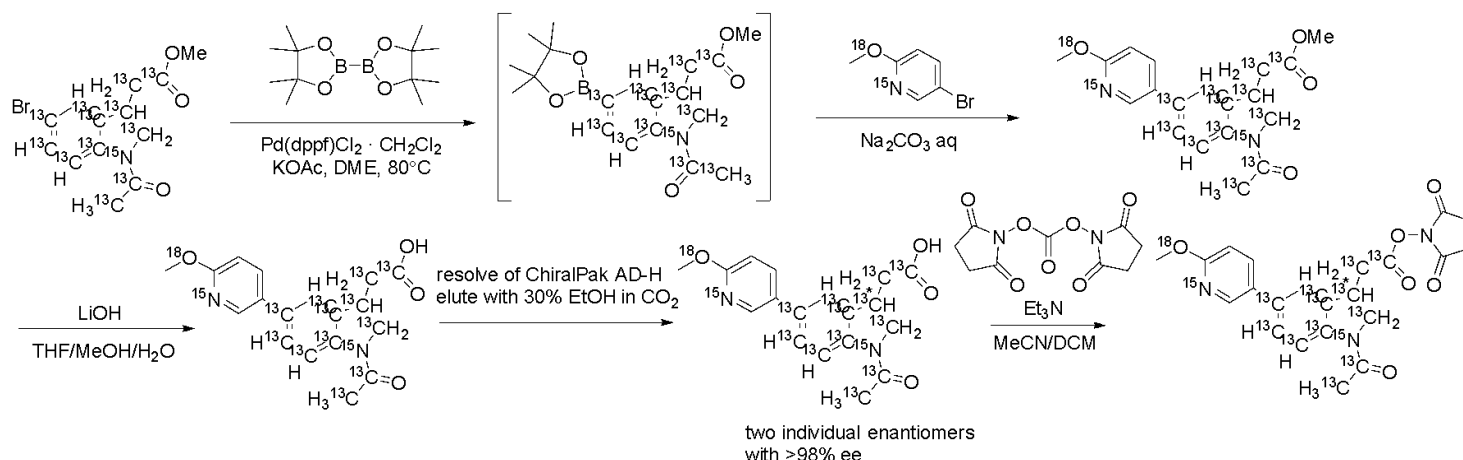
Peak 1: prepared according to the protocol used for 2,5-Dioxopyrrolidin-1-yl

2-(1-(acetyl-¹³C2)-5-(6-methoxypyridin-3-yl)indolin-3-yl)-3a,4,5,6,7,7a-¹³C6)acetate 33.3 mg (71%) LCMS rt 5.25 min, [M+H]⁺: expected 438.2, observed 437.8

Peak 2: 2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹³C)pyridin-3-yl)indolin-3-yl)-¹³C8-¹⁵N)acetic-¹³C2 acid (33 mg, 0.097 mmol), N,N'-Disuccinimidyl Carbonate (27 mg, 0.11 mmol), and TEA (42 μ L, 0.30 mmol) were dissolved in 0.5 mL of CH₃CN. The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of DCM and 10 mL of 0.1 M KH₂PO₄ solution, followed by extraction with DCM (3 $\times$ 20 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na₂SO₄. The solvent was removed under reduced pressure to give crude product. The crude residue was purified by column chromatography (DCM/EtOAc, 5:5) to provide **T14-peak2**.

The product was dissolved in 5 mL of acetonitrile and lyophilized to give **T14-peak2** (0.027 g, 64 %) as off-white solid. ¹H NMR (300 MHz, CDCl₃) δ 8.36 (s, 1H), 8.25 (bd, *J* = 167.1 Hz, 1 H), 7.76 – 7.67 (m, 2H), 7.19 – 7.12 (m, 1H), 6.83 (dd, *J* = 7.8 Hz, 1H), 4.23 – 4.152 (m, 3H), 3.80– 3.70 (m, 2H), 3.3 – 3.17 (m, 1H), 2.90 – 2.70 (m, 5H), 2.25 (bd, *J* = 128.7 Hz, 3 H); LCMS rt 5.60 min [M+H]⁺: expected 438.2, observed 437.7

T16:



Methyl 2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl-1-¹⁵N)indolin-3-yl-¹³C8-¹⁵N)acetate-¹³C2 was prepared according to the protocol used for methyl

2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl)indolin-3-yl)acetate 85.4 mg (78%) [M+H]⁺ 357.2 expected, observed 357.0

2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl-1-¹⁵N)indolin-3-yl-¹³C8-¹⁵N)acetic-¹³C2 acid was prepared according to the protocol used for 2-(1-(acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C6)acetic acid. 72.7 mg (89%) LCMS RT: 5.37min, [M+H]⁺ 343.2 expected, observed 343.2

Resolution: ChiralPak AD-H 21x250 mm, with 30% ethanol in CO₂ and a flow rate of 70 mL/min. Sample dissolved at 2 mg/mL in 50:50 dichloromethane:methanol and loaded at 2 mL per injection to afford 26.5 mg of peak 1, >99.5% purity and ee, 26.8 mg of peak 2 with >99.5% purity and 98.36% ee.

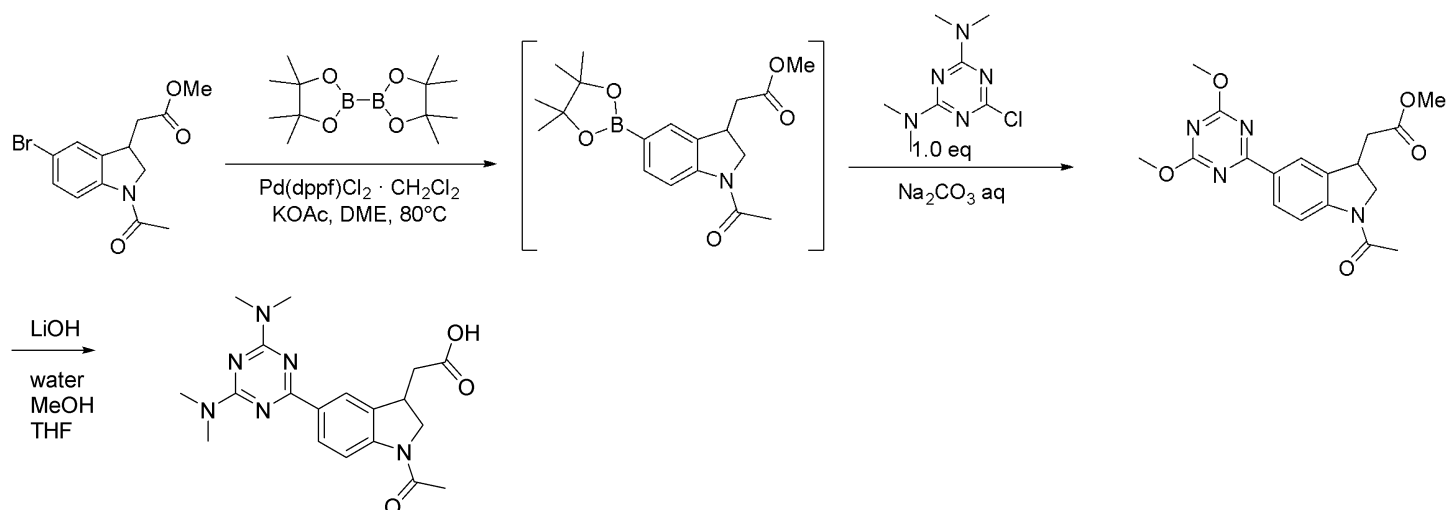
2,5-Dioxopyrrolidin-1-yl

2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl-1-¹⁵N)indolin-3-yl-¹³C8-¹⁵N)acetate-¹³C2

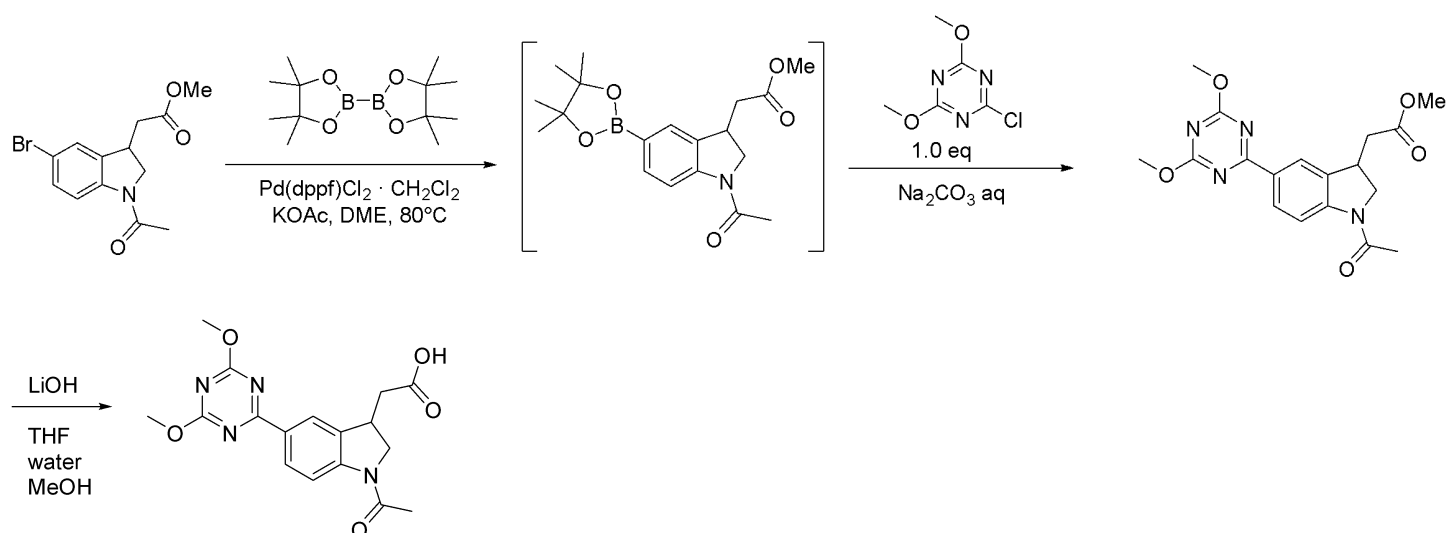
Peak 1: prepared according to the protocol used for 2,5-Dioxopyrrolidin-1-yl

2-(1-(acetyl-¹³C2)-5-(6-methoxypyridin-3-yl)indolin-3-yl-3a,4,5,6,7,7a-¹³C6)acetate 26.2 mg (77%) LCMS rt 5.48 min, [M+H]⁺ 440.2 expected, 439.9 observed

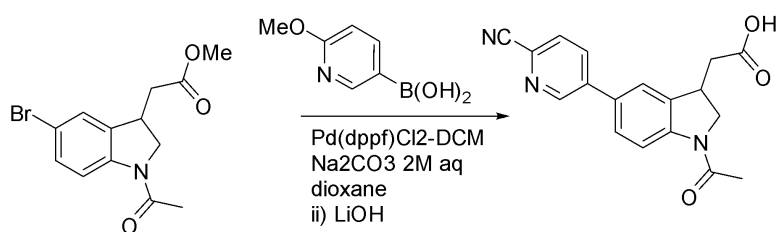
Peak 2: 2-(1-(acetyl-¹³C2)-5-(6-(methoxy-¹⁸O)pyridin-3-yl-1-¹⁵N)indolin-3-yl-¹³C8-¹⁵N)acetic-¹³C2 acid (26 mg, 0.076 mmol), N,N'-Disuccinimidyl Carbonate (22 mg, 0.084 mmol), and TEA (33 μL, 0.24 mmol) were dissolved in 0.5 mL of CH₃CN. The reaction was stirred for 1 hour at room temperature. LCMS indicated the consumption of the starting material. The reaction mixture was then diluted with 10 mL of DCM and 10 mL of 0.1 M KH₂PO₄ solution, followed by extraction with DCM (3 × 20 mL). The combined organic layers were washed brine solution (20 mL) and dried over Na₂SO₄. The solvent was removed under reduced pressure to give crude product. The crude residue was purified by column chromatography (DCM/EtOAc, 5:5) to provide **T16-peak2**. The product was dissolved in 5 mL of acetonitrile and lyophilized to give **T16-peak2** (0.0261 g, 78 %) as off-white solid. ¹H NMR (300 MHz, CDCl₃) δ 8.36 (s, 1H), 8.25 (bd, *J* = 167.1 Hz, 1 H), 7.76 – 7.67 (m, 2H), 7.19 – 7.12 (m, 1H), 6.83 (dd, *J* = 7.8 Hz, 1H), 4.20 – 4.10 (m, 4H), 3.80– 3.70 (m, 1H), 3.3 – 3.14 (m, 1H), 2.90 – 2.68 (m, 5H), 2.26 (bd, *J* = 128.7 Hz, 3 H); LCMS rt 5.51 min, [M+H]⁺ 440.2 expected, 439.9 observed



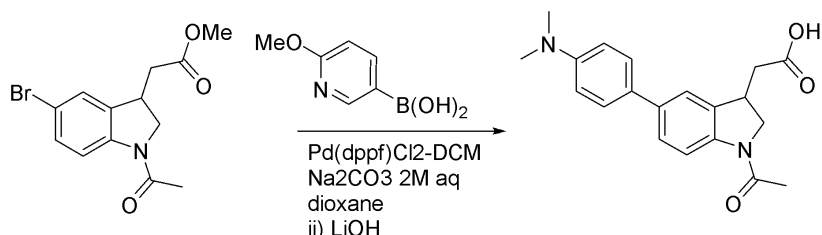
2-(1-acetyl-5-(4,6-bis(dimethylamino)-1,3,5-triazin-2-yl)indolin-3-yl)acetic acid was prepared according to the protocol used to make 2-(1-(Acetyl-¹³C2)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid. 73 mg LCMS rt 5.32 min, [M+H]⁺ 385.2 expected, 385.2 observed



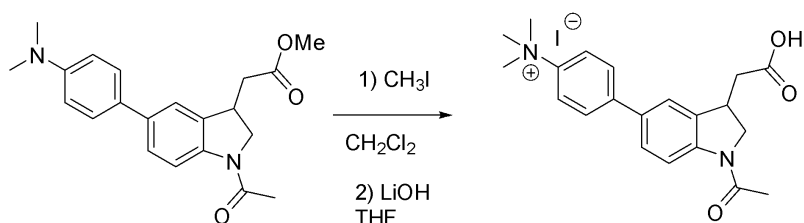
2-(1-acetyl-5-(4,6-dimethoxy-1,3,5-triazin-2-yl)indolin-3-yl)acetic acid was prepared according to the protocol used to make 2-(1-(Acetyl-¹³C2)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid. LCMS rt 4.76 min, [M+H]⁺ 359.1 expected, 359.3 observed



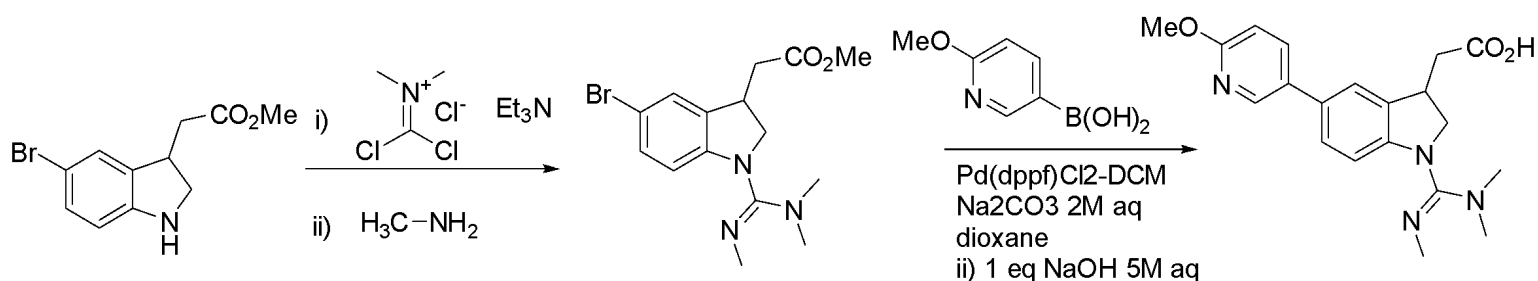
2-(1-acetyl-5-(6-cyanopyridin-3-yl)indolin-3-yl)acetic acid was prepared according to the protocol used to make 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid 12.3 mg LCMS rt 4.74 min, [M+H]⁺ 322.1 expected, 322.3 observed



2-(1-acetyl-5-(4-(dimethylamino)phenyl)indolin-3-yl)acetic acid was prepared according to the protocol used to make 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid 12.3 mg LCMS rt 4.41 min, [M-H]⁻ 339.2 expected, 339.1 observed

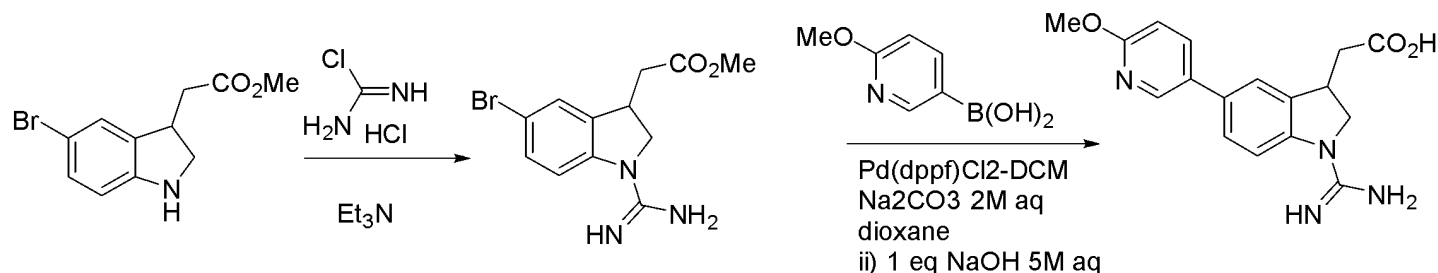


To methyl 2-(1-acetyl-5-(4-(dimethylamino)phenyl)indolin-3-yl)acetate (29.7 mg, 84.3 μmol) in DCM 1 mL was added iodomethane 23.9 mg (169 μmol) at room temp. After stirring overnight, the resulting precipitate was collected to afford 37.6 mg (90%) of 4-(1-acetyl-3-(2-methoxy-2-oxoethyl)indolin-5-yl)-N,N,N-trimethylbenzenaminium, Iodide (37.6 mg, 76.1 μmol) which was stirred with LiOH (4.0 mg, 170 μmol) in water (0.5 mL) with enough THF to solubilize (0.2 mL) for 3 days at 4 C. LCMS showed complete conversion to product. The THF was evaporated under a balloon with Ar. HCl (4 M, 42 μL, 170 μmol) was added. No ppt formed, so the mixture was evaporated for 10 min under a balloon of N₂ to remove any excess HCl, then frozen and lyophilized. The resulting solid was triturated with Et₂O. LCMS shows two peaks by UV and ES⁺/ES⁻ indicative of iodide (ES⁻ 127, very fast-eluting) and the quaternary ammonium salt. 22.3 mg (83%). LCMS rt 0.73 min, M⁻ 126.9 expected, 126.9 observed, rt 3.68 min, M⁺ 354.2 expected, 354.1 observed



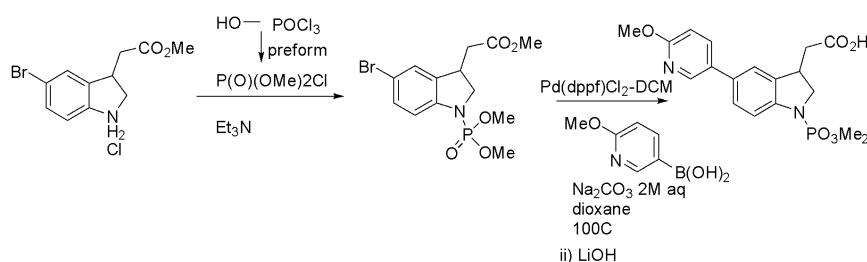
Methyl 2-(5-bromoindolin-3-yl)acetate (mg, 0. mmol) and Dichloromethylenedimethyliminium Chloride (67.4 mg, 0.52 mmol) were stirred in 2 mL ACN, and triethylamine was added. The reaction mixture was stirred for 2h, and an LCMS was taken by quenching an aliquot into MeNH₂/THF. After this LCMS showed predominant conversion to desired product, the rest of the reaction mix was quenched by the addition of 415 uL of 2M MeNH₂ in THF (2 eq, 830 umol). The reaction mix was purified by C18 chromatography with 0.1% formic acid modifier to afford 24.4 mg (17%) product.

2-(5-(6-methoxypyridin-3-yl)-1-(N,N,N'-trimethylcarbamimidoyl)indolin-3-yl)acetic acid was prepared by the protocol used to make 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid, with the exception that the ester hydrolysis step was conducted in situ with 1 eq 5M NaOH aq to afford 20.6 mg LCMS rt 4.52 min [M-H]⁻ 367.2 expected, 367.3 observed



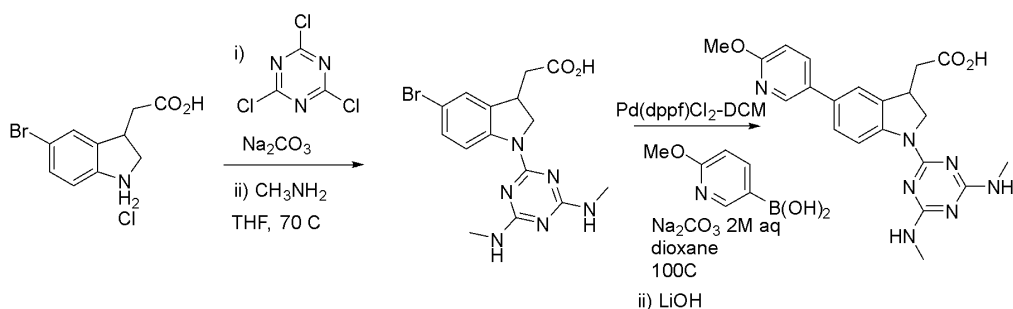
To methyl 2-(5-bromoindolin-3-yl)acetate (107 mg, 396 umol) in DCM 1 mL, at room temp, was added triethylamine (60 uL, 1.1 eq) and aminomethanecarbonimidoyl chloride hydrochloride (50 mg, 1.1 eq). LCMS after overnight shows substantial product formation (ES 312/4+), along with recovered starting material and a more polar unknown peak. The reaction mix was purified on Yamazen with formic acid-modified water/ACN on 14 g C18, 20 mL/min, 3% ACN over 4 min, ramp to 13% over 30 min, hold 10 min then ramp to 20% over 5 min, hold 5 min, then 3 min to 100% ACN and hold 2 min. Product elutes barely into first 3-13% ramp. 36.3 mg (29%).

2-(1-carbamimidoyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid was prepared using the protocol to make 2-(5-(6-methoxypyridin-3-yl)-1-(N,N,N'-trimethylcarbamimidoyl)indolin-3-yl)acetic acid 8.2 mg (21%) LCMS rt 5.54 min, [M-H]⁻ expected, 325.3 observed



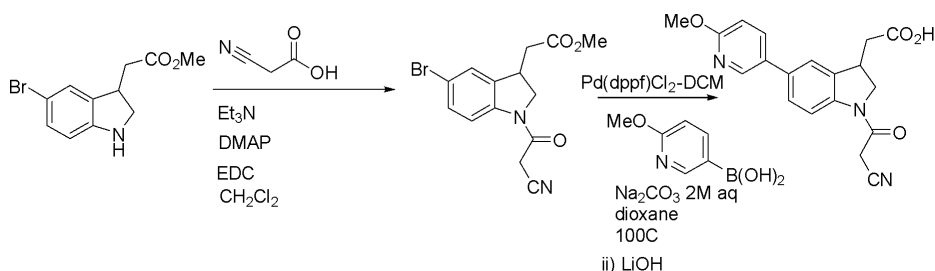
To POCl₃ in 1 mL DCM at 0 C was added dropwise a 1:1 premixed solution of MeOH and triethylamine in 1 mL DCM. A white precipitate formed immediately. The reaction mix was stirred for 1 h, and methyl 2-(5-bromo-1-chloro-1H-indolin-3-yl)acetate hydrochloride (50 mg, 0.16 mmol) was added, followed by triethylamine. The reaction mix was stirred overnight. Chromatography on a 25 g silica cartridge from 0-100% EtOAc/Hex at 20 mL/min over 1 h afforded product elution at around 70% EtOAc: 12 mg.

2-(1-(dimethoxyphosphoryl)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid was prepared using the protocol to make 2-(5-(6-methoxypyridin-3-yl)-1-(N,N,N'-trimethylcarbamimidoyl)indolin-3-yl)acetic acid. 3.5 mg LCMS rt 4.94 min, [M+H]⁺ 393.1 expected, 393.1 observed



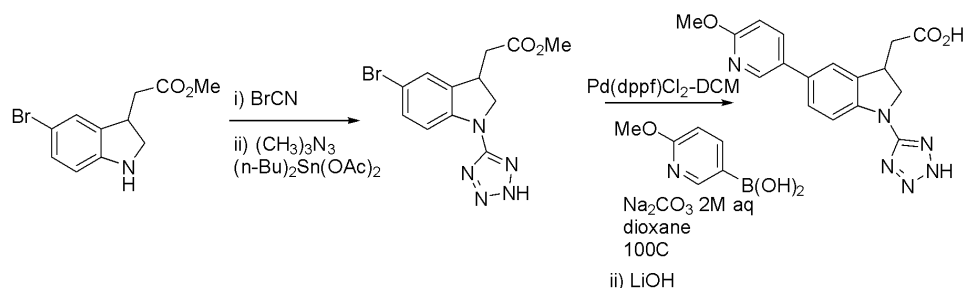
2-(5-bromo-1-chloro-1H-indolin-3-yl)acetic acid hydrochloride (57.1 mg, 0.195 mmol), sodium carbonate (42.2 mg, 0.781 mmol), and 2,4,6-trichloro-1,3,5-triazine (72.0 mg, 0.390 mmol) were stirred in THF 0.77 mL for 3 days (over the weekend). LCMS shows the major peak has ES+ 401/403/405, corresponding to the dichloro intermediate. 1 eq, 7 μ L of water was added in case any ester adduct of the trichlorotriazine with the acid was present, and the mix was stirred at rt for 20 min, followed by 4 eq, 781 μ L of methylamine 2M in THF. After stirring for 5h at 23 C, LCMS showed a major peak of the mono-NHMe adduct along with a minor peak of the corresponding methylamide, which indicates that either the time or water equivalents was insufficient with the 20 min stirring with 1 eq water. The mixture was heated at 70C overnight, after which LCMS indicated approx 1:1 of mono and bis MeNH₂-addition products, so an additional 10 eq of methylamine 2M in THF = 2 mL was added and heating at 70C was continued. LCMS showed complete conversion to desired product, and the mixture was evaporated to dryness under Ar at 50C. The mixture was purified on formic acid-modified Yamazen flash C18 to obtain 57.0 mg of pure product.

2-(1-(4,6-bis(methylamino)-1,3,5-triazin-2-yl)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid was prepared using the protocol to make 2-(5-(6-methoxypyridin-3-yl)-1-(N,N,N'-trimethylcarbamimidoyl)indolin-3-yl)acetic acid. 27.6 mg LCMS rt 5.27 min, [M+H]⁺ 422.1 expected, 422.2 observed



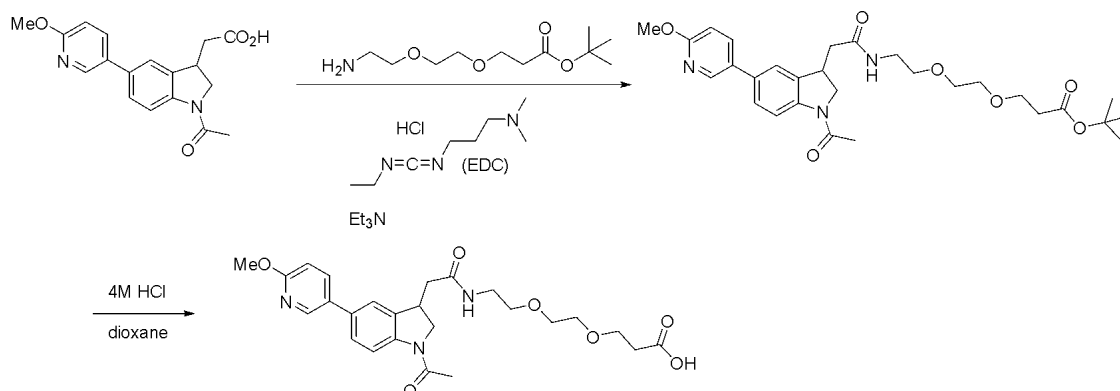
Methyl 2-(5-bromo-1-chloro-1H-indolin-3-yl)acetate (50 mg, 0.16 mmol), 2-cyanoacetic acid (18 mg, 0.21 mmol), triethylamine (68 μ L, 0.49 mmol), EDC-HCl (34 mg 0.18 mmol), and DMAP 2 mg (0.02 mmol) were stirred in 1 mL DCM at room temp overnight. LCMS showed product formation. The reaction mix was extracted between EtOAc and 1N HCl, then sat aq NaCl. The organic layer was dried over sodium sulfate and concentrated to dryness. Chromatography on a 25 g silica cartridge from 0-100% EtOAc/Hex at 20 mL/min over 1 h afforded product elution at around 70% EtOAc: 20 mg.

2-(1-(2-cyanoacetyl)-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid was prepared using the protocol to make 2-(5-(6-methoxypyridin-3-yl)-1-(N,N,N'-trimethylcarbamimidoyl)indolin-3-yl)acetic acid. 48 mg LCMS rt 4.98 min, [M+H]⁺ 352.1 expected, 352.1 observed



To a solution of BrCN (70.6 mg, 0.666 mmol) in THF (3 mL), indoline (180 mg, 0.666 mmol) was added at 0°C. The reaction mixture was stirred at room temperature for 12 hours. Afterward, dimethyl acetate and TMS azide were added, and the mixture was stirred for an additional 12 hours. LCMS analysis confirmed the formation of the desired tetrazole. The reaction mixture was directly loaded onto a CombiFlash instrument for purification, using a gradient from 10% hexane in ethyl acetate to 100% ethyl acetate over one hour. The compound eluted at 70% ethyl acetate in hexane, using a 25 g column at a flow rate of 20 mL per minute under pressure. The collected fractions were concentrated to yield 65 mg of the desired tetrazole.

2-(5-(6-methoxypyridin-3-yl)-1-(2H-tetrazol-5-yl)indolin-3-yl)acetic acid was prepared according to the method used to make 2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid. 65 mg LCMS rt 4.85 min, $[M+H]^+$ 353.1 expected, 353.3 observed



3-(2-(2-(2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetamido)ethoxy)ethoxy)propanoic acid

2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetic acid 19.0 mg (0.0582 mmol, 1 equivalent), tert-butyl 3-(2-(2-aminoethoxy)ethoxy)propanoate 17.7 mg (0.0757 mmol, 1.3 equivalents, cas 756525-95-8), and triethylamine (10.5 uL, 0.0757 mmol, 1.3 eq) were dissolved in DCM 0.3 mL.

N-(3-Dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride 13.4 mg (0.0699 mmol, 1.3 equivalents, Sigma 379115) was added. LCMS at 2h shows 44:32 starting material to product ratio. After overnight, similar profile; 6 mg more EDC added, after overnight, LCMS indicates no remaining acid starting material. The mixture was purified on silica gel chromatography on a 7g cartridge, 0-70% EtOAc/hex over 70 min at 8 mL/min - nothing eluted. The cartridge was flushed with 10% MeOH/DCM, and the material was repurified: 1-3% MeOH/DCM, then ramp quickly to 5% & hold: product elutes at 5%, but 1 fraction pure (1.5 mg), 1 fraction mixed which was repurified: 1-5% MeOH/DCM, product elutes quickly but got pure product 2.5 mg (total 3.9 mg, 12%).

To tert-butyl 3-(2-(2-(2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetamido)ethoxy)ethoxy)propanoate 2.4 mg was added 0.2 mL 4M HCl/dioxane. After 30 min, LCMS showed complete conversion to product, at

which point the mixture was evaporated to dryness to afford 1.5 mg of
3-(2-(2-(2-(1-acetyl-5-(6-methoxypyridin-3-yl)indolin-3-yl)acetamido)ethoxy)ethoxy)propanoic acid [M+H]⁺
486.2 expected, observed 486.3